JSS2 MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: BASIC OPERATION OF INTEGERS/WHOLE NUMBERS I

TOPIC: Whole Numbers and Decimal Numbers in Standard Form

CONTENT:

A. INTRODUCTION TO STANDARD FORM

Standard form (also called scientific notation) is a method of writing very large or very small numbers in a compact form. It is expressed as:

$A \times 10^n$

Where:

- A is a number between 1 and 10 ($1 \leq A < 10$)
- n is an integer (positive for large numbers, negative for small numbers)

Why Standard Form is Important:

- Makes very large and very small numbers easier to write
- Simplifies calculations with such numbers
- Used extensively in science, engineering, astronomy, and economics
- Reduces errors in reading and writing numbers

B. POWERS OF TEN

Understanding powers of 10 is fundamental to standard form:

Power	Value	Number
10^0	1	1

Power	Value	Number
10^1	10	10
10^2	10×10	100
10^3	$10 \times 10 \times 10$	1,000
10^4		10,000
10^5		100,000
10^6		1,000,000
10^9		1,000,000,000
10^{-1}	1/10	0.1
10^{-2}	1/100	0.01
10^{-3}	1/1,000	0.001
10^{-6}	1/1,000,000	0.000001

C. CONVERTING LARGE NUMBERS TO STANDARD FORM

Method:

1. Identify the first non-zero digit
2. Place a decimal point immediately after this digit
3. Count how many places the decimal point has moved from its original position
4. Write as $A \times 10^n$ where n is the number of places moved

Worked Examples:

Example 1: Express 5,000 in standard form

5,000 = 5.000
Decimal moved 3 places to the left
Answer: 5×10^3

Example 2: Express 456,000 in standard form

456,000 = 4.56000

Decimal moved 5 places to the left

Answer: 4.56×10^5

Example 3: Express 78,900,000 in standard form

78,900,000 = 7.89

Decimal moved 7 places to the left

Answer: 7.89×10^7

Example 4: Express 234 in standard form

234 = 2.34

Decimal moved 2 places to the left

Answer: 2.34×10^2

Example 5: Express 9,500,000,000 in standard form

9,500,000,000 = 9.5

Decimal moved 9 places to the left

Answer: 9.5×10^9

D. CONVERTING FROM STANDARD FORM TO ORDINARY FORM

Rule: When n is positive in $A \times 10^n$, move the decimal point n places to the RIGHT.

Worked Examples:

Example 6: Express 3.5×10^4 in ordinary form

Move decimal 4 places right

3.5 → 35 → 350 → 3,500 → 35,000

Answer: 35,000

Example 7: Express 6.78×10^6 in ordinary form

Move decimal 6 places right

Answer: 6,780,000

Example 8: Express 9.2×10^3 in ordinary form

$9.2 \times 1{,}000 = 9{,}200$

Answer: 9,200

Example 9: Express 2.34×10^8 in ordinary form

Answer: 234,000,000

Example 10: Express 7×10^5 in ordinary form

Answer: 700,000

E. SMALL NUMBERS IN STANDARD FORM (NEGATIVE INDICES)

For numbers less than 1, we use negative powers of 10.

Method:

1. Count how many places the decimal point must move RIGHT to get a number between 1 and 10
2. This count becomes the negative index

Worked Examples:

Example 11: Express 0.005 in standard form

$0.005 \rightarrow 5$ (decimal moved 3 places right)

Answer: 5×10^{-3}

Example 12: Express 0.000067 in standard form

$0.000067 \rightarrow 6.7$ (decimal moved 5 places right)

Answer: 6.7×10^{-5}

Example 13: Express 0.00000234 in standard form

0.00000234 → 2.34 (decimal moved 6 places right)
Answer: 2.34×10^{-6}

Example 14: Express 0.78 in standard form

0.78 → 7.8 (decimal moved 1 place right)
Answer: 7.8×10^{-1}

Example 15: Express 0.000000089 in standard form

Answer: 8.9×10^{-8}

F. CONVERTING NEGATIVE INDEX STANDARD FORM TO ORDINARY FORM

Rule: When n is negative in $A \times 10^{-n}$, move the decimal point n places to the LEFT.

Worked Examples:

Example 16: Express 3.5×10^{-4} in ordinary form

Move decimal 4 places left
3.5 → 0.35 → 0.035 → 0.0035 → 0.00035
Answer: 0.00035

Example 17: Express 7.8×10^{-3} in ordinary form

Answer: 0.0078

Example 18: Express 2.34×10^{-6} in ordinary form

Answer: 0.00000234

Example 19: Express 9×10^{-2} in ordinary form

Answer: 0.09

G. OPERATIONS WITH NUMBERS IN STANDARD FORM

1. MULTIPLICATION

Rule: $(a \times 10^{m}) \times (b \times 10^{n}) = (a \times b) \times 10^{(m+n)}$

Worked Examples:

Example 20: Calculate $(2 \times 10^{3}) \times (3 \times 10^{4})$

Solution:

$= (2 \times 3) \times 10^{(3+4)}$

$= 6 \times 10^{7}$

Example 21: Calculate $(4 \times 10^{5}) \times (2 \times 10^{2})$

Solution:

$= (4 \times 2) \times 10^{(5+2)}$

$= 8 \times 10^{7}$

Example 22: Calculate $(5 \times 10^{6}) \times (3 \times 10^{-2})$

Solution:

$= (5 \times 3) \times 10^{(6+(-2))}$

$= 15 \times 10^{4}$

$= 1.5 \times 10^{1} \times 10^{4}$

$= 1.5 \times 10^{5}$

Example 23: Calculate $(6.5 \times 10^{4}) \times (2 \times 10^{3})$

Solution:

$= (6.5 \times 2) \times 10^{(4+3)}$

$= 13 \times 10^{7}$

$= 1.3 \times 10^{8}$

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2. DIVISION

Rule: $(a \times 10^m) \div (b \times 10^n) = (a \div b) \times 10^{(m-n)}$

Worked Examples:

Example 24: Calculate $(8 \times 10^5) \div (2 \times 10^2)$

Solution:

$= (8 \div 2) \times 10^{(5-2)}$

$= 4 \times 10^3$

Example 25: Calculate $(9 \times 10^7) \div (3 \times 10^4)$

Solution:

$= (9 \div 3) \times 10^{(7-4)}$

$= 3 \times 10^3$

Example 26: Calculate $(6 \times 10^3) \div (2 \times 10^5)$

Solution:

$= (6 \div 2) \times 10^{(3-5)}$

$= 3 \times 10^{-2}$

Example 27: Calculate $(4 \times 10^5) \div (8 \times 10^2)$

Solution:

$= (4 \div 8) \times 10^{(5-2)}$

$= 0.5 \times 10^3$

$= 5 \times 10^{-1} \times 10^3$

$= 5 \times 10^2$

---

3. ADDITION AND SUBTRACTION

Important: Numbers must have the same power of 10.

Worked Examples:

Example 28: Calculate $(3 \times 10^4) + (5 \times 10^4)$

Solution:

$= (3 + 5) \times 10^4$

$= 8 \times 10^4$

Example 29: Calculate $(4 \times 10^5) + (2 \times 10^4)$

Solution:

Convert to same power:

$4 \times 10^5 = 40 \times 10^4$

$= (40 + 2) \times 10^4$

$= 42 \times 10^4$

$= 4.2 \times 10^5$

Example 30: Calculate $(7 \times 10^6) - (3 \times 10^6)$

Solution:

$= (7 - 3) \times 10^6$

$= 4 \times 10^6$

H. REAL-WORLD APPLICATIONS

Example 31: The distance from Earth to the Sun is 1.5×10^8 km. If light travels at 3×10^5 km/s, how long does light take to reach Earth?

Solution:

Time = Distance ÷ Speed

$= (1.5 \times 10^8) \div (3 \times 10^5)$

$= (1.5 \div 3) \times 10^{(8-5)}$

$= 0.5 \times 10^3$

$= 5 \times 10^2$ seconds

= 500 seconds or 8 minutes 20 seconds

Example 32: A country's population is 2×10^7 and land area is 4×10^5 km². Find the population density.

Solution:

Density = Population ÷ Area

$= (2 \times 10^7) \div (4 \times 10^5)$

$= 0.5 \times 10^2$

= 50 people per km²

Example 33: A bacteria colony has 3×10^5 bacteria. Each bacteria divides into 4. How many bacteria will there be?

Solution:

$= (3 \times 10^5) \times 4$

$= 12 \times 10^5$

$= 1.2 \times 10^6$ bacteria

EVALUATION:

1. Define standard form
2. Express in standard form:
   - a) 67,000
   - b) 0.00045
   - c) 345,000,000
3. Express in ordinary form:
   - a) 4.5×10^6
   - b) 7.8×10^{-4}
4. Calculate: $(3 \times 10^4) \times (2 \times 10^5)$
5. Calculate: $(8 \times 10^7) \div (4 \times 10^3)$

REVISION QUESTIONS:

1. What is the coefficient in 6.78×10^5?
2. What is the index in 9.2×10^{-3}?

3. Is 12.5×10^4 in correct standard form? If not, correct it.
4. Arrange in ascending order: 4.5×10^3, 6.7×10^2, 2.3×10^3
5. Which is larger: 5.6×10^{-3} or 7.8×10^{-4}? Explain.
6. Why is standard form useful in scientific calculations?
7. Convert 5.6×10^3 meters to kilometers
8. The speed of sound is 3.4×10^2 m/s. How far does sound travel in 5×10^2 seconds?
9. A computer can perform 4×10^9 calculations per second. How many in 3×10^2 seconds?
10. Compare: Number of stars in Milky Way (2×10^{11}) vs cells in human body (3.7×10^{13}). Which is larger and by how many times?

ASSIGNMENT:

Section A: Convert to standard form:

1. 8,900,000
2. 0.000456
3. 567,000,000
4. 0.00000078
5. 45,600
6. 0.0234
7. 789,000,000,000
8. 0.000000345

Section B: Convert to ordinary form:

1. 7.8×10^6
2. 4.56×10^{-4}
3. 9.2×10^8
4. 3.4×10^{-6}
5. 6.78×10^5
6. 2.3×10^{-2}

7. 8.9×10^9
8. 5.67×10^{-5}

Section C: Perform the following operations:

1. $(5 \times 10^4) \times (3 \times 10^6)$
2. $(8 \times 10^8) \div (2 \times 10^5)$
3. $(6 \times 10^5) + (4 \times 10^4)$
4. $(9 \times 10^7) - (3 \times 10^6)$
5. $(2.5 \times 10^4) \times (4 \times 10^{-2})$
6. $(7.2 \times 10^6) \div (3 \times 10^2)$
7. $(3.6 \times 10^5) \times (2.5 \times 10^3)$
8. $(4.8 \times 10^{-3}) \div (1.2 \times 10^{-5})$

Section D: Word Problems:

1. The Andromeda Galaxy is 2.4×10^6 light-years away. If one light-year equals 9.5×10^{12} km, how far is Andromeda in kilometers?

2. A human body contains 3.7×10^{13} cells. If each cell has a mass of 1×10^{-9} grams, what is the total mass of all cells?

3. A country's debt is 1.5×10^{12} dollars with a population of 3×10^8 people. What is the debt per person?

4. The mass of Earth is 6×10^{24} kg and the moon is 7×10^{22} kg. How many times heavier is Earth than the moon?

5. A water treatment plant processes 4.8×10^6 liters daily. Each liter contains 2.5×10^8 bacteria before treatment. How many bacteria are processed daily?

WEEK 2: BASIC OPERATION OF INTEGERS/WHOLE NUMBERS II

TOPIC: Prime Factors and Operations on Whole Numbers

CONTENT:

A. PRIME AND COMPOSITE NUMBERS

Definition of Prime Number: A prime number is a whole number greater than 1 that has exactly two factors: 1 and itself.

Examples of Prime Numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97...

Definition of Composite Number: A composite number is a whole number greater than 1 that has more than two factors.

Examples of Composite Numbers: 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21, 22, 24, 25, 26, 27, 28, 30...

Special Cases:

- **0:** Neither prime nor composite
- **1:** Neither prime nor composite (has only one factor)
- **2:** The only even prime number (all other even numbers are divisible by 2)

B. TESTS FOR DIVISIBILITY

Divisibility by 2: Last digit is 0, 2, 4, 6, or 8

- Examples: 24, 56, 108, 230

Divisibility by 3: Sum of digits is divisible by 3

- Example: 147 → 1+4+7 = 12 → 12÷3 = 4 ✓

Divisibility by 4: Last two digits form a number divisible by 4

- Example: 1,324 → 24÷4 = 6 ✓

Divisibility by 5: Last digit is 0 or 5

- Examples: 45, 80, 125, 1000

Divisibility by 6: Divisible by both 2 and 3

- Example: 48 (even, and 4+8=12 is divisible by 3)

Divisibility by 9: Sum of digits is divisible by 9

- Example: 729 → 7+2+9 = 18 → 18÷9 = 2 ✓

Divisibility by 10: Last digit is 0

- Examples: 50, 120, 1000

Divisibility by 11: Difference between sum of digits at odd positions and sum at even positions is 0 or divisible by 11

- Example: 121 → (1+1) - 2 = 0 ✓

C. PRIME FACTORIZATION

Definition: Prime factorization is expressing a composite number as a product of its prime factors.

Method 1: Division Method (Ladder Method)

Example 1: Express 60 as a product of prime factors

2 | 60
2 | 30
3 | 15
5 | 5
| 1

$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Example 2: Express 72 as a product of prime factors

2 | 72

2 | 36

2 | 18

3 | 9

3 | 3

| 1

$72 = 2 \times 2 \times 2 \times 3 \times 3 = 2^3 \times 3^2$

Example 3: Express 120 as a product of prime factors

2 | 120

2 | 60

2 | 30

3 | 15

5 | 5

| 1

$120 = 2 \times 2 \times 2 \times 3 \times 5 = 2^3 \times 3 \times 5$

Example 4: Express 144 as a product of prime factors

2 | 144

2 | 72

2 | 36

2 | 18

3 | 9

3 | 3

| 1

$144 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 = 2^4 \times 3^2$

Method 2: Factor Tree Method

Example 5: Express 60 as a product of prime factors

```
     60
    /  \
   6    10
  / \   / \
 2   3 2   5
```

$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5$

Example 6: Express 84 as a product of prime factors

```
     84
    /  \
   4    21
  / \   / \
 2   2 3   7
```

$84 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$

Example 7: Express 90 as a product of prime factors

```
     90
    /  \
   9    10
  / \   / \
 3   3 2   5
```

$90 = 2 \times 3 \times 3 \times 5 = 2 \times 3^2 \times 5$

Example 8: Express 150 as a product of prime factors

```
     150
    /   \
   15    10
  / \    / \
 3   5  2   5
```

$150 = 2 \times 3 \times 5 \times 5 = 2 \times 3 \times 5^2$

D. WRITING IN INDEX FORM

When prime factors repeat, we use indices (powers):

Examples:

- $2 \times 2 \times 2 = 2^3$
- $3 \times 3 = 3^2$
- $5 \times 5 \times 5 \times 5 = 5^4$
- $2 \times 2 \times 3 \times 3 \times 3 = 2^2 \times 3^3$

Example 9: Express 200 in prime factor form with indices

```
2 | 200
2 | 100
2 | 50
5 | 25
5 | 5
  | 1
```

$200 = 2 \times 2 \times 2 \times 5 \times 5 = 2^3 \times 5^2$

Example 10: Express 360 in prime factor form with indices

```
2 | 360
2 | 180
```

2 | 90

3 | 45

3 | 15

5 | 5

| 1

$360 = 2 \times 2 \times 2 \times 3 \times 3 \times 5 = 2^3 \times 3^2 \times 5$

E. APPLICATIONS OF PRIME FACTORIZATION

Application 1: Finding All Factors of a Number

Example 11: Find all factors of 60 using prime factorization

$60 = 2^2 \times 3 \times 5$

Factors: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60
(Combine prime factors in all possible ways)

Application 2: Determining if a Number is a Perfect Square

A number is a perfect square if all indices in its prime factorization are even.

Example 12: Is 144 a perfect square?

$144 = 2^4 \times 3^2$
All indices are even (4 and 2)
Therefore, 144 is a perfect square
$\sqrt{144} = 2^2 \times 3 = 12$

Example 13: Is 72 a perfect square?

$72 = 2^3 \times 3^2$
Index of 2 is odd (3)
Therefore, 72 is NOT a perfect square

F. ADDITION OF WHOLE NUMBERS

Rules:

1. Align numbers by place value (units under units, tens under tens, etc.)
2. Add from right to left
3. Carry over when sum exceeds 9

Example 14:

34,567
+ 12,845

47,412

Example 15:

456,789
+ 234,567
+ 123,456

814,812

Example 16: Word Problem

A school has 12,345 students in the morning shift and 8,976 in the afternoon shift. How many students in total?

Solution:

12,345
+ 8,976

21,321 students

G. SUBTRACTION OF WHOLE NUMBERS

Rules:

1. Align numbers by place value
2. Subtract from right to left
3. Borrow from the next column when necessary

Example 17:

  50,000
- 23,456

____

  26,544

Example 18:

  100,000
- 67,845

____

  32,155

Example 19: Word Problem

A town had a population of 456,789. After 10 years,
123,456 people migrated. What is the new population?

Solution:

  456,789
- 123,456

____

  333,333 people

H. MULTIPLICATION OF WHOLE NUMBERS

Method 1: Long Multiplication

Example 20:

$$
\begin{array}{rl}
456 & \\
\times\ 34 & \\
\hline
1824 & (456 \times 4) \\
13680 & (456 \times 30) \\
\hline
15,504 &
\end{array}
$$

Example 21:

$$
\begin{array}{rl}
789 & \\
\times\ 123 & \\
\hline
2367 & (789 \times 3) \\
15780 & (789 \times 20) \\
78900 & (789 \times 100) \\
\hline
97,047 &
\end{array}
$$

Method 2: Using Prime Factorization

Example 22: Find the product of 12 × 18 using prime factors

$12 = 2^2 \times 3$

$18 = 2 \times 3^2$

$12 \times 18 = 2^2 \times 3 \times 2 \times 3^2$

$= 2^3 \times 3^3$

$= 8 \times 27$

$= 216$

I. DIVISION OF WHOLE NUMBERS

Method 1: Long Division

Example 23: 4,536 ÷ 12

```
    378
  ______
12 | 4536
     36
     __
      93
      84
      __
       96
       96
       __
        0
```

Answer: 378

Example 24: 5,678 ÷ 23

```
    246 remainder 20
  _______________
23 | 5678
     46
     __
     107
      92
      __
      158
```

138

——

20

Answer: 246 R 20 or $246\frac{20}{23}$

Method 2: Division Using Prime Factors

Example 25: Divide 144 by 12 using prime factorization

$144 = 2^4 \times 3^2$

$12 = 2^2 \times 3$

$144 \div 12 = (2^4 \times 3^2) \div (2^2 \times 3)$

$= 2^{(4-2)} \times 3^{(2-1)}$

$= 2^2 \times 3^1$

$= 4 \times 3$

$= 12$

J. ORDER OF OPERATIONS (BODMAS/PEDMAS)

B - Brackets

O - Of (Orders/Powers/Indices)

D - Division

M - Multiplication

A - Addition

S - Subtraction

Example 26: Calculate: $24 \div 4 + 3 \times 2$

Solution:

$= 24 \div 4 + 3 \times 2$

$= 6 + 6$ (division and multiplication first)

$= 12$

Example 27: Calculate: $(15 + 5) \times 3 - 8 \div 2$

Solution:

$= (15 + 5) \times 3 - 8 \div 2$

$= 20 \times 3 - 8 \div 2$ (brackets first)

$= 60 - 4$ (multiplication and division)

$= 56$

Example 28: Calculate: $48 \div \{12 - (3 + 1)\}$

Solution:

$= 48 \div \{12 - (3 + 1)\}$

$= 48 \div \{12 - 4\}$ (inner bracket)

$= 48 \div 8$ (outer bracket)

$= 6$

Example 29: Calculate: $2^3 \times 5 + 10 \div 2$

Solution:

$= 2^3 \times 5 + 10 \div 2$

$= 8 \times 5 + 10 \div 2$ (indices first)

$= 40 + 5$ (multiplication and division)

$= 45$

Example 30: Calculate: $100 - \{50 \div (2 \times 5) + 15\}$

Solution:

$= 100 - \{50 \div (2 \times 5) + 15\}$

$= 100 - \{50 \div 10 + 15\}$

$= 100 - \{5 + 15\}$

$= 100 - 20$

$= 80$

K. WORD PROBLEMS INVOLVING WHOLE NUMBERS

Example 31: A factory produces 12,345 items daily. How many items in 15 days?

Solution:

12,345 × 15 = 185,175 items

Example 32: A school bought 240 books at ₦350 each. What is the total cost?

Solution:

240 × 350 = ₦84,000

Example 33: A farmer harvested 5,600 kg of rice from 8 equal plots. How much rice per plot?

Solution:

5,600 ÷ 8 = 700 kg per plot

Example 34: A bus travels 85 km per hour. How far in 6 hours?

Solution:

85 × 6 = 510 km

Example 35: A trader bought goods for ₦145,000 and sold them for ₦198,000. What is the profit?

Solution:

Profit = Selling Price - Cost Price

= 198,000 - 145,000

= ₦53,000

EVALUATION:

1. Define prime and composite numbers with examples
2. List the first 15 prime numbers
3. Express 180 as a product of prime factors in index form
4. Find all factors of 48 using prime factorization

5. Calculate: 23,456 + 17,889
6. Calculate: 100,000 - 47,856
7. Calculate: 456 × 78
8. Calculate: 5,832 ÷ 24
9. Is 196 a perfect square? Use prime factorization to explain
10. Evaluate: 48 ÷ (8 - 2) + 5 × 3

REVISION QUESTIONS:

1. Why is 2 the only even prime number?
2. Why is 1 neither prime nor composite?
3. State the divisibility rule for 3, 6, and 9
4. Express 360 as a product of its prime factors
5. How many prime numbers are there between 1 and 50?
6. What is the largest 2-digit prime number?
7. Express $2^5 \times 3^2 \times 5$ as an ordinary number
8. A number when divided by 2, 3, and 5 leaves no remainder. What is the smallest such number?
9. Calculate: 2,456 + 3,789 - 1,234
10. Multiply 234 by 56 and express the answer in standard form
11. Using BODMAS, evaluate: 120 ÷ {15 - (3 × 2)} + 8
12. Find the prime factorization of 1,000
13. How can you use prime factorization to find the LCM and HCF of numbers?
14. Express $2^4 \times 3^3 \times 5^2$ in ordinary form
15. A number has prime factorization $2^3 \times 3^2 \times 7$. Is it a perfect square? Why?

ASSIGNMENT:

Section A: Prime and Composite Numbers

1. Classify as prime or composite: 17, 21, 29, 33, 37, 41, 51, 53, 57, 59
2. List all prime numbers between 50 and 100
3. Find all factors of: 36, 48, 64, 100
4. Why is 91 not a prime number even though it seems like one?
5. What is the sum of all prime numbers between 1 and 20?

Section B: Prime Factorization

1. Express in prime factor form using the division method:
   - a) 56
   - b) 96
   - c) 132
   - d) 210
   - e) 315
2. Express in prime factor form using the factor tree method:
   - a) 64
   - b) 100
   - c) 180
   - d) 240
   - e) 450
3. Write in index form:
   - a) $2 \times 2 \times 2 \times 3 \times 3 \times 5$
   - b) $2 \times 2 \times 2 \times 2 \times 5 \times 5 \times 5$
   - c) $3 \times 3 \times 3 \times 7 \times 7$
   - d) $2 \times 2 \times 3 \times 3 \times 3 \times 11$
4. Express as ordinary numbers:
   - a) $2^3 \times 3^2$
   - b) $2^4 \times 5^2$
   - c) $3^3 \times 5 \times 7$

- d) $2^5 \times 3 \times 11$

5. Determine if the following are perfect squares using prime factorization:

    - a) 225
    - b) 288
    - c) 400
    - d) 150

Section C: Basic Operations

1. Add: 456,789 + 234,567 + 123,456
2. Subtract: 1,000,000 - 678,945
3. Multiply: 567 × 89
4. Divide: 8,456 ÷ 28
5. Calculate using BODMAS:
    - a) $72 \div 9 + 4 \times 5$
    - b) $(18 + 12) \div 6 \times 3$
    - c) $100 - \{25 + (15 - 8)\}$
    - d) $2^4 + 3^2 \times 5$
    - e) $144 \div \{12 \div (9 - 6)\} + 20$

Section D: Word Problems

1. A school has 1,245 students. If 678 are boys, how many are girls?
2. A factory produces 2,500 bottles per hour. How many bottles in 24 hours?
3. A farmer has 15,680 kg of yam. He sells them equally to 8 traders. How much does each trader get?
4. A trader bought 120 bags of rice at ₦25,000 each. What is the total cost?
5. A car travels 456 km in 6 hours. What is its average speed per hour?
6. The product of two numbers is 432. One number is 18. Find the other number.

7. A book costs ₦1,250. How many books can be bought with ₦50,000?

8. Three friends shared ₦360,000 equally. How much did each receive?

9. A rectangular farm has length 125m and width 80m. Find:
    - a) The perimeter
    - b) The area

10. Using prime factorization, express 2,520 as a product of prime factors in index form, then:
    - a) Find all the prime factors
    - b) Determine if it's a perfect square
    - c) Find the smallest number it must be multiplied by to make it a perfect square

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WEEK 3: LCM & HCF

TOPIC: Lowest Common Multiple (LCM), Highest Common Factor (HCF), Square and Square Roots

CONTENT:

A. MULTIPLES AND FACTORS (REVIEW)

Multiples: The multiples of a number are obtained by multiplying that number by 1, 2, 3, 4, 5, ...

Example 1: Multiples of 4

$4 \times 1 = 4$

$4 \times 2 = 8$

$4 \times 3 = 12$

$4 \times 4 = 16$

$4 \times 5 = 20$

...

Multiples of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40...

Example 2: Multiples of 6

Multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48, 54, 60...

Factors: Factors of a number are all the whole numbers that divide exactly into that number.

Example 3: Factors of 24

$24 = 1 \times 24$

$24 = 2 \times 12$

$24 = 3 \times 8$

$24 = 4 \times 6$

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24

B. LOWEST COMMON MULTIPLE (LCM)

Definition: The Lowest Common Multiple (LCM) of two or more numbers is the smallest number that is a multiple of all the given numbers.

Method 1: Listing Method

Example 4: Find the LCM of 4 and 6

Multiples of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36...
Multiples of 6: 6, 12, 18, 24, 30, 36, 42...

Common multiples: 12, 24, 36...
LCM = 12 (the smallest)

Example 5: Find the LCM of 8 and 12

Multiples of 8: 8, 16, 24, 32, 40, 48...
Multiples of 12: 12, 24, 36, 48...

LCM = 24

Method 2: Prime Factorization Method

Steps:

1. Express each number as a product of prime factors
2. Write all prime factors that appear in any number
3. For each prime, use the highest power that appears
4. Multiply these together

Example 6: Find the LCM of 12 and 18

$12 = 2^2 \times 3$
$18 = 2 \times 3^2$

LCM = $2^2 \times 3^2$ (highest powers)

$= 4 \times 9$

$= 36$

Example 7: Find the LCM of 24, 36, and 48

$24 = 2^3 \times 3$

$36 = 2^2 \times 3^2$

$48 = 2^4 \times 3$

LCM = $2^4 \times 3^2$ (highest powers of each prime)

$= 16 \times 9$

$= 144$

Example 8: Find the LCM of 15, 20, and 25

$15 = 3 \times 5$

$20 = 2^2 \times 5$

$25 = 5^2$

LCM = $2^2 \times 3 \times 5^2$

$= 4 \times 3 \times 25$

$= 300$

Method 3: Division Method

Example 9: Find the LCM of 12, 18, and 24

2 | 12, 18, 24

2 | 6, 9, 12

3 | 3, 9, 6

| 1, 3, 2

3 | 1, 3, 2

2 | 1, 1, 2
  | 1, 1, 1

LCM = 2 × 2 × 3 × 3 × 2 = 72

C. HIGHEST COMMON FACTOR (HCF)

Definition: The Highest Common Factor (HCF) of two or more numbers is the largest number that divides exactly into all the given numbers. Also called Greatest Common Divisor (GCD).

Method 1: Listing Method

Example 10: Find the HCF of 12 and 18

Factors of 12: 1, 2, 3, 4, 6, 12
Factors of 18: 1, 2, 3, 6, 9, 18

Common factors: 1, 2, 3, 6
HCF = 6 (the largest)

Example 11: Find the HCF of 24 and 36

Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24
Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36

Common factors: 1, 2, 3, 4, 6, 12
HCF = 12

Method 2: Prime Factorization Method

Steps:

1. Express each number as a product of prime factors
2. Identify common prime factors

3. For each common prime, use the lowest power
4. Multiply these together

Example 12: Find the HCF of 24 and 36

$24 = 2^3 \times 3$

$36 = 2^2 \times 3^2$

Common primes: 2 and 3

HCF = $2^2 \times 3$ (lowest powers)

$= 4 \times 3$

$= 12$

Example 13: Find the HCF of 60, 84, and 108

$60 = 2^2 \times 3 \times 5$

$84 = 2^2 \times 3 \times 7$

$108 = 2^2 \times 3^3$

Common primes: 2 and 3

HCF = $2^2 \times 3$ (lowest powers)

$= 4 \times 3$

$= 12$

Example 14: Find the HCF of 48, 72, and 96

$48 = 2^4 \times 3$

$72 = 2^3 \times 3^2$

$96 = 2^5 \times 3$

Common primes: 2 and 3

HCF = $2^3 \times 3$

$= 8 \times 3$

$= 24$

Method 3: Division Method (Euclid's Algorithm)

For two numbers:

1. Divide the larger by the smaller
2. Divide the divisor by the remainder
3. Continue until remainder is 0
4. The last divisor is the HCF

Example 15: Find the HCF of 48 and 18

48 ÷ 18 = 2 remainder 12
18 ÷ 12 = 1 remainder 6
12 ÷ 6 = 2 remainder 0

HCF = 6

D. RELATIONSHIP BETWEEN LCM AND HCF

For any two numbers a and b: a × b = LCM(a, b) × HCF(a, b)

Example 16: Verify for 12 and 18

12 × 18 = 216

LCM(12, 18) = 36
HCF(12, 18) = 6

LCM × HCF = 36 × 6 = 216 ✓

Example 17: The LCM of two numbers is 180 and their HCF is 12. If one number is 36, find the other.

Solution:

Let the other number be x

$36 \times x = 180 \times 12$

$36x = 2{,}160$

$x = 2{,}160 \div 36$

$x = 60$

The other number is 60

E. APPLICATIONS OF LCM AND HCF

LCM Applications:

- Finding when events coincide
- Least quantity needed for equal distribution
- Common denominators for fractions

Example 18: Two bells ring at intervals of 12 minutes and 18 minutes. If they ring together at 8:00 AM, when will they ring together again?

Solution:

Find LCM of 12 and 18

$12 = 2^2 \times 3$

$18 = 2 \times 3^2$

$\text{LCM} = 2^2 \times 3^2 = 36$ minutes

They will ring together after 36 minutes

Answer: 8:36 AM

Example 19: Three pipes fill a tank at intervals of 15, 20, and 25 minutes. If they start together, after how many minutes will they fill together again?

Solution:
LCM of 15, 20, 25

$15 = 3 \times 5$
$20 = 2^2 \times 5$
$25 = 5^2$

LCM = $2^2 \times 3 \times 5^2 = 4 \times 3 \times 25 = 300$ minutes = 5 hours

---

HCF Applications:

- Dividing things into equal parts
- Arranging items in rows/columns
- Finding largest common measure

Example 20: A carpenter has two wooden planks: 72 cm and 96 cm long. He wants to cut them into equal pieces of maximum length. What should be the length of each piece?

Solution:
Find HCF of 72 and 96

$72 = 2^3 \times 3^2$
$96 = 2^5 \times 3$

HCF = $2^3 \times 3 = 24$ cm

Each piece should be 24 cm long
Number of pieces = $(72 \div 24) + (96 \div 24) = 3 + 4 = 7$ pieces

Example 21: Three containers hold 48, 64, and 80 liters. Find the capacity of the largest measure that can exactly measure all three.

Solution:
HCF of 48, 64, 80

$48 = 2^4 \times 3$

$64 = 2^6$

$80 = 2^4 \times 5$

HCF = 2^4 = 16 liters

F. PERFECT SQUARES

Definition: A perfect square is a number that can be expressed as the product of two equal integers.

Perfect Squares from 1 to 15:

$1^2 = 1$

$2^2 = 4$

$3^2 = 9$

$4^2 = 16$

$5^2 = 25$

$6^2 = 36$

$7^2 = 49$

$8^2 = 64$

$9^2 = 81$

$10^2 = 100$

$11^2 = 121$

$12^2 = 144$

$13^2 = 169$

$14^2 = 196$

$15^2 = 225$

Identifying Perfect Squares:

Method 1: Prime Factorization A number is a perfect square if all the indices in its prime factorization are even.

Example 22: Is 144 a perfect square?

$144 = 2^4 \times 3^2$

All indices are even (4 and 2)

Therefore, 144 is a perfect square

$\sqrt{144} = 2^2 \times 3 = 12$

Example 23: Is 180 a perfect square?

$180 = 2^2 \times 3^2 \times 5$

Index of 5 is 1 (odd)

Therefore, 180 is NOT a perfect square

Method 2: Last Digit Perfect squares can only end in: 0, 1, 4, 5, 6, or 9 Perfect squares NEVER end in: 2, 3, 7, or 8

Example 24: Can 342 be a perfect square?

No, because it ends in 2

G. SQUARE ROOTS

Definition: The square root of a number is a value that, when multiplied by itself, gives the original number.

Symbol: √ (radical sign)

Example 25: Find $\sqrt{64}$

$\sqrt{64} = 8$ because $8 \times 8 = 64$

Example 26: Find $\sqrt{225}$

$\sqrt{225} = 15$ because $15 \times 15 = 225$

Finding Square Roots by Prime Factorization:

Example 27: Find $\sqrt{324}$

$324 = 2^2 \times 3^4$

$\sqrt{324} = \sqrt{(2^2 \times 3^4)}$

$= 2 \times 3^2$

$= 2 \times 9$

$= 18$

Example 28: Find $\sqrt{576}$

$576 = 2^6 \times 3^2$

$\sqrt{576} = \sqrt{(2^6 \times 3^2)}$

$= 2^3 \times 3$

$= 8 \times 3$

$= 24$

Example 29: Find $\sqrt{1,296}$

$1,296 = 2^4 \times 3^4$

$\sqrt{1,296} = \sqrt{(2^4 \times 3^4)}$

$= 2^2 \times 3^2$

$= 4 \times 9$

$= 36$

Square Root of Non-Perfect Squares:

Use square root tables or calculator for approximation.

Example 30: Find $\sqrt{50}$ (approximately)

$\sqrt{49} = 7$
$\sqrt{50} \approx 7.071$
$\sqrt{64} = 8$

$\sqrt{50}$ is between 7 and 8, closer to 7

H. APPLICATION IN QUANTITATIVE REASONING

Example 31: The product of two numbers is 180 and their HCF is 6. Find the numbers if their sum is 42.

Solution:
Let the numbers be 6a and 6b (since HCF = 6)

$6a \times 6b = 180$
$36ab = 180$
$ab = 5$

Also: $6a + 6b = 42$
$6(a + b) = 42$
$a + b = 7$

Numbers with sum 7 and product 5: a = 5, b = 2 or a = 2, b = 5

Therefore, the numbers are:
$6 \times 5 = 30$ and $6 \times 2 = 12$

Verification: $30 \times 12 = 360$... wait, this doesn't work.

Let me recalculate:
$6a \times 6b = 180$
$36ab = 180$
$ab = 5$

But ab can't be 5 with integers. Let me try a different approach.

Actually, if HCF = 6, let numbers be 6m and 6n where HCF(m,n) = 1
$6m \times 6n = 180$
$36mn = 180$
$mn = 5$

Since m and n are coprime with product 5: $m=5$, $n=1$ or $m=1$, $n=5$
Numbers: 30 and 6

Check: $30 + 6 = 36$ (not 42)

The problem may have an error. Let's verify with:
If LCM×HCF = product: $LCM \times 6 = 180$, so $LCM = 30$
Numbers are 6 and 30 (HCF=6, LCM=30, product=180)

Example 32: A square field has an area of 1,296 m². Find the length of one side.

Solution:
Area = side²
1,296 = side²
side = $\sqrt{1{,}296}$

$1{,}296 = 2^4 \times 3^4$
$\sqrt{1{,}296} = 2^2 \times 3^2 = 4 \times 9 = 36$ m

Answer: 36 meters

Example 33: Find the smallest number that must be multiplied to 180 to make it a perfect square.

Solution:
$180 = 2^2 \times 3^2 \times 5$

For a perfect square, all indices must be even.
5 has index 1 (odd)

Must multiply by 5 to make it 5^2

Smallest number = 5
New number = $180 \times 5 = 900 = 2^2 \times 3^2 \times 5^2 = 30^2$

EVALUATION:

1. Define LCM and HCF
2. Find the LCM of 12, 18, and 24
3. Find the HCF of 36, 48, and 60
4. The LCM of two numbers is 60 and their HCF is 5. If one number is 15, find the other
5. Is 196 a perfect square? Find its square root
6. Find $\sqrt{441}$ using prime factorization
7. Two bells ring at intervals of 15 and 25 minutes. If they ring together at 9:00 AM, when will they ring together again?
8. Find the smallest number that must be divided from 252 to make it a perfect square
9. A rectangular hall is 18 m long and 12 m wide. Square tiles of equal size are to be laid. Find the maximum size of each tile
10. Evaluate: $\sqrt{144} + \sqrt{225} - \sqrt{100}$

REVISION QUESTIONS:

1. What is the difference between LCM and HCF?
2. Can the HCF of two numbers be greater than their LCM? Explain
3. List all perfect squares between 50 and 150
4. What digits can a perfect square end with?
5. Find LCM and HCF of 24 and 36. Verify that their product equals 24×36
6. Three buses leave a station at 8:00 AM at intervals of 20, 30, and 40 minutes. When will they next leave together?
7. Express 1,764 as a perfect square
8. Find $\sqrt{625}$ without using a calculator
9. A rope 120 m long is cut into equal pieces of maximum length. If no piece is wasted, what is the length of each piece?
10. Find the smallest perfect square divisible by 2, 3, and 5
11. Calculate: $(\sqrt{81} + \sqrt{49}) \times \sqrt{25}$
12. Two numbers have HCF 7 and LCM 140. If one number is 28, find the other
13. What is the smallest number that when multiplied by 200 gives a perfect square?
14. Between which two consecutive integers does $\sqrt{85}$ lie?
15. A square garden has perimeter 144 m. Find its area

ASSIGNMENT:

Section A: LCM

1. Find the LCM using listing method:
   - a) 6 and 8
   - b) 9 and 12
   - c) 10 and 15
2. Find the LCM using prime factorization:
   - a) 16 and 24

- b) 18, 24, and 30
- c) 20, 25, and 30
- d) 12, 15, and 18

3. Find the LCM using division method:
   - a) 24, 36, and 48
   - b) 15, 20, 25, and 30

Section B: HCF

1. Find the HCF using listing method:
   - a) 12 and 18
   - b) 24 and 36
   - c) 30 and 45
2. Find the HCF using prime factorization:
   - a) 48 and 72
   - b) 60, 84, and 108
   - c) 36, 54, and 72
   - d) 45, 60, and 75
3. Find the HCF of 84 and 126 using Euclid's algorithm

Section C: Combined Problems

1. Find both LCM and HCF of:
   - a) 12 and 20
   - b) 18 and 24
   - c) 24, 36, and 48
2. The LCM of two numbers is 180 and their HCF is 12. If one number is 60, find the other
3. Two numbers have product 1,440 and HCF 12. Find their LCM

4. The HCF and LCM of two numbers are 8 and 240 respectively. If one number is 48, find the other
5. Show that: LCM(24, 36) × HCF(24, 36) = 24 × 36

Section D: Perfect Squares and Square Roots

1. Identify which are perfect squares: 64, 75, 81, 100, 125, 144, 150, 169, 200, 225
2. Find the square roots using prime factorization:
   - a) $\sqrt{256}$
   - b) $\sqrt{400}$
   - c) $\sqrt{529}$
   - d) $\sqrt{784}$
   - e) $\sqrt{1,024}$
3. Find the smallest number that must be:
   - a) Multiplied to 72 to make it a perfect square
   - b) Multiplied to 108 to make it a perfect square
   - c) Divided from 200 to make it a perfect square
   - d) Divided from 432 to make it a perfect square
4. Without calculating, state why these cannot be perfect squares:
   - a) 123
   - b) 1,458
   - c) 2,347
5. Evaluate:
   - a) $\sqrt{169} + \sqrt{121}$
   - b) $\sqrt{225} - \sqrt{64}$
   - c) $\sqrt{144} \times \sqrt{25}$
   - d) $\sqrt{400} \div \sqrt{16}$
   - e) $(\sqrt{81} + \sqrt{49})^2$

Section E: Word Problems

1. Three electric bulbs flash at intervals of 12, 18, and 24 seconds. If they flash together at 6:00 PM, at what time will they flash together again?
2. A farmer wants to plant trees in rows such that each row has the same number of trees. He has 96 mango trees, 72 orange trees, and 120 banana trees. What is the maximum number of trees he can plant in each row?
3. A rectangular courtyard is 18 m by 24 m. What is the size of the largest square tile that can be used to pave it exactly?
4. A square playground has an area of 2,025 m^2. Find:
   - a) The length of one side
   - b) The perimeter
5. Two trains leave a station at 7:00 AM. One returns every 40 minutes, the other every 60 minutes. At what time will both trains be at the station together again?
6. Three bells toll at intervals of 9, 12, and 15 minutes. If they toll together at noon, when will they toll together again?
7. The product of two numbers is 864 and their HCF is 12. Find the possible pairs of numbers.
8. A hall measuring 36 m by 24 m is to be paved with identical square marble tiles. Find:
   - a) The size of the largest tile that can be used
   - b) The number of tiles needed
9. What is the smallest 4-digit perfect square? What is the largest 3-digit perfect square?
10. A square field needs to be fenced. If the area is 2,304 m^2 and fencing costs ₦500 per meter, find the total cost.

WEEK 4: FRACTIONS – TRANSACTIONS IN HOMES AND OFFICES

TOPIC: Expressing Fractions as Ratios, Decimals, and Percentages; Applications of Fractions in Daily Transactions

CONTENT:

A. REVIEW OF FRACTIONS

Definition: A fraction represents a part of a whole. It is written as a/b where:

- **a** = numerator (top number) - parts we have
- **b** = denominator (bottom number) - total parts

Types of Fractions:

1. **Proper Fraction:** Numerator < Denominator
    - Examples: 3/4, 5/8, 2/7, 11/15
2. **Improper Fraction:** Numerator ≥ Denominator
    - Examples: 7/4, 9/5, 15/8, 23/12
3. **Mixed Number:** Whole number + Proper fraction
    - Examples: $2\frac{1}{4}$, $3\frac{2}{5}$, $5\frac{3}{8}$
4. **Equivalent Fractions:** Fractions with the same value
    - Examples: 1/2 = 2/4 = 3/6 = 4/8

B. CONVERTING BETWEEN FRACTIONS, DECIMALS, AND PERCENTAGES

1. FRACTION TO DECIMAL

Method: Divide the numerator by the denominator

Example 1: Convert 3/4 to decimal

3 ÷ 4 = 0.75

Example 2: Convert 7/8 to decimal

$7 \div 8 = 0.875$

Example 3: Convert 5/6 to decimal

$5 \div 6 = 0.8333... = 0.8\bar{3}$ (recurring)

Example 4: Convert $2^3/_5$ to decimal

First convert to improper fraction:
$2^3/_5 = (2 \times 5 + 3)/5 = 13/5$
$13 \div 5 = 2.6$

Example 5: Convert 11/20 to decimal

$11 \div 20 = 0.55$

---

2. DECIMAL TO FRACTION

Method:

- Count decimal places
- Write as fraction with appropriate denominator (10, 100, 1000, etc.)
- Simplify

Example 6: Convert 0.75 to fraction

$0.75 = 75/100$
$= (75 \div 25)/(100 \div 25)$
$= 3/4$

Example 7: Convert 0.625 to fraction

$0.625 = 625/1000$
$= (625 \div 125)/(1000 \div 125)$
$= 5/8$

Example 8: Convert 0.4 to fraction

0.4 = 4/10 = 2/5

Example 9: Convert 2.35 to fraction

2.35 = 2 + 0.35
= 2 + 35/100
= 2 + 7/20
= $2^7/_{20}$

Example 10: Convert 0.125 to fraction

0.125 = 125/1000 = 1/8

---

3. FRACTION TO PERCENTAGE

Method: Convert to decimal, then multiply by 100%

Example 11: Convert 3/5 to percentage

3/5 = 0.6
0.6 × 100% = 60%

Example 12: Convert 7/8 to percentage

7/8 = 0.875
0.875 × 100% = 87.5%

Example 13: Convert 2/3 to percentage

2/3 = 0.6667
0.6667 × 100% = 66.67% (or $66^2/_3$%)

Example 14: Convert 5/4 to percentage

5/4 = 1.25
1.25 × 100% = 125%

4. PERCENTAGE TO FRACTION

Method: Write as fraction over 100, then simplify

Example 15: Convert 75% to fraction

75% = 75/100
= 3/4

Example 16: Convert 45% to fraction

45% = 45/100
= 9/20

Example 17: Convert 12.5% to fraction

12.5% = 12.5/100
= 125/1000
= 1/8

Example 18: Convert 150% to fraction

150% = 150/100
= 3/2
= 1½

5. DECIMAL TO PERCENTAGE

Method: Multiply by 100%

Example 19: Convert 0.85 to percentage

$0.85 \times 100\% = 85\%$

Example 20: Convert 0.375 to percentage

0.375 × 100% = 37.5%

Example 21: Convert 1.5 to percentage

1.5 × 100% = 150%

---

6. PERCENTAGE TO DECIMAL

Method: Divide by 100

Example 22: Convert 68% to decimal

68% = 68 ÷ 100 = 0.68

Example 23: Convert 7.5% to decimal

7.5% = 7.5 ÷ 100 = 0.075

Example 24: Convert 125% to decimal

125% = 125 ÷ 100 = 1.25

C. EXPRESSING FRACTIONS AS RATIOS

Definition of Ratio: A ratio compares two or more quantities of the same kind. Written as a:b or a/b

Converting Fraction to Ratio:

Example 25: Express 3/5 as a ratio

3/5 = 3:5

Example 26: Express 7/12 as a ratio

7/12 = 7:12

Example 27: A class has 15 boys and 20 girls. Express as ratio.

Boys:Girls = 15:20 = 3:4 (simplified)

Converting Ratio to Fraction:

Example 28: Express the ratio 4:7 as a fraction

4:7 = 4/11 (4 parts out of 11 total parts)
First quantity: 4/(4+7) = 4/11
Second quantity: 7/(4+7) = 7/11

Example 29: Divide ₦550 in the ratio 3:2

Total parts = 3 + 2 = 5
First share = 3/5 × 550 = ₦330
Second share = 2/5 × 550 = ₦220

Quick Conversion Table:

Fraction	Decimal	Percentage	Ratio
1/2	0.5	50%	1:2
1/4	0.25	25%	1:4
3/4	0.75	75%	3:4
1/5	0.2	20%	1:5
2/5	0.4	40%	2:5
3/5	0.6	60%	3:5
1/8	0.125	12.5%	1:8
1/10	0.1	10%	1:10
1/3	$0.3\overline{3}$	$33.\overline{3}\%$	1:3
2/3	$0.6\overline{6}$	$66.\overline{6}\%$	2:3

D. APPLICATION OF FRACTIONS IN DAILY TRANSACTIONS

1. SHOPPING AND DISCOUNTS

Example 30: A shirt costs ₦2,400. If there's a 25% discount, how much do you pay?

Solution:

Discount = 25% of ₦2,400

= 25/100 × 2,400

= 0.25 × 2,400

= ₦600

Amount paid = Original price - Discount

= 2,400 - 600

= ₦1,800

OR directly:

Amount paid = 75% of ₦2,400 (since 100% - 25% = 75%)

= 0.75 × 2,400

= ₦1,800

Example 31: A phone originally costs ₦45,000. After a 15% discount and 7.5% VAT on the discounted price, what's the final cost?

Solution:

Step 1: Calculate discount

Discount = 15% of 45,000 = 0.15 × 45,000 = ₦6,750

Discounted price = 45,000 - 6,750 = ₦38,250

Step 2: Calculate VAT

VAT = 7.5% of 38,250 = 0.075 × 38,250 = ₦2,868.75

Step 3: Final price

Final price = 38,250 + 2,868.75 = ₦41,118.75

2. PROFIT AND LOSS

Example 32: A trader bought goods for ₦12,000 and sold them for ₦15,000. Find: a) Profit b) Profit percentage

Solution:

a) Profit = Selling Price - Cost Price

= 15,000 - 12,000

= ₦3,000

b) Profit % = (Profit/Cost Price) × 100%

= (3,000/12,000) × 100%

= 0.25 × 100%

= 25%

Example 33: A shopkeeper sold a bag for ₦8,500 at a loss of 15%. What was the cost price?

Solution:

If loss = 15%, then Selling Price = 85% of Cost Price

Let Cost Price = x

85% of x = 8,500

0.85x = 8,500

x = 8,500 ÷ 0.85

x = ₦10,000

Cost Price = ₦10,000

Example 34: A trader sells goods at a 20% markup. If the cost price is ₦5,000, find the selling price.

Solution:

Markup = 20% of 5,000 = 0.2 × 5,000 = ₦1,000

Selling Price = Cost Price + Markup
= 5,000 + 1,000
= ₦6,000

OR:
Selling Price = 120% of Cost Price
= 1.2 × 5,000
= ₦6,000

---

3. SIMPLE INTEREST

Formula: I = (P × R × T)/100

Where:

- I = Interest
- P = Principal (original amount)
- R = Rate (percentage per year)
- T = Time (in years)

Example 35: Find the simple interest on ₦20,000 at 5% per annum for 3 years.

Solution:
I = (P × R × T)/100
= (20,000 × 5 × 3)/100
= 300,000/100
= ₦3,000

Amount = Principal + Interest
= 20,000 + 3,000
= ₦23,000

Example 36: A man borrowed ₦50,000 at 8% per annum. If he repaid ₦58,000, for how long did he borrow the money?

Solution:

Interest = Amount - Principal

= 58,000 - 50,000

= ₦8,000

$I = (P \times R \times T)/100$

$8{,}000 = (50{,}000 \times 8 \times T)/100$

$8{,}000 = 400T$

$T = 8{,}000 \div 400$

T = 2 years

---

4. SHARING IN RATIOS

Example 37: Three brothers share ₦120,000 in the ratio 2:3:5. How much does each receive?

Solution:

Total parts = 2 + 3 + 5 = 10 parts

First brother = 2/10 × 120,000 = ₦24,000

Second brother = 3/10 × 120,000 = ₦36,000

Third brother = 5/10 × 120,000 = ₦60,000

Verification: 24,000 + 36,000 + 60,000 = 120,000 ✓

Example 38: A father divides his property among his three children in the ratio of their ages: 5:7:8. If the youngest child gets ₦2,500,000, find: a) Total value of property b) What the other children receive

Solution:

Youngest child's share = 5 parts = ₦2,500,000

Therefore, 1 part = 2,500,000 ÷ 5 = ₦500,000

a) Total parts = 5 + 7 + 8 = 20 parts
Total property = 20 × 500,000 = ₦10,000,000

b) Second child = 7 × 500,000 = ₦3,500,000
Eldest child = 8 × 500,000 = ₦4,000,000

---

5. HOUSEHOLD BUDGETING

Example 39: A family's monthly income is ₦150,000. They spend:

- Food: 30%
- Rent: 25%
- Transport: 15%
- Utilities: 10%
- Savings: 20%

Calculate each amount.

Solution:
Food = 30% of 150,000 = 0.30 × 150,000 = ₦45,000
Rent = 25% of 150,000 = 0.25 × 150,000 = ₦37,500
Transport = 15% of 150,000 = 0.15 × 150,000 = ₦22,500
Utilities = 10% of 150,000 = 0.10 × 150,000 = ₦15,000
Savings = 20% of 150,000 = 0.20 × 150,000 = ₦30,000

Total = 45,000 + 37,500 + 22,500 + 15,000 + 30,000 = ₦150,000 ✓

Example 40: A woman saves 3/8 of her salary. If her salary is ₦80,000, how much does she: a) Save b) Spend

Solution:
a) Savings = 3/8 × 80,000 = ₦30,000

b) Spending = 80,000 - 30,000 = ₦50,000
OR: 5/8 × 80,000 = ₦50,000

---

6. MEASURING AND RECIPES

Example 41: A recipe for 4 people requires 2½ cups of flour. How much flour for: a) 6 people b) 10 people

Solution:
For 4 people = 2½ = 5/2 cups

a) For 6 people:
6 is 1.5 times 4
Flour needed = 5/2 × 1.5 = 5/2 × 3/2 = 15/4 = 3¾ cups

b) For 10 people:
10 is 2.5 times 4
Flour needed = 5/2 × 2.5 = 5/2 × 5/2 = 25/4 = 6¼ cups

Example 42: A 5-liter container is ⅔ full of water. How much water is in it?

Solution:
Water = 2/3 × 5 = 10/3 = 3⅓ liters

---

7. COMMISSION AND SALARY

Example 43: A salesperson earns a basic salary of ₦30,000 plus 5% commission on sales. If monthly sales are ₦400,000, calculate total earnings.

Solution:

Commission = 5% of 400,000

= 0.05 × 400,000

= ₦20,000

Total earnings = Basic salary + Commission

= 30,000 + 20,000

= ₦50,000

Example 44: An agent sells a house for ₦15,000,000 and earns 2.5% commission. How much does he earn?

Solution:

Commission = 2.5% of 15,000,000

= 0.025 × 15,000,000

= ₦375,000

---

8. TAX CALCULATIONS

Example 45: A worker's gross salary is ₦60,000. If 7.5% is deducted as pension, how much is: a) The pension deduction b) Net salary

Solution:

a) Pension = 7.5% of 60,000

= 0.075 × 60,000

= ₦4,500

b) Net salary = Gross salary - Deductions

= 60,000 - 4,500

= ₦55,500

Example 46: Income tax is calculated as follows:

- First ₦20,000: Tax-free
- Next ₦30,000: 10%
- Balance: 15%

Calculate tax on ₦80,000.

Solution:
First ₦20,000: ₦0 (tax-free)
Next ₦30,000: 10% of 30,000 = ₦3,000
Balance (₦30,000): 15% of 30,000 = ₦4,500

Total tax = 0 + 3,000 + 4,500 = ₦7,500

---

9. FUEL CONSUMPTION

Example 47: A car travels 240 km on 20 liters of fuel. Express fuel consumption as: a) Kilometers per liter b) Liters per 100 km

Solution:
a) km/liter = 240 ÷ 20 = 12 km/liter

b) For 100 km:
Liters needed = (20/240) × 100
= 100/12
= 8.33 liters per 100 km

---

10. MIXTURES AND SOLUTIONS

Example 48: A 60-liter solution is 40% alcohol. How much pure alcohol does it contain?

Solution:
Alcohol = 40% of 60

= 0.40 × 60

= 24 liters

Example 49: How much water must be added to 80 liters of 25% salt solution to make it 20% solution?

Solution:

Salt in original solution = 25% of 80 = 20 liters

Let x = liters of water to be added

New volume = 80 + x

20% of (80 + x) = 20

0.20(80 + x) = 20

16 + 0.20x = 20

0.20x = 4

x = 20 liters

Add 20 liters of water

E. QUANTITATIVE REASONING WITH FRACTIONS

Example 50: If ⅔ of a number is 42, find: a) The number b) ¾ of the number

Solution:

a) Let the number be x

$2/3 \times x = 42$

$x = 42 \times 3/2$

$x = 63$

b) 3/4 of 63 = 3/4 × 63 = 189/4 = 47.25 or 47¼

Example 51: A man spent ⅓ of his money on food, ¼ on transport, and saved the rest. If he saved ₦5,000, how much did he have initially?

Solution:
Spent on food = 1/3
Spent on transport = 1/4
Total spent = 1/3 + 1/4 = 4/12 + 3/12 = 7/12

Saved = 1 - 7/12 = 5/12

5/12 of money = 5,000
Money = 5,000 × 12/5 = ₦12,000

Example 52: A tank is ⅗ full. After drawing out 60 liters, it becomes ⅖ full. Find the capacity of the tank.

Solution:
Fraction drawn = 3/5 - 2/5 = 1/5

1/5 of capacity = 60 liters
Capacity = 60 × 5 = 300 liters

Example 53: Three-fifths of a class are girls. If there are 18 boys, how many students in total?

Solution:
If 3/5 are girls, then 2/5 are boys

2/5 of total = 18
Total = 18 × 5/2 = 45 students

EVALUATION:

1. Convert to decimal: a) 5/8 b) 3¾
2. Convert to fraction (simplest form): a) 0.625 b) 0.45

3. Convert to percentage: a) 7/20 b) 1.35
4. Express 35% as: a) decimal b) fraction
5. A dress costs ₦8,000. After 20% discount, what do you pay?
6. Find simple interest on ₦15,000 at 6% for 2 years
7. Share ₦90,000 in ratio 2:3:4
8. A trader buys goods for ₦24,000 and sells at 25% profit. Find selling price
9. If ⅘ of a number is 56, find the number
10. Convert the ratio 3:7 to a fraction

REVISION QUESTIONS:

1. Why is percentage useful in everyday life?
2. How do you convert a recurring decimal to a fraction?
3. Arrange in ascending order: 0.6, 60%, 3/5, 6:10
4. What's the difference between markup and discount?
5. A shirt was marked ₦5,000. After 30% discount and 5% VAT on discounted price, find final cost
6. Calculate profit % if CP = ₦800 and SP = ₦1,000
7. At what rate will ₦10,000 yield ₦2,500 interest in 5 years?
8. Three partners invest in ratio 5:3:2. If smallest share is ₦40,000, find total investment
9. A recipe uses ingredients in ratio flour:sugar:butter = 5:3:2. How much sugar if flour is 250g?
10. Express $^{7}/_{8}$ as percentage, decimal, and ratio
11. A man spends ⅓ on rent, ¼ on food. What fraction is left?
12. Find the ratio of 45 minutes to 2 hours
13. A 25% discount reduces price by ₦1,500. What was original price?
14. If 60% of students passed and 24 failed, how many students in total?
15. Convert 2:5:3 to percentages of the whole

ASSIGNMENT:

Section A: Conversions (20 marks)

1. Convert to decimals:
   - a) 3/8
   - b) 7/20
   - c) $2\frac{3}{5}$
   - d) 11/16
2. Convert to fractions (simplest form):
   - a) 0.375
   - b) 0.85
   - c) 1.75
   - d) 0.225
3. Convert to percentages:
   - a) 9/25
   - b) 0.68
   - c) 3/8
   - d) 1.4
4. Convert to fractions:
   - a) 45%
   - b) 7.5%
   - c) 135%
   - d) 62.5%
5. Convert to decimals:
   - a) 55%
   - b) 8.5%
   - c) 120%

- d) 0.5%

Section B: Ratios (10 marks)

1. Express as ratios:
   - a) 3/7
   - b) 0.6 (as ratio in simplest form)
   - c) 45%
2. Share ₦240,000 in ratio:
   - a) 3:5
   - b) 2:3:7
3. If a:b = 2:3 and b:c = 4:5, find a:b:c
4. Simplify the ratio 25:15:10
5. Two numbers are in ratio 5:8. If their sum is 91, find the numbers

Section C: Shopping and Discounts (15 marks)

1. A television costs ₦95,000. Find the selling price after:
   - a) 15% discount
   - b) 25% discount
   - c) 10% discount and 7.5% VAT on discounted price
2. After 20% discount, a phone costs ₦48,000. Find original price
3. A trader gives successive discounts of 10% and 5%. Find net discount
4. An item marked ₦12,000 is sold at ₦9,600. Find discount percentage
5. During a sale, prices are reduced by ⅓. Find new price of items costing:
   - a) ₦6,000
   - b) ₦15,000

Section D: Profit, Loss, and Interest (20 marks)

1. Calculate profit or loss and percentage:
   - a) CP = ₦5,000, SP = ₦6,500
   - b) CP = ₦8,000, SP = ₦7,200
2. A trader sells goods at 30% profit. If CP is ₦15,000, find SP
3. Goods sold for ₦21,000 at 25% loss. Find CP
4. Calculate simple interest:
   - a) P = ₦25,000, R = 8%, T = 3 years
   - b) P = ₦40,000, R = 5%, T = 2½ years
5. A man borrowed ₦80,000 and repaid ₦92,000 after 3 years. Find interest rate
6. At what rate will ₦20,000 amount to ₦23,000 in 2 years?
7. If ₦12,000 yields ₦2,160 interest in 3 years, find the rate

Section E: Sharing and Budgeting (15 marks)

1. Three children share ₦45,000 in ratio 2:3:5. Find each share
2. A man's salary is ₦120,000. He spends:
   - Rent: 30%
   - Food: 25%
   - Transport: 15%
   - Others: 20%
   - Saves the rest

   Calculate: a) Each amount b) Savings
3. A woman saves ⅗ of her income. If she saves ₦36,000, find:
   - a) Her income
   - b) Her spending

4. Divide ₦180,000 among A, B, C such that A gets ⅓, B gets ½ of remainder, C gets the rest

5. Partners contribute in ratio 5:3:2. If total capital is ₦500,000, find each contribution

Section F: Quantitative Reasoning (20 marks)

1. If ⅔ of a number is 48, find:
   - a) The number
   - b) ¾ of the number
2. A man spent ¼ on clothes, ⅓ on food, and saved ₦10,000. Find his total money
3. A tank ¾ full contains 90 liters. Find:
   - a) Tank capacity
   - b) Amount when ⅖ full
4. ⅗ of a class are girls. If there are 28 boys, find total students
5. A number increased by 25% becomes 75. Find the number
6. After spending 40% of her money, Mary has ₦12,000 left. How much initially?
7. A car uses 15 liters for 180 km. How much fuel for 300 km?
8. A 80-liter solution is 30% alcohol. How much pure alcohol?
9. In a school, ⅖ play football, ⅓ play basketball, 60 play volleyball. Find total students if no one plays more than one game
10. A recipe for 6 people uses 2¼ cups sugar. How much for 10 people?

WEEK 6: APPROXIMATION

TOPIC: Approximation of Numbers to Decimal Places, Significant Figures, and Various Place Values

CONTENT:

A. INTRODUCTION TO APPROXIMATION

Definition: Approximation means expressing a number to a required degree of accuracy by rounding off.

Why Approximate?

- Makes calculations easier
- Provides reasonable estimates
- Used when exact values are unnecessary
- Common in real-life situations (money, measurements, statistics)

General Rounding Rule:

- If the next digit is 5 or more, round UP
- If the next digit is less than 5, round DOWN

B. APPROXIMATION TO THE NEAREST WHOLE NUMBER

Rule: Look at the first decimal place (tenths)

Example 1: Approximate to nearest whole number:

a) $45.7 \rightarrow 46$ ($7 \geq 5$, round up)
b) $67.3 \rightarrow 67$ ($3 < 5$, round down)
c) $89.5 \rightarrow 90$ ($5 = 5$, round up)
d) $234.49 \rightarrow 234$ ($4 < 5$, round down)
e) $156.8 \rightarrow 157$ ($8 \geq 5$, round up)

Example 2: Round to nearest whole number:

a) 23.456 → 23

b) 78.912 → 79

c) 145.501 → 146

d) 999.499 → 999

e) 0.678 → 1

C. APPROXIMATION TO DECIMAL PLACES (d.p.)

Decimal Places count the number of digits after the decimal point.

1 Decimal Place (1 d.p.):

Example 3: Approximate to 1 d.p.:

a) 45.67 → 45.7 (look at 7, it's ≥5, round up 6 to 7)

b) 23.42 → 23.4 (look at 2, it's <5, keep 4)

c) 78.95 → 79.0 or 79 (look at 5, round up 9 to 10)

d) 0.678 → 0.7 (look at 7, round up)

e) 12.349 → 12.3 (look at 4, round down)

2 Decimal Places (2 d.p.):

Example 4: Approximate to 2 d.p.:

a) 45.678 → 45.68 (look at 8, round up)

b) 23.423 → 23.42 (look at 3, round down)

c) 0.6789 → 0.68 (look at 8, round up)

d) 156.9951 → 156.00 or 156 (look at 5, round up 99 to 100)

e) 3.14159 → 3.14 (look at 1, round down)

3 Decimal Places (3 d.p.):

Example 5: Approximate to 3 d.p.:

a) 12.34567 → 12.346 (look at 6, round up)

b) 0.123456 → 0.123 (look at 4, round down)

c) 67.89954 → 67.900 (look at 5, round up)
d) 2.71828 → 2.718 (look at 2, round down)
e) 3.141592 → 3.142 (look at 5, round up)

D. APPROXIMATION TO SIGNIFICANT FIGURES (s.f.)

Significant Figures count from the first non-zero digit.

Rules for Identifying Significant Figures:

11. All non-zero digits are significant (1-9)
12. Zeros between non-zero digits are significant
13. Leading zeros (before first non-zero digit) are NOT significant
14. Trailing zeros after decimal point ARE significant
15. Trailing zeros in whole numbers without decimal point may or may not be significant

Example 6: Count significant figures:

a) 345 → 3 s.f. (all non-zero)
b) 3,405 → 4 s.f. (zero between 4 and 5 counts)
c) 0.0045 → 2 s.f. (leading zeros don't count)
d) 0.04050 → 4 s.f. (trailing zero after decimal counts)
e) 3,400 → 2 s.f. (trailing zeros ambiguous, likely not significant)
f) 3,400.0 → 5 s.f. (decimal makes all zeros significant)

---

1 Significant Figure (1 s.f.):

Example 7: Approximate to 1 s.f.:

a) 456 → 500 (look at 5, round 4 up to 5, rest become zeros)
b) 7,834 → 8,000 (look at 8, round up)
c) 0.0678 → 0.07 (first s.f. is 6, look at 7, round up)

d) 0.00345 → 0.003 (first s.f. is 3, look at 4, round down)

e) 9,876 → 10,000 (look at 8, round 9 up to 10)

2 Significant Figures (2 s.f.):

Example 8: Approximate to 2 s.f.:

a) 4,567 → 4,600 (keep 4 and 5, look at 6, round up)

b) 0.0678 → 0.068 (keep 6 and 7, look at 8, round up)

c) 234.56 → 230 (keep 2 and 3, look at 4, round down)

d) 0.003456 → 0.0035 (keep 3 and 4, look at 5, round up)

e) 98,765 → 99,000 (keep 9 and 8, look at 7, round up)

3 Significant Figures (3 s.f.):

Example 9: Approximate to 3 s.f.:

a) 45,678 → 45,700 (keep 4, 5, 6; look at 7, round up)

b) 0.012345 → 0.0123 (keep 1, 2, 3; look at 4, round down)

c) 789.456 → 789 (keep 7, 8, 9; look at 4, round down)

d) 0.0009876 → 0.000988 (keep 9, 8, 7; look at 6, round up)

e) 123,456 → 123,000 (keep 1, 2, 3; look at 4, round down)

4 Significant Figures (4 s.f.):

Example 10: Approximate to 4 s.f.:

a) 567,890 → 567,900

b) 0.0123456 → 0.01235

c) 98.7654 → 98.77

d) 1,234,567 → 1,235,000

e) 0.00098765 → 0.0009877

E. APPROXIMATION TO NEAREST TEN, HUNDRED, THOUSAND

Nearest Ten:

Example 11: Approximate to nearest ten:

a) 234 → 230 (look at units digit 4, <5, round down)

b) 567 → 570 (look at units digit 7, ≥5, round up)

c) 1,895 → 1,900 (look at units digit 5, round up)

d) 3,444 → 3,440 (look at units digit 4, round down)

e) 9,996 → 10,000 (round up 999 to 1000)

Nearest Hundred:

Example 12: Approximate to nearest hundred:

a) 2,345 → 2,300 (look at tens digit 4, round down)

b) 6,789 → 6,800 (look at tens digit 8, round up)

c) 19,850 → 19,900 (look at tens digit 5, round up)

d) 123,449 → 123,400 (look at tens digit 4, round down)

e) 99,950 → 100,000 (round up)

Nearest Thousand:

Example 13: Approximate to nearest thousand:

a) 12,345 → 12,000 (look at hundreds digit 3, round down)

b) 67,890 → 68,000 (look at hundreds digit 8, round up)

c) 195,500 → 196,000 (look at hundreds digit 5, round up)

d) 1,234,499 → 1,234,000 (look at hundreds digit 4, round down)

e) 999,500 → 1,000,000 (round up)

Nearest Ten Thousand:

Example 14: Approximate to nearest ten thousand:

a) 234,567 → 230,000

b) 678,900 → 680,000

c) 1,995,000 → 2,000,000

d) 123,444 → 120,000

e) 55,678 → 60,000

F. APPROXIMATION TO NEAREST TENTH, HUNDREDTH, THOUSANDTH

Nearest Tenth (1 decimal place):

Example 15: Same as 1 d.p. (shown earlier)

a) 45.67 → 45.7

b) 0.349 → 0.3

c) 123.95 → 124.0 or 124

Nearest Hundredth (2 decimal places):

Example 16: Same as 2 d.p.

a) 0.6789 → 0.68

b) 12.3456 → 12.35

c) 0.0995 → 0.10

Nearest Thousandth (3 decimal places):

Example 17: Same as 3 d.p.

a) 0.12345 → 0.123

b) 3.14159 → 3.142

c) 0.00789 → 0.008

G. ESTIMATING CALCULATIONS

Using Approximation to Check Answers:

Example 18: Estimate 49×21

Approximation:

$49 \approx 50$

$21 \approx 20$

Estimate: $50 \times 20 = 1{,}000$

Actual: $49 \times 21 = 1{,}029$ ✓ (close to estimate)

Example 19: Estimate $789 \div 38$

Approximation (1 s.f.):

$789 \approx 800$

$38 \approx 40$

Estimate: $800 \div 40 = 20$

Actual: $789 \div 38 = 20.76...$ ✓

Example 20: Estimate 6.78×9.23

Approximation (1 d.p.):

$6.78 \approx 7$

$9.23 \approx 9$

Estimate: $7 \times 9 = 63$

Actual: $6.78 \times 9.23 = 62.58$ ✓

Example 21: Estimate $\sqrt{98}$

$\sqrt{98}$ is between $\sqrt{81} = 9$ and $\sqrt{100} = 10$

Since 98 is close to 100:

Estimate: $\sqrt{98} \approx 9.9$

Actual: $\sqrt{98} = 9.899...$ ✓

Example 22: Estimate $(37.8 \times 19.6) \div 5.1$

Approximation:

$37.8 \approx 40$

$19.6 \approx 20$

$5.1 \approx 5$

Estimate: $(40 \times 20) \div 5 = 800 \div 5 = 160$

Actual: $(37.8 \times 19.6) \div 5.1 = 145.2$ (reasonably close)

H. UPPER AND LOWER BOUNDS

Definition:

- **Lower Bound:** The smallest value a rounded number could have been
- **Upper Bound:** The largest value a rounded number could have been (just less than the next rounded value)

Example 23: A length is given as 45 cm to nearest cm. Find bounds.

Lower bound = 44.5 cm
Upper bound = 45.5 cm (but not including 45.5)
Range: $44.5 \leq \text{length} < 45.5$

Example 24: A mass is 350 g to nearest 10 g. Find bounds.

Lower bound = 345 g
Upper bound = 355 g (not including 355)
Range: $345 \leq \text{mass} < 355$

Example 25: A measurement is 4.5 m to 1 d.p. Find bounds.

Lower bound = 4.45 m
Upper bound = 4.55 m (not including 4.55)
Range: $4.45 \leq \text{measurement} < 4.55$

Example 26: A value is 3,500 to 2 s.f. Find bounds.

Lower bound = 3,450

Upper bound = 3,550 (not including 3,550)

Range: 3,450 ≤ value < 3,550

I. ERROR IN APPROXIMATION

Absolute Error = |Actual Value - Approximate Value|

Relative Error = (Absolute Error / Actual Value) × 100%

Example 27: Actual value = 45.78, Approximated to 46. Find errors.

Absolute error = |45.78 - 46| = |-0.22| = 0.22

Relative error = (0.22/45.78) × 100% = 0.48%

Example 28: π = 3.14159... approximated as 3.14. Find absolute error.

Absolute error = |3.14159 - 3.14| = 0.00159

J. PRACTICAL APPLICATIONS

Example 29: Money Approximation A bill shows ₦4,567.89. Approximate to:

a) Nearest naira: ₦4,568

b) Nearest ten naira: ₦4,570

c) Nearest hundred naira: ₦4,600

Example 30: Population Lagos state population: 15,378,259. Express to:

a) Nearest thousand: 15,378,000

b) Nearest million: 15,000,000

c) 2 s.f.: 15,000,000

d) 3 s.f.: 15,400,000

Example 31: Scientific Measurements Speed of light: 299,792,458 m/s. Express to:

a) 3 s.f.: 300,000,000 m/s
b) 4 s.f.: 299,800,000 m/s
c) Standard form (3 s.f.): 3.00×10^8 m/s

Example 32: Shopping Items cost: ₦234, ₦567, ₦891. Estimate total:

Approximate to nearest hundred:
₦200 + ₦600 + ₦900 = ₦1,700

Actual: ₦1,692 (close estimate)

Example 33: Fuel Consumption Car travels 247 km on 18.7 liters. Estimate km/liter:

Approximate:
$247 \approx 250$
$18.7 \approx 20$

Estimate: $250 \div 20 = 12.5$ km/liter

Actual: $247 \div 18.7 = 13.2$ km/liter ✓

K. QUANTITATIVE REASONING WITH APPROXIMATION

Example 34: The area of a rectangle is given as 45.6 m² to 1 d.p. Find maximum and minimum possible areas.

Lower bound = 45.55 m²
Upper bound < 45.65 m²
Range: $45.55 \leq \text{area} < 45.65$

Example 35: Two measurements are 23 cm (to nearest cm) and 15 cm (to nearest cm). Find: a) Maximum possible sum b) Minimum possible product

a) Maximum sum:

23 upper bound = 23.5 cm

15 upper bound = 15.5 cm

Max sum = 23.5 + 15.5 = 39 cm

b) Minimum product:

23 lower bound = 22.5 cm

15 lower bound = 14.5 cm

Min product = 22.5 × 14.5 = 326.25 cm²

Example 36: A calculator shows 45.678912. Write to: a) 2 d.p. b) 3 s.f. c) Nearest hundredth

a) 45.68

b) 45.7

c) 45.68 (same as 2 d.p.)

EVALUATION:

1. Approximate 67.89 to: a) Nearest whole number b) 1 d.p.
2. Approximate 4,567 to: a) Nearest hundred b) 2 s.f.
3. Approximate 0.06789 to: a) 2 d.p. b) 3 s.f.
4. Estimate: (48 × 22) ÷ 5
5. A length is 45 cm to nearest cm. Find lower and upper bounds
6. Approximate 234,567 to nearest thousand
7. Count significant figures in: a) 4,050 b) 0.00450
8. Approximate 3.14159 to 3 s.f.
9. Estimate √122
10. Express 0.0004567 to 2 s.f.

REVISION QUESTIONS:

1. What is the difference between decimal places and significant figures?

2. Why do we approximate in everyday life?
3. When rounding, what do you do if the next digit is exactly 5?
4. Are leading zeros significant? Explain
5. Approximate 9,876.54 to: a) 2 d.p. b) 3 s.f. c) Nearest ten
6. A measurement is 250 g to 2 s.f. What are the bounds?
7. Estimate (38.7 × 19.3) ÷ 4.8
8. Which is more accurate: 3 s.f. or 3 d.p.? Explain
9. Approximate ₦4,567.89 to nearest ₦100
10. How many significant figures in 0.00304?
11. Round 99.97 to 1 d.p.
12. Express 123,456,789 to nearest million
13. Why is approximation useful in estimating calculations?
14. A value 45.0 to 1 d.p. has how many significant figures?
15. Find bounds for 3.5 m given to nearest 0.1 m

ASSIGNMENT:

Section A: Decimal Places (15 marks)

Approximate to the specified number of decimal places:

1. To 1 d.p.:
   - a) 45.67
   - b) 23.349
   - c) 0.678
   - d) 156.95
2. To 2 d.p.:
   - a) 34.567
   - b) 0.6789
   - c) 123.995

- d) 8.0049

3. To 3 d.p.:

   - a) 12.34567
   - b) 0.123456
   - c) 67.89954

Section B: Significant Figures (20 marks)

1. Count the significant figures in:

   - a) 3,405
   - b) 0.00670
   - c) 3,400
   - d) 0.04050
   - e) 120.00

2. Approximate to 1 s.f.:

   - a) 456
   - b) 7,834
   - c) 0.0678
   - d) 9,876

3. Approximate to 2 s.f.:

   - a) 4,567
   - b) 0.0678
   - c) 234.56
   - d) 98,765

4. Approximate to 3 s.f.:

   - a) 45,678
   - b) 0.012345
   - c) 789,456

- d) 123,456

5. Approximate to 4 s.f.:
    - a) 567,890
    - b) 0.0123456
    - c) 98.7654

Section C: Place Values (15 marks)

1. Approximate to nearest:

Nearest ten:

- a) 234
- b) 1,895
- c) 9,996

Nearest hundred:

- a) 2,345
- b) 19,850
- c) 99,950

Nearest thousand:

- a) 12,345
- b) 195,500
- c) 999,500

2. Approximate to nearest:

Tenth:

- a) 45.67
- b) 123.95

Hundredth:

- a) 0.6789
- b) 0.0995

Thousandth:

- a) 0.12345
- b) 3.14159

Section D: Estimation (20 marks)

2. Estimate by approximating to 1 s.f.:
   - a) 49 × 21
   - b) 789 ÷ 38
   - c) 6.78 × 9.23
   - d) (47 × 18) ÷ 5
3. Estimate by approximating suitably:
   - a) 234 + 567 + 891
   - b) 5,678 - 2,345
   - c) 45.6 × 3.9
   - d) 897.5 ÷ 29.8
4. Estimate:
   - a) √99
   - b) √48
   - c) √82
5. Estimate the cost of 23 books at ₦487 each
6. Estimate distance if speed is 62 km/h for 4.8 hours

Section E: Bounds (15 marks)

Find the lower and upper bounds for:

1. 45 cm to nearest cm

2. 250 g to nearest 10 g
3. 4.5 m to 1 d.p.
4. 3,500 to 2 s.f.
5. 67.8 to 1 d.p.

Find the maximum and minimum values:

6. Sum of 23 cm and 15 cm (both to nearest cm)
7. Product of 12 m and 8 m (both to nearest m)
8. Difference between 45 kg and 23 kg (both to nearest kg)

Section F: Mixed Problems (15 marks)

1. The population of a city is 3,456,789. Express to:
    - a) Nearest thousand
    - b) Nearest hundred thousand
    - c) 2 s.f.
    - d) 3 s.f.
2. A calculator shows 234.567891234. Write to:
    - a) 2 d.p.
    - b) 4 d.p.
    - c) 3 s.f.
    - d) 5 s.f.
3. Approximate ₦12,345.67 to:
    - a) Nearest naira
    - b) Nearest ₦10
    - c) Nearest ₦100
4. Distance from Earth to Moon: 384,400 km. Express to:
    - a) 2 s.f.
    - b) 3 s.f.

- c) Nearest thousand km

5. Find absolute error when:

    - a) 45.78 is approximated to 46
    - b) π (3.14159...) is approximated to 3.14

WEEK 8: DIRECTED AND NON-DIRECTED NUMBERS

TOPIC: Definition and Examples of Directed and Non-Directed Numbers; Use of Square and Square Root Tables

CONTENT:

A. INTRODUCTION TO DIRECTED NUMBERS

Definition of Directed Numbers: Directed numbers (also called integers or signed numbers) are numbers with direction indicated by a + (positive) or - (negative) sign.

Number Line:

... -5 -4 -3 -2 -1 0 +1 +2 +3 +4 +5 ...
← Negative Zero Positive →

Types of Numbers:

6. **Positive Numbers (+):** Numbers greater than zero
    - Examples: +1, +5, +10, +234
    - Usually written without the + sign: 1, 5, 10, 234
7. **Negative Numbers (-):** Numbers less than zero
    - Examples: -1, -5, -10, -234
    - MUST be written with the - sign
8. **Zero (0):** Neither positive nor negative
    - The point of reference

B. NON-DIRECTED NUMBERS

Definition: Non-directed numbers are positive numbers without signs. They represent magnitude only, with no direction.

Examples:

- Counting numbers: 1, 2, 3, 4, 5...
- Natural numbers
- Absolute values

Difference:

Directed Numbers	Non-Directed Numbers
+5, -3, +7, -10	5, 3, 7, 10
Have direction	Only magnitude
Can be negative	Always positive
Used for: temperature, profit/loss, above/below	Used for: counting, measuring quantities

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C. REAL-LIFE EXAMPLES OF DIRECTED NUMBERS

1. Temperature:

- Above freezing: +5°C, +10°C, +25°C

- Below freezing: -2°C, -10°C, -20°C

- Freezing point: 0°C

Example 1: Temperature drops from +5°C to -3°C. What's the change?

Change = Final - Initial

= (-3) - (+5)

= -3 - 5

= -8°C (a drop of 8 degrees)

2. Sea Level:

- Above sea level: +500m, +2,000m

- Below sea level: -50m, -200m

- Sea level: 0m

Example 2: A submarine at -120m rises to -45m. How far did it rise?

Distance = |-45 - (-120)|

= |-45 + 120|

= |75|

= 75m

3. Profit and Loss:

- Profit: +₦5,000, +₦10,000
- Loss: -₦3,000, -₦7,000
- Break-even: ₦0

Example 3: A trader had losses of ₦2,000, ₦3,000, then profit of ₦8,000. What's net position?

Net = -2,000 + (-3,000) + 8,000

= -5,000 + 8,000

= +₦3,000 (profit)

4. Time Zones:

- Ahead of GMT: +1, +2, +5
- Behind GMT: -5, -7, -10
- GMT: 0

5. Bank Account:

- Credit (deposit): +₦5,000
- Debit (withdrawal): -₦2,000
- Overdraft: Negative balance

6. Elevation/Depth:

- Mountain height: +3,000m
- Ocean depth: -5,000m
- Ground level: 0m

7. Football/Sports:

- Goals for: +3
- Goals against: -2
- Goal difference: +1

D. ORDERING DIRECTED NUMBERS

Rule: On a number line, numbers to the right are greater.

Example 4: Arrange in ascending order: -5, +3, -2, 0, +7, -10

Number line: -10, -5, -2, 0, +3, +7

Ascending order (smallest to largest):
-10, -5, -2, 0, +3, +7

Example 5: Arrange in descending order: -8, +4, -3, +9, 0, -1

Descending order (largest to smallest):
+9, +4, 0, -1, -3, -8

Comparison Rules:

- Any positive number > 0 > any negative number
- For negative numbers: the one with smaller absolute value is greater Example: -3 > -7 (because -3 is closer to zero)

Example 6: Which is greater?

a) -5 or -12? Answer: -5 (closer to zero)
b) +8 or -15? Answer: +8 (positive > negative)
c) 0 or -3? Answer: 0 (zero > negative)
d) -7 or +2? Answer: +2 (positive > negative)

E. ABSOLUTE VALUE

Definition: The absolute value of a number is its distance from zero on the number line, regardless of direction.

Symbol: $|n|$ (vertical bars)

Example 7: Find absolute values:

a) $|+5| = 5$
b) $|-5| = 5$
c) $|0| = 0$
d) $|+234| = 234$
e) $|-234| = 234$
f) $|-45.7| = 45.7$

Properties:

- $|x| \geq 0$ (always non-negative)
- $|x| = |-x|$
- $|x + y| \leq |x| + |y|$ (triangle inequality)

Example 8: Compare using absolute values:

$|-8|$ and $|+5|$
$|-8| = 8$
$|+5| = 5$
Therefore: $|-8| > |+5|$

F. ADDITION OF DIRECTED NUMBERS

Rules:

1. Same Signs: Add absolute values, keep the sign

(+3) + (+5) = +8

(-3) + (-5) = -8

2. Different Signs: Subtract smaller absolute value from larger, take sign of larger

(+7) + (-3) = +4 (7-3=4, sign of 7)

(-7) + (+3) = -4 (7-3=4, sign of -7)

Example 9: Addition problems:

a) (+8) + (+5) = +13

b) (-6) + (-4) = -10

c) (+12) + (-7) = +5

d) (-15) + (+9) = -6

e) (+20) + (-20) = 0

f) (-8) + (+15) = +7

g) (+7) + (-12) = -5

h) (-3) + (-7) + (+5) = -10 + (+5) = -5

Example 10: Word problem: Temperature is +8°C, drops 15°C. Final temperature?

(+8) + (-15) = -7°C

G. SUBTRACTION OF DIRECTED NUMBERS

Rule: Change subtraction to addition of the opposite

a - b = a + (-b)

Example 11: Subtraction problems:

a) (+8) - (+5) = (+8) + (-5) = +3

b) (+8) - (-5) = (+8) + (+5) = +13

c) (-8) - (+5) = (-8) + (-5) = -13

d) (-8) - (-5) = (-8) + (+5) = -3

e) 0 - (+7) = 0 + (-7) = -7

f) 0 - (-7) = 0 + (+7) = +7

Example 12: Complex problems:

a) (+12) - (+8) - (-5)

= (+12) + (-8) + (+5)

= +4 + (+5)

= +9

b) (-15) - (-7) - (+3)

= (-15) + (+7) + (-3)

= -8 + (-3)

= -11

Example 13: Word problem: Account has +₦5,000. After withdrawal of ₦8,000, what's balance?

(+5,000) - (+8,000)

= (+5,000) + (-8,000)

= -₦3,000 (overdraft)

H. COMBINED OPERATIONS

Example 14: Evaluate:

(+8) + (-5) - (+3) - (-7)

= (+8) + (-5) + (-3) + (+7)

= +8 - 5 - 3 + 7

= +7

Example 15: Evaluate:

(-12) - (-8) + (+5) - (+15)

= (-12) + (+8) + (+5) + (-15)

= -12 + 8 + 5 - 15

= -14

Example 16: Simplify:

(+7) - [(+3) - (-5)]

= (+7) - [(+3) + (+5)]

= (+7) - (+8)

= (+7) + (-8)

= -1

I. SQUARE AND SQUARE ROOT TABLES

Using Square Tables:

Square tables show squares of numbers from 1 to 100 (or more).

Example 17: Find using tables:

a) $7^2 = 49$ (lookup 7 in table)

b) $23^2 = 529$

c) $45^2 = 2{,}025$

d) $67^2 = 4{,}489$

Reading Square Tables:

- Locate the number in the leftmost column
- Read across to find its square

Example 18: Find squares of:

a) 12 = 144

b) 35 = 1,225

c) 58 = 3,364

d) 89 = 7,921

Using Square Root Tables:

Square root tables show square roots of numbers.

Example 19: Find using tables:

a) $\sqrt{49} = 7$
b) $\sqrt{144} = 12$
c) $\sqrt{529} = 23$
d) $\sqrt{2,025} = 45$

Example 20: Approximate square roots:

a) $\sqrt{50}$ (between $\sqrt{49}=7$ and $\sqrt{64}=8$)
$\sqrt{50} \approx 7.07$ (from table)

b) $\sqrt{150} \approx 12.25$ (from table)

c) $\sqrt{300} \approx 17.32$ (from table)

Interpolation for Non-Perfect Squares:

Example 21: Find $\sqrt{54}$ using tables

From table:
$\sqrt{49} = 7.000$
$\sqrt{64} = 8.000$

$\sqrt{54}$ is between 7 and 8
From detailed table: $\sqrt{54} \approx 7.348$

Example 22: Find $\sqrt{72}$

$\sqrt{72} \approx 8.485$ (from table)

Using Tables for Larger Numbers:

Example 23: Find $\sqrt{2,500}$

$\sqrt{2,500} = \sqrt{(25 \times 100)}$

$= \sqrt{25} \times \sqrt{100}$

$= 5 \times 10$

$= 50$

Example 24: Find $\sqrt{4,900}$

$\sqrt{4,900} = \sqrt{(49 \times 100)}$

$= 7 \times 10$

$= 70$

Using Tables with Decimals:

Example 25: Find $\sqrt{0.64}$

$\sqrt{0.64} = \sqrt{(64/100)}$

$= \sqrt{64}/\sqrt{100}$

$= 8/10$

$= 0.8$

Example 26: Find $\sqrt{0.0016}$

$\sqrt{0.0016} = \sqrt{(16/10,000)}$

$= 4/100$

$= 0.04$

J. PRACTICAL APPLICATIONS

Example 27: Temperature Problems Day temperatures: +15°C, -5°C, +8°C, -12°C, +3°C Find: a) Average temperature b) Range

a) Average = Sum ÷ Count

= [15 + (-5) + 8 + (-12) + 3] ÷ 5

= [15 - 5 + 8 - 12 + 3] ÷ 5

= 9 ÷ 5

= +1.8°C

b) Range = Highest - Lowest

= (+15) - (-12)

= 15 + 12

= 27°C

Example 28: Finance Weekly profit/loss: -₦2,000, +₦5,000, -₦1,500, +₦8,000, -₦3,000 Find net position:

Net = -2,000 + 5,000 - 1,500 + 8,000 - 3,000

= +₦6,500 (net profit)

Example 29: Elevation A helicopter at +1,200m altitude descends to -50m (below sea level). What's the change?

Change = (-50) - (+1,200)

= -50 - 1,200

= -1,250m (descended 1,250m)

Example 30: Square Root Application A square field has area 2,304 m². Find the length of one side.

Side = √2,304

= √(23.04 × 100)

= √23.04 × 10

From table: $\sqrt{23.04} \approx 4.8$

Side = $4.8 \times 10 = 48$ m

EVALUATION:

1. Define directed and non-directed numbers with examples
2. Arrange in ascending order: -8, +3, -1, 0, +5, -12
3. Calculate: a) (-7) + (+12) b) (+15) - (-8)
4. Find: a) |-15| b) |+23|
5. Temperature drops from +12°C to -5°C. What's the change?
6. Using tables, find: a) 34^2 b) $\sqrt{1,225}$
7. Simplify: (+8) - (+5) + (-3) - (-7)
8. Find $\sqrt{0.81}$ using square root tables
9. Which is greater: -15 or -8?
10. Calculate: (-12) + (+7) - (-5) + (-3)

REVISION QUESTIONS:

1. What is the difference between directed and non-directed numbers?
2. Give 5 real-life examples where directed numbers are used
3. What is the absolute value of a number?
4. Is zero a directed number? Explain
5. How do you add two negative numbers?
6. How do you subtract a negative from a positive?
7. Arrange: -20, +15, 0, -5, +8, -15, +20
8. Temperature is -8°C, rises 15°C. Final temperature?
9. Account balance: +₦3,000, withdraw ₦5,000. New balance?
10. Find absolute value: |-45.7|
11. Calculate: (+12) - (-12)

12. What's the rule for subtracting directed numbers?
13. Using tables, find 47^2
14. Find $\sqrt{196}$ without tables
15. Explain how to use square root tables for non-perfect squares

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ASSIGNMENT:

Section A: Concepts (10 marks)

1. Define:
    - a) Directed numbers
    - b) Non-directed numbers
    - c) Absolute value
2. Give 3 real-life situations where:
    - a) Positive numbers are used
    - b) Negative numbers are used
3. Explain why zero is neither positive nor negative

Section B: Ordering and Comparing (15 marks)

4. Arrange in ascending order:
    - a) -12, +8, -5, 0, +3, -1
    - b) -20, -15, -10, -25, -5
    - c) +15, -18, +12, -8, 0, +5
5. Arrange in descending order:
    - a) -7, +9, -3, +2, 0, -12
    - b) +20, -15, +5, -25, +10, 0
6. Which is greater?
    - a) -15 or -8

- b) -25 or +3
- c) 0 or -5
- d) |-7| or |+5|
- e) |-12| or |-20|

Section C: Absolute Value (10 marks)

Find the absolute value:

1. |+25|
2. |-34|
3. |0|
4. |-125.5|
5. |+45.78|

Evaluate:

6. |-8| + |+5|
7. |-12| - |+7|
8. |-15| × |+3|
9. |-20| ÷ |-5|
10. |(-8) + (15)|

Section D: Addition (20 marks)

Calculate:

1. (+7) + (+12)
2. (-8) + (-15)
3. (+20) + (-12)
4. (-15) + (+8)
5. (+25) + (-25)
6. (-18) + (+30)
7. (+45) + (-67)

8. (-123) + (+89)

Multiple operations:

9. (+8) + (-5) + (+12)
10. (-15) + (+7) + (-3)
11. (+20) + (-12) + (+8) + (-15)
12. (-7) + (-5) + (+18) + (-6)

Word problems:

13. Temperature +12°C, rises 8°C. Final temperature?
14. Account +₦5,000, deposit ₦3,000, withdraw ₦7,000. Balance?
15. Profit ₦8,000, loss ₦12,000, profit ₦15,000. Net position?

Section E: Subtraction (20 marks)

Calculate:

1. (+15) - (+8)
2. (+12) - (-5)
3. (-18) - (+7)
4. (-20) - (-12)
5. 0 - (+15)
6. 0 - (-20)
7. (+45) - (+67)
8. (-34) - (-56)

Complex problems:

9. (+12) - (+8) - (-5)
10. (-15) - (-7) - (+3)
11. (+20) - (-12) - (+8)
12. (-25) - (+15) - (-10)

Word problems:

13. Temperature +5°C, drops 18°C. Final temperature?
14. Mountain +2,500m, valley -200m. Difference?
15. Account +₦12,000, withdrawal ₦18,000. Balance?

Section F: Combined Operations (15 marks)

Evaluate:

1. (+8) + (-5) - (+3)
2. (-12) - (+7) + (-3)
3. (+15) + (-8) - (-5) + (+3)
4. (-20) + (+12) - (-8) - (+15)
5. (+25) - (-15) + (-8) - (+12)

Simplify:

6. (+7) - [(+3) - (-5)]
7. (-12) + [(-8) - (+5)]
8. (+20) - [(+12) - (-7) + (+3)]

Word problems:

9. Weekly profit/loss: -₦3,000, +₦7,000, -₦2,000, +₦5,000. Net?
10. Temperature changes: +8°C, -12°C, +5°C, -3°C. Net change?

Section G: Square and Square Root Tables (10 marks)

Using tables, find:

1. 15^2
2. 38^2
3. 62^2
4. 89^2
5. $\sqrt{225}$

6. $\sqrt{676}$
7. $\sqrt{1,521}$
8. $\sqrt{3,844}$

Without tables:

9. $\sqrt{2,500}$
10. $\sqrt{0.64}$

WEEK 9: MULTIPLICATION AND DIVISION OF DIRECTED NUMBERS

TOPIC: Multiplication and Division of Directed and Non-Directed Numbers with Practical Exercises

CONTENT:

A. RULES FOR MULTIPLICATION OF DIRECTED NUMBERS

Sign Rules:

1. **Positive × Positive = Positive**
   - (+) × (+) = (+)
2. **Negative × Negative = Positive**
   - (-) × (-) = (+)
3. **Positive × Negative = Negative**
   - (+) × (-) = (-)
4. **Negative × Positive = Negative**
   - (-) × (+) = (-)

Memory Aid:

- Same signs → Positive result
- Different signs → Negative result

B. MULTIPLICATION OF DIRECTED NUMBERS

Example 1: Simple multiplication:

a) (+5) × (+3) = +15

b) (-5) × (-3) = +15

c) (+5) × (-3) = -15

d) (-5) × (+3) = -15

e) (+8) × (+7) = +56

f) $(-8) \times (-7) = +56$

g) $(+8) \times (-7) = -56$

h) $(-8) \times (+7) = -56$

Example 2: Multiplication with larger numbers:

a) $(+12) \times (+5) = +60$

b) $(-12) \times (-5) = +60$

c) $(+12) \times (-5) = -60$

d) $(-12) \times (+5) = -60$

e) $(+25) \times (+4) = +100$

f) $(-25) \times (-4) = +100$

g) $(+25) \times (-4) = -100$

h) $(-25) \times (+4) = -100$

Example 3: Multiplication with zero:

a) $(+5) \times 0 = 0$

b) $(-5) \times 0 = 0$

c) $0 \times (+8) = 0$

d) $0 \times (-8) = 0$

Rule: Any number $\times$ 0 = 0

Example 4: Multiplication with one:

a) $(+7) \times (+1) = +7$

b) $(-7) \times (+1) = -7$

c) $(+7) \times (-1) = -7$

d) $(-7) \times (-1) = +7$

Rule: $n \times 1 = n$

$n \times (-1) = -n$

C. MULTIPLE MULTIPLICATIONS

Example 5: Three numbers:

a) (+2) × (+3) × (+4)
= (+6) × (+4)
= +24

b) (-2) × (-3) × (-4)
= (+6) × (-4)
= -24

c) (+2) × (-3) × (+4)
= (-6) × (+4)
= -24

d) (-2) × (+3) × (-4)
= (-6) × (-4)
= +24

Rule for Multiple Multiplications:

- **Even number of negative signs → Positive result**
- **Odd number of negative signs → Negative result**

Example 6: Count negative signs:

a) (-2) × (-3) × (-4) × (-5)
4 negative signs (even) → Positive
= +120

b) (-2) × (+3) × (-4) × (+5)
2 negative signs (even) → Positive
= +120

c) (-2) × (-3) × (+4) × (-5)

3 negative signs (odd) → Negative

= -120

d) (+2) × (-3) × (-4) × (-5)

3 negative signs (odd) → Negative

= -120

Example 7: More complex:

a) (-1) × (-2) × (-3) × (-4) × (-5)

5 negative signs (odd) → Negative

= -120

b) (-3) × (-3) × (-3)

3 negative signs (odd) → Negative

= -27

c) (-2) × (-2) × (-2) × (-2)

4 negative signs (even) → Positive

= +16

D. POWERS OF DIRECTED NUMBERS

Example 8: Powers of positive numbers:

a) $(+2)^2 = (+2) \times (+2) = +4$

b) $(+3)^3 = (+3) \times (+3) \times (+3) = +27$

c) $(+5)^2 = +25$

d) $(+10)^3 = +1,000$

Example 9: Powers of negative numbers:

a) $(-2)^2 = (-2) \times (-2) = +4$ (even power → positive)

b) $(-2)^3 = (-2) \times (-2) \times (-2) = -8$ (odd power → negative)

c) $(-3)^2 = +9$ (even power)

d) $(-3)^3 = -27$ (odd power)

e) $(-5)^4 = +625$ (even power)

f) $(-5)^5 = -3{,}125$ (odd power)

Rule:

- **$(-n)^{even}$ = positive**
- **$(-n)^{odd}$ = negative**

Example 10: Compare:

a) $(-4)^2 = +16$

b) $-(4)^2 = -(16) = -16$

Note the difference:

$(-4)^2$ means square the negative number

$-(4)^2$ means negate the squared number

E. DIVISION OF DIRECTED NUMBERS

Sign Rules (Same as Multiplication):

5. **Positive ÷ Positive = Positive**
   - (+) ÷ (+) = (+)
6. **Negative ÷ Negative = Positive**
   - (-) ÷ (-) = (+)
7. **Positive ÷ Negative = Negative**
   - (+) ÷ (-) = (-)
8. **Negative ÷ Positive = Negative**

- (-) ÷ (+) = (-)

Example 11: Simple division:

a) (+12) ÷ (+3) = +4

b) (-12) ÷ (-3) = +4

c) (+12) ÷ (-3) = -4

d) (-12) ÷ (+3) = -4

e) (+20) ÷ (+5) = +4

f) (-20) ÷ (-5) = +4

g) (+20) ÷ (-5) = -4

h) (-20) ÷ (+5) = -4

Example 12: Larger numbers:

a) (+56) ÷ (+8) = +7

b) (-56) ÷ (-8) = +7

c) (+56) ÷ (-8) = -7

d) (-56) ÷ (+8) = -7

e) (+144) ÷ (+12) = +12

f) (-144) ÷ (-12) = +12

g) (+144) ÷ (-12) = -12

h) (-144) ÷ (+12) = -12

Example 13: Division with remainders:

a) (+25) ÷ (+4) = +6 R 1 or +6.25

b) (-25) ÷ (-4) = +6 R 1 or +6.25

c) (+25) ÷ (-4) = -6 R 1 or -6.25

d) (-25) ÷ (+4) = -6 R 1 or -6.25

Example 14: Division involving zero:

a) 0 ÷ (+5) = 0

b) 0 ÷ (-5) = 0

c) (+5) ÷ 0 = undefined (cannot divide by zero)

d) (-5) ÷ 0 = undefined

Rule: 0 ÷ n = 0 (for any n ≠ 0)

n ÷ 0 = undefined

F. COMBINED MULTIPLICATION AND DIVISION

Example 15: Multiple operations:

a) (+24) ÷ (+6) × (+2)

= (+4) × (+2)

= +8

b) (-24) ÷ (-6) × (-2)

= (+4) × (-2)

= -8

c) (+48) ÷ (-8) ÷ (+2)

= (-6) ÷ (+2)

= -3

d) (-48) ÷ (-8) ÷ (-2)

= (+6) ÷ (-2)

= -3

Example 16: With brackets:

a) [(-12) × (+5)] ÷ (-6)

= (-60) ÷ (-6)

= +10

b) (-15) × [(+20) ÷ (-4)]

= (-15) × (-5)

= +75

c) [(+36) ÷ (-9)] × [(-8) ÷ (+4)]

= (-4) × (-2)

= +8

Example 17: Complex expressions:

a) (-3) × (+4) ÷ (-2) × (+5)

= (-12) ÷ (-2) × (+5)

= (+6) × (+5)

= +30

b) (+48) ÷ (-6) × (-2) ÷ (+4)

= (-8) × (-2) ÷ (+4)

= (+16) ÷ (+4)

= +4

G. BODMAS WITH DIRECTED NUMBERS

Remember: BODMAS/PEDMAS order:

- **B**rackets
- **O**rders (Powers)
- **D**ivision and **M**ultiplication (left to right)
- **A**ddition and **S**ubtraction (left to right)

Example 18: With addition and subtraction:

a) (+8) + (-5) × (+2)

= (+8) + (-10) [multiply first]

= -2

b) $(-12) \div (+4) - (-3)$

$= (-3) - (-3)$ [divide first]

$= (-3) + (+3)$

$= 0$

c) $(+15) - (-6) \times (+2)$

$= (+15) - (-12)$

$= (+15) + (+12)$

$= +27$

Example 19: With brackets:

a) $[(-8) + (+3)] \times (+5)$

$= (-5) \times (+5)$

$= -25$

b) $(+20) \div [(-12) + (+7)]$

$= (+20) \div (-5)$

$= -4$

c) $[(-15) - (+5)] \div [(-8) + (+3)]$

$= (-20) \div (-5)$

$= +4$

Example 20: With powers:

a) $(-2)^3 + (+5) \times (-3)$

$= (-8) + (+5) \times (-3)$ [power first]

$= (-8) + (-15)$ [multiply]

$= -23$ [add]

b) $(+3)^2 - (-4)^2$

$= (+9) - (+16)$

$= (+9) + (-16)$

$= -7$

c) $(-5)^2 \times (+2) - (-3)^3$

$= (+25) \times (+2) - (-27)$

$= (+50) - (-27)$

$= (+50) + (+27)$

$= +77$

Example 21: Complex BODMAS:

$(-6) \times [(-8) \div (+4) + (-3)] - (-2)^3$

Step 1: Brackets - Division first

$= (-6) \times [(-2) + (-3)] - (-2)^3$

Step 2: Brackets - Addition

$= (-6) \times (-5) - (-2)^3$

Step 3: Power

$= (-6) \times (-5) - (-8)$

Step 4: Multiplication

$= (+30) - (-8)$

Step 5: Subtraction

$= (+30) + (+8)$

$= +38$

H. WORD PROBLEMS

Example 22: Temperature problem: Temperature drops 3°C every hour for 5 hours. If it starts at +10°C, what's the final temperature?

Solution:

Total drop = $5 \times (-3) = -15°C$

Final temperature = $(+10) + (-15) = -5°C$

Example 23: Financial problem: A business loses ₦2,500 daily for 6 days. What's the total loss?

Solution:

Total loss = $6 \times (-2{,}500) =$ -₦15,000

Example 24: Profit/Loss problem: A trader's weekly profit/loss pattern: gains ₦4,000 on 3 days, loses ₦6,000 on 2 days. What's the net position?

Solution:

Gains: $3 \times (+4{,}000) =$ +₦12,000

Losses: $2 \times (-6{,}000) =$ -₦12,000

Net: $(+12{,}000) + (-12{,}000) =$ ₦0 (break-even)

Example 25: Depth problem: A submarine descends 15 meters every minute for 8 minutes. If it started at sea level (0m), what's its final depth?

Solution:

Total descent = $8 \times (-15) = -120$ meters

Final position = $0 + (-120) = -120$m (120m below sea level)

Example 26: Sharing problem: A debt of ₦24,000 is shared equally among 6 people. How much does each person owe?

Solution:

Each person's share = $(-24{,}000) \div 6 =$ -₦4,000

(Each owes ₦4,000)

Example 27: Temperature change: Temperature changes by -3°C per hour. After how many hours will a temperature of +18°C reach -6°C?

Solution:
Total change needed = (-6) - (+18) = -24°C
Number of hours = (-24) ÷ (-3) = +8 hours

Example 28: Investment problem: An investor loses ₦5,000 monthly for 4 months, then gains ₦12,000 for 2 months. What's the net result?

Solution:
Losses: 4 × (-5,000) = -₦20,000
Gains: 2 × (+12,000) = +₦24,000
Net: (-20,000) + (+24,000) = +₦4,000 (profit)

Example 29: Elevator problem: An elevator descends 4 floors (each floor = -3m) then ascends 7 floors. Starting at ground level (0m), what's the final position?

Solution:
Descent: 4 × (-3) = -12m
Ascent: 7 × (+3) = +21m
Final position: 0 + (-12) + (+21) = +9m (9m above ground)

Example 30: Multiplication pattern: If (-2) × (-3) × (-4) = -24, explain why the answer is negative.

Explanation:
Three negative numbers (odd count)
Odd number of negatives → Negative result
(-2) × (-3) = +6
(+6) × (-4) = -24

I. RECIPROCALS AND DIVISION

Definition: The reciprocal of a number n is 1/n.

Example 31: Find reciprocals:

a) Reciprocal of +5 = 1/5 or +0.2

b) Reciprocal of -4 = -1/4 or -0.25

c) Reciprocal of +1/3 = +3

d) Reciprocal of -2/5 = -5/2 or -2.5

Division using reciprocals: $a \div b = a \times (1/b)$

Example 32: Using reciprocals:

a) $(+12) \div (+3) = (+12) \times (1/3) = +4$

b) $(-20) \div (-5) = (-20) \times (-1/5) = +4$

c) $(+15) \div (-3) = (+15) \times (-1/3) = -5$

J. PROPERTIES OF MULTIPLICATION AND DIVISION

1. Commutative Property (Multiplication only): $a \times b = b \times a$

Example 33:

$(-5) \times (+3) = (+3) \times (-5) = -15$ ✓

But: $(+12) \div (+3) \neq (+3) \div (+12)$

$4 \neq 0.25$

Division is NOT commutative

2. Associative Property (Multiplication only): $(a \times b) \times c = a \times (b \times c)$

Example 34:

$[(-2) \times (+3)] \times (-4) = (-2) \times [(+3) \times (-4)]$

$(-6) \times (-4) = (-2) \times (-12)$

$+24 = +24$ ✓

Division is NOT associative

3. Distributive Property: $a \times (b + c) = (a \times b) + (a \times c)$

Example 35:

$(-3) \times [(+5) + (-2)]$
$= (-3) \times (+3)$
$= -9$

OR:
$[(-3) \times (+5)] + [(-3) \times (-2)]$
$= (-15) + (+6)$
$= -9$ ✓

4. Identity Elements:

- Multiplication identity: 1 (any number × 1 = itself)
- Division identity: 1 (any number ÷ 1 = itself)

Example 36:

$(-7) \times 1 = -7$
$(-7) \div 1 = -7$

K. RATIO AND DIRECTED NUMBERS

Example 37: Simplify ratios with directed numbers:

a) $(+12) : (-4) = -3 : 1$ or $-3/1$
b) $(-15) : (+5) = -3 : 1$
c) $(-20) : (-4) = +5 : 1$ or $5 : 1$

Example 38: Sharing with directed numbers: Divide -₦60 (debt) in ratio 2:3

Total parts = 2 + 3 = 5

First share = (2/5) × (-60) = -₦24

Second share = (3/5) × (-60) = -₦36

Verification: -24 + (-36) = -60 ✓

L. PATTERNS AND SEQUENCES

Example 39: Find the pattern:

(+2), (-4), (+8), (-16), (+32), ...

Pattern: Multiply by -2 each time

Next term: (+32) × (-2) = -64

Example 40: Complete the sequence:

(-3), (+6), (-12), (+24), ...

Pattern: Multiply by -2

Next term: (+24) × (-2) = -48

Example 41: Find the missing term:

(+5), (-10), (+20), ___, (+80)

Pattern: Multiply by -2

Missing term: (+20) × (-2) = -40

EVALUATION:

1. Calculate: a) (-7) × (+8) b) (+15) × (-4) c) (-9) × (-6)
2. Calculate: a) (+48) ÷ (-6) b) (-72) ÷ (-9) c) (-56) ÷ (+8)

3. Evaluate: $(-2) \times (-3) \times (-4)$
4. Find: a) $(-5)^2$ b) $(-3)^3$ c) $(-2)^4$
5. Simplify: $[(-15) + (+8)] \times (-3)$
6. Evaluate: $(-4)^2 - (+3) \times (-2)$
7. Temperature drops 5°C hourly for 6 hours from +12°C. Final temperature?
8. Divide -₦120 debt among 8 people. How much each?
9. Calculate: $(+48) \div (-6) \times (-2)$
10. Evaluate: $(-2)^3 \times (+3) - (-5)^2$

REVISION QUESTIONS:

1. State the rules for multiplying directed numbers
2. What is the sign of the product of 5 negative numbers? Why?
3. Explain why $(-3)^2 \neq -(3)^2$
4. Can you divide a number by zero? Explain
5. What's the sign of $(-2)^{100}$?
6. Calculate: $(-5) \times (+6) \div (-2)$
7. If $a \times b = -24$ and $a = -6$, find b
8. True or False: $(-5) \times (-5) = -25$? Explain
9. Simplify: $[(-20) \div (+5)] \times [(-3) \times (+2)]$
10. What's the reciprocal of -8?
11. Evaluate: $(-1)^5 + (-1)^6$
12. Pattern: (+3), (-6), (+12), (-24), ... Find next 2 terms
13. A loss of ₦500 per day for 12 days. Total loss?
14. Temperature -8°C, rises 3°C per hour for 5 hours. Final?
15. Explain why odd number of negatives gives negative result

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ASSIGNMENT:

Section A: Multiplication (25 marks)

1. Calculate:
    - a) $(+8) \times (+7)$
    - b) $(-8) \times (-7)$
    - c) $(+8) \times (-7)$
    - d) $(-8) \times (+7)$
    - e) $(+15) \times (+4)$
    - f) $(-15) \times (-4)$
    - g) $(+15) \times (-4)$
    - h) $(-15) \times (+4)$
2. Multiply:
    - a) $(-12) \times (+5)$
    - b) $(+25) \times (-3)$
    - c) $(-18) \times (-6)$
    - d) $(+32) \times (+4)$
3. Evaluate:
    - a) $(-2) \times (-3) \times (+4)$
    - b) $(+5) \times (-2) \times (-3)$
    - c) $(-4) \times (+3) \times (-2)$
    - d) $(-2) \times (-2) \times (-2)$
    - e) $(-3) \times (-3) \times (-3) \times (-3)$
4. Find:
    - a) $(-4)^2$
    - b) $(-5)^3$
    - c) $(-2)^5$
    - d) $(-1)^7$
    - e) $(-3)^4$
5. Calculate:

- a) (-1) × (+2) × (-3) × (+4) × (-5)
- b) $(-2)^2 \times (-3)^2$
- c) $(-5)^3 \times (+2)^2$

Section B: Division (25 marks)

1. Calculate:
    - a) (+36) ÷ (+6)
    - b) (-36) ÷ (-6)
    - c) (+36) ÷ (-6)
    - d) (-36) ÷ (+6)
    - e) (+72) ÷ (+8)
    - f) (-72) ÷ (-8)
    - g) (+72) ÷ (-8)
    - h) (-72) ÷ (+8)
2. Divide:
    - a) (-84) ÷ (+12)
    - b) (+96) ÷ (-16)
    - c) (-108) ÷ (-9)
    - d) (+144) ÷ (+12)
3. Calculate:
    - a) (+48) ÷ (-6) ÷ (+2)
    - b) (-60) ÷ (-5) ÷ (-3)
    - c) (+72) ÷ (-8) ÷ (-3)
    - d) (-96) ÷ (+8) ÷ (-3)
4. Evaluate:
    - a) [(-24) × (+3)] ÷ (-6)
    - b) (+40) ÷ [(-15) + (+5)]
    - c) [(-32) ÷ (+8)] × [(-5) × (+2)]

5. Find the reciprocal:

    - a) +8
    - b) -5
    - c) +1/4
    - d) -3/7

Section C: Combined Operations (20 marks)

1. Evaluate:

    - a) $(-8) + (+5) \times (-2)$
    - b) $(+12) \div (-3) - (+5)$
    - c) $(-15) \times (+2) + (-8) \div (+4)$
    - d) $(+20) - (-6) \times (+3)$

2. Simplify:

    - a) $[(-12) + (+5)] \times (-3)$
    - b) $(+18) \div [(-8) + (+2)]$
    - c) $[(-15) - (+7)] \div [(-6) + (+4)]$
    - d) $[(-3) \times (+4)] - [(-20) \div (+5)]$

3. Calculate using BODMAS:

    - a) $(-3)^2 + (+5) \times (-2)$
    - b) $(+4)^2 - (-3)^2 \times (+2)$
    - c) $(-2)^3 \times (+3) - (-4)^2$
    - d) $(+5)^2 \div (-5) + (-3)^3$

4. Evaluate:

    - a) $(-5) \times [(-8) \div (+2) + (-3)]$
    - b) $(+6) \div [(-12) \div (-4) - (+1)]$
    - c) $[(-3)^2 - (+2)^3] \times (-5)$

5. Complex expressions:

- a) $(-2) \times [(+15) \div (-3) + (-7)] - (+4)^2$
- b) $[(-5)^2 - (-3)^3] \div [(-8) + (+2)]$

Section D: Word Problems (20 marks)

1. Temperature drops 4°C per hour for 7 hours. Starting at +15°C, what's the final temperature?
2. A business loses ₦3,500 daily for 9 days. What's the total loss?
3. A debt of ₦48,000 is shared equally among 12 people. How much does each owe?
4. A submarine descends 20 meters every 2 minutes for 10 minutes. Starting at sea level, what's its final depth?
5. Temperature at midnight is -8°C. It rises 2°C per hour for 5 hours. What's the temperature at 5:00 AM?
6. A trader gains ₦5,000 daily for 4 days, then loses ₦8,000 daily for 3 days. What's the net position?
7. An elevator starting at ground level (0m) descends 6 floors (each = -4m), then ascends 10 floors. Final position?
8. If temperature drops 6°C every 2 hours, how long to go from +18°C to -12°C?
9. Share a loss of ₦84,000 among three partners in ratio 2:3:7. How much does each lose?
10. A stock price falls ₦25 per day for 6 days, then rises ₦30 per day for 5 days. Starting at ₦500, what's the final price?

Section E: Patterns and Properties (10 marks)

1. Find the next two terms:
    - a) (+4), (-8), (+16), (-32), ...
    - b) (-5), (+10), (-20), (+40), ...
2. Find the missing term:

- a) (-2), (+6), (-18), ___, (+162)
- b) (+3), (-12), (+48), ___, (+768)

3. True or False (Explain):

- a) $(-5) \times (-3) = +15$
- b) $(-4)^2 = -16$
- c) $(-2)^3 = -8$
- d) $0 \div (-5) = 0$
- e) $(-8) \div 0$ = undefined

4. Verify:

- a) $(-3) \times [(+5) + (-2)] = [(-3) \times (+5)] + [(-3) \times (-2)]$
- b) Does $(+12) \div (+3) = (+3) \div (+12)$?

WEEK 10: ALGEBRAIC EXPRESSIONS I

TOPIC: Definition of Algebraic Expressions; Expression of Algebraic Forms; Factorization of Simple Algebraic Expressions

CONTENT:

A. INTRODUCTION TO ALGEBRA

Definition: Algebra is a branch of mathematics that uses letters (variables) to represent numbers in equations and expressions.

Why Use Algebra?

- To represent unknown values
- To generalize mathematical relationships
- To solve problems more efficiently
- To express patterns and formulas

Variables and Constants:

- **Variable:** A letter that represents an unknown or changing value (x, y, a, b, etc.)
- **Constant:** A fixed numerical value (5, -3, 0.5, etc.)

B. ALGEBRAIC EXPRESSIONS

Definition: An algebraic expression is a mathematical phrase that contains numbers, variables, and operation symbols (+, -, ×, ÷).

Examples of Algebraic Expressions:

a) $3x + 5$

b) $2y - 7$

c) $4a + 3b - 2$

d) $x^2 + 5x - 6$

e) $7m - 4n + 8$

Not Algebraic Expressions (these are equations):

$3x + 5 = 11$ (has = sign)

$2y - 7 = 15$ (has = sign)

C. TERMS IN ALGEBRAIC EXPRESSIONS

Definition of a Term: A term is a single number, variable, or the product of numbers and variables.

Example 1: Identify terms in: $5x + 3y - 7$

Terms: $5x$, $3y$, -7

(3 terms separated by + or - signs)

Example 2: Identify terms in: $3x^2 - 4xy + 2y^2 - 8x + 5$

Terms: $3x^2$, $-4xy$, $2y^2$, $-8x$, 5

(5 terms)

Types of Terms:

1. Constant Term: A term without a variable

In $3x + 5y - 7$, the constant term is -7

In $2a^2 - 3a + 8$, the constant term is 8

2. Variable Term: A term containing a variable

In $5x - 3y + 2$, variable terms are $5x$ and $-3y$

3. Coefficient: The numerical part of a variable term

In $5x$: coefficient is 5

In $-3y$: coefficient is -3

In x (which is $1x$): coefficient is 1

In $-y$ (which is $-1y$): coefficient is -1

Example 3: Identify coefficients:

a) In 7a: coefficient = 7

b) In -5b: coefficient = -5

c) In x: coefficient = 1

d) In -y: coefficient = -1

e) In ½x: coefficient = ½

D. LIKE AND UNLIKE TERMS

Like Terms: Terms that have exactly the same variable(s) raised to the same power.

Example 4: Identify like terms:

In: 3x, 5y, -2x, 4x, 7y, -8

Like terms:

- 3x, -2x, 4x (all have x)
- 5y, 7y (all have y)
- -8 is a constant (different from variable terms)

Example 5: Which are like terms?

a) 5x and 3x → Like terms ✓

b) 4y and -7y → Like terms ✓

c) 3x and 3y → Unlike terms ✗ (different variables)

d) $2x^2$ and 5x → Unlike terms ✗ (different powers)

e) 7ab and -3ab → Like terms ✓

f) 4xy and 4yx → Like terms ✓ (order doesn't matter)

Unlike Terms: Terms with different variables or different powers.

Example 6:

5x and 3y are unlike terms (different variables)

$2x^2$ and 3x are unlike terms (different powers)

4ab and 5bc are unlike terms (different variable combinations)

E. SIMPLIFYING ALGEBRAIC EXPRESSIONS

Rule: Only like terms can be added or subtracted.

Example 7: Simplify by collecting like terms:

$3x + 5x = 8x$

$7y - 2y = 5y$

$4a - 7a = -3a$

$-3b - 5b = -8b$

Example 8: Simplify: $5x + 3y - 2x + 7y$

Solution:

Collect like terms:

x terms: $5x - 2x = 3x$

y terms: $3y + 7y = 10y$

Answer: $3x + 10y$

Example 9: Simplify: $7a - 4b + 3a + 8b - 2a$

Solution:

a terms: $7a + 3a - 2a = 8a$

b terms: $-4b + 8b = 4b$

Answer: $8a + 4b$

Example 10: Simplify: $4x + 5 - 2x + 3 + 7x - 8$

Solution:

x terms: $4x - 2x + 7x = 9x$

Constants: $5 + 3 - 8 = 0$

Answer: $9x$

Example 11: Simplify: $3x^2 + 5x - 2x^2 + 7x + 4$

Solution:

x^2 terms: $3x^2 - 2x^2 = x^2$

x terms: $5x + 7x = 12x$

Constants: 4

Answer: $x^2 + 12x + 4$

Example 12: Simplify: $5xy - 3x + 7xy + 2y - 8xy + x$

Solution:

xy terms: $5xy + 7xy - 8xy = 4xy$

x terms: $-3x + x = -2x$

y terms: $2y$

Answer: $4xy - 2x + 2y$

F. REMOVING BRACKETS

Rule 1: Positive sign before bracket: Remove bracket, keep all signs $+(a + b) = a + b$

Example 13:

a) $+(3x + 5) = 3x + 5$

b) $+(7a - 4b) = 7a - 4b$

c) $x + (2y + 3) = x + 2y + 3$

Rule 2: Negative sign before bracket: Remove bracket, change all signs $-(a + b) = -a - b$

Example 14:

a) $-(3x + 5) = -3x - 5$

b) $-(7a - 4b) = -7a + 4b$

c) $5x - (2x + 3) = 5x - 2x - 3 = 3x - 3$

Example 15: Simplify: $3(x + 2y) - 2(2x - y)$

Solution:

$= 3x + 6y - 4x + 2y$

$= 3x - 4x + 6y + 2y$

$= -x + 8y$

Example 16: Simplify: $5(2a - 3b) + 3(4a + 5b)$

Solution:

$= 10a - 15b + 12a + 15b$

$= 10a + 12a - 15b + 15b$

$= 22a$

Example 17: Remove brackets and simplify:

$4x - [3x - (2x + 5)]$

Solution:

$= 4x - [3x - 2x - 5]$

$= 4x - [x - 5]$

$= 4x - x + 5$

$= 3x + 5$

G. MULTIPLICATION OF ALGEBRAIC EXPRESSIONS

Multiplying a Term by a Term:

Example 18:

a) $3 \times 5x = 15x$

b) $4a \times 7 = 28a$

c) $2x \times 3y = 6xy$

d) $5a \times 4b = 20ab$

e) $(-3x) \times 4y = -12xy$

f) $(-2a) \times (-5b) = 10ab$

Multiplying with Powers:

Rule: $x^m \times x^n = x^{m+n}$

Example 19:

a) $x \times x = x^2$ (or $x^{1+1} = x^2$)

b) $x^2 \times x^3 = x^5$

c) $2x \times 3x^2 = 6x^3$

d) $5a^2 \times 3a = 15a^3$

e) $4y^3 \times 2y^2 = 8y^5$

Example 20: Multiply:

a) $3x \times 5x = 15x^2$

b) $4a \times 2a = 8a^2$

c) $2x^2 \times 3x = 6x^3$

d) $5y^3 \times 2y^2 = 10y^5$

e) $(-3a^2) \times 4a^3 = -12a^5$

Expanding Brackets (Distributive Property):

Rule: $a(b + c) = ab + ac$

Example 21:

a) $3(x + 5) = 3x + 15$

b) $4(2a - 3) = 8a - 12$

c) $5(3y + 7) = 15y + 35$

d) $-2(x + 4) = -2x - 8$

e) $-3(2a - 5) = -6a + 15$

Example 22: Expand:

a) $x(x + 3) = x^2 + 3x$

b) $2a(a + 5) = 2a^2 + 10a$

c) $3y(2y - 4) = 6y^2 - 12y$

d) $4x(3x - 7) = 12x^2 - 28x$

e) $-2a(3a + 5) = -6a^2 - 10a$

Example 23: Expand and simplify:

$3(x + 4) + 2(x - 5)$

Solution:

$= 3x + 12 + 2x - 10$

$= 3x + 2x + 12 - 10$

$= 5x + 2$

Example 24: Expand and simplify:

$5(2a - 3) - 2(a + 7)$

Solution:

$= 10a - 15 - 2a - 14$

$= 10a - 2a - 15 - 14$

$= 8a - 29$

H. SUBSTITUTION IN ALGEBRAIC EXPRESSIONS

Substitution: Replacing variables with given numbers.

Example 25: If $x = 3$, find the value of $2x + 5$

Solution:

$2x + 5 = 2(3) + 5$

$= 6 + 5$

$= 11$

Example 26: If $a = -2$, find $3a^2 - 5a + 4$

Solution:

$3a^2 - 5a + 4 = 3(-2)^2 - 5(-2) + 4$

$= 3(4) + 10 + 4$

$= 12 + 10 + 4$

$= 26$

Example 27: If $x = 4$ and $y = 2$, find $3x - 2y + 7$

Solution:

$3x - 2y + 7 = 3(4) - 2(2) + 7$

$= 12 - 4 + 7$

$= 15$

Example 28: If $a = 5$ and $b = -3$, find $2a^2 + 3ab - b^2$

Solution:

$2a^2 + 3ab - b^2 = 2(5)^2 + 3(5)(-3) - (-3)^2$

$= 2(25) + 3(-15) - 9$

$= 50 - 45 - 9$

$= -4$

Example 29: If $p = -1$ and $q = 4$, find $p^3 - 2pq + q^2$

Solution:

$p^3 - 2pq + q^2 = (-1)^3 - 2(-1)(4) + (4)^2$

$= -1 - 2(-4) + 16$

$= -1 + 8 + 16$

$= 23$

I. INTRODUCTION TO FACTORIZATION

Definition: Factorization is the process of expressing an algebraic expression as a product of its factors.

Why Factorize?

- To simplify expressions
- To solve equations
- To identify patterns

Types of Factorization:

5. Common factor
6. Grouping
7. Difference of two squares
8. Trinomials

J. FACTORIZATION BY COMMON FACTOR

Method: Find the highest common factor (HCF) of all terms.

Example 30: Factorize: $6x + 9$

Solution:

HCF of 6 and 9 = 3

$6x + 9 = 3(2x + 3)$

Check: $3(2x + 3) = 6x + 9$ ✓

Example 31: Factorize: $8a - 12$

Solution:

HCF = 4

$8a - 12 = 4(2a - 3)$

Example 32: Factorize: $15x + 20y$

Solution:

HCF = 5

$15x + 20y = 5(3x + 4y)$

Example 33: Factorize: $4x^2 + 6x$

Solution:

HCF = $2x$

$4x^2 + 6x = 2x(2x + 3)$

Check: $2x(2x + 3) = 4x^2 + 6x$ ✓

Example 34: Factorize: $12a^2 - 18ab$

Solution:

HCF = $6a$

$12a^2 - 18ab = 6a(2a - 3b)$

Example 35: Factorize: $10x^3 + 15x^2 - 5x$

Solution:

HCF = $5x$

$10x^3 + 15x^2 - 5x = 5x(2x^2 + 3x - 1)$

Example 36: Factorize: $9xy^2 - 12x^2y + 6xy$

Solution:

HCF = $3xy$

$9xy^2 - 12x^2y + 6xy = 3xy(3y - 4x + 2)$

Example 37: Factorize: $-8a - 12b$

Solution:

HCF = -4 (can take negative factor)

-8a - 12b = -4(2a + 3b)

OR: 4(-2a - 3b)

K. FACTORIZATION BY GROUPING

Method: Group terms with common factors, then factorize each group.

Example 38: Factorize: ax + ay + bx + by

Solution:

Group: (ax + ay) + (bx + by)

Factor each: a(x + y) + b(x + y)

Common factor: (x + y)(a + b)

Example 39: Factorize: 2x + 2y + ax + ay

Solution:

= (2x + 2y) + (ax + ay)

= 2(x + y) + a(x + y)

= (x + y)(2 + a)

Example 40: Factorize: 3a - 3b + xa - xb

Solution:

= (3a - 3b) + (xa - xb)

= 3(a - b) + x(a - b)

= (a - b)(3 + x)

Example 41: Factorize: xy + 3x + 2y + 6

Solution:

= (xy + 3x) + (2y + 6)

$= x(y + 3) + 2(y + 3)$

$= (y + 3)(x + 2)$

Example 42: Factorize: $pq - 4p + 5q - 20$

Solution:

$= (pq - 4p) + (5q - 20)$

$= p(q - 4) + 5(q - 4)$

$= (q - 4)(p + 5)$

L. APPLYING FACTORIZATION

Example 43: Factorize completely: $6x^2 + 9x$

Solution:

HCF = $3x$

$6x^2 + 9x = 3x(2x + 3)$

Example 44: Factorize: $4ab + 6ac - 10ad$

Solution:

HCF = $2a$

$4ab + 6ac - 10ad = 2a(2b + 3c - 5d)$

Example 45: Factorize: $x(a + b) + y(a + b)$

Solution:

Common factor: $(a + b)$

$x(a + b) + y(a + b) = (a + b)(x + y)$

EVALUATION:

1. Define: a) Variable b) Constant c) Coefficient
2. Identify terms in: $5x^2 - 3xy + 7y^2 - 4x + 9$

3. Simplify: $7a + 4b - 3a + 9b - 5a$
4. Expand: $4(3x - 5) - 2(x + 7)$
5. If $x = -2$, find $3x^2 - 5x + 8$
6. Factorize: a) $12x + 18$ b) $8a^2 - 12ab$
7. Simplify: $5x - (3x - 2y) + 4y$
8. Multiply: $3x^2 \times 4x^3$
9. Factorize by grouping: $ax + ay + 2x + 2y$
10. Expand and simplify: $3(2a - 5) + 2(3a + 4)$

REVISION QUESTIONS:

1. What is the difference between an expression and an equation?
2. What are like terms? Give examples
3. Why can't we add $3x$ and $4y$?
4. When removing a negative bracket, what happens to signs inside?
5. What's the coefficient in: a) $-5x$ b) y c) $-z$
6. Simplify: $8x - 3y + 2x - 7y + 5x - y$
7. What is meant by factorization?
8. How do you check if factorization is correct?
9. Expand: $a(b + c) - a(b - c)$
10. If $a = 3$, find: $2a^2 - a + 5$
11. Factorize: $15xy - 10xz$
12. What's the HCF of $12x^2$ and $18x$?
13. Simplify: $3(x + 2) - 2(x - 4) + 5(2x - 1)$
14. Multiply: $(-3a^2) \times 5a^3$
15. Factorize: $ab + 3a + 2b + 6$

ASSIGNMENT:

Section A: Basic Concepts (15 marks)

1. Define algebraic expression with 3 examples
2. Identify:
    - a) Variables in: $3x^2 - 5xy + 7z$
    - b) Constants in: $4a - 7b + 12$
    - c) Coefficients in: $8x, -3y, z, -w$
3. State whether like or unlike terms:
    - a) $5x$ and $3x$
    - b) $4y$ and $4z$
    - c) $2ab$ and $-3ab$
    - d) $3x^2$ and $2x$
4. Identify all terms in:
    - a) $7x - 3y + 5$
    - b) $4a^2 - 6ab + 2b^2 - 8$

Section B: Simplification (20 marks)

1. Collect like terms:
    - a) $5x + 3x$
    - b) $8y - 5y$
    - c) $7a - 9a$
    - d) $-4b - 6b$
2. Simplify:
    - a) $4x + 7y - 2x + 5y$
    - b) $9a - 3b + 2a - 8b + 5a$
    - c) $6x + 4 - 3x + 8 - 2x - 5$
    - d) $5x^2 + 3x - 2x^2 + 7x - x^2 + 5$
3. Remove brackets and simplify:

- a) $3x + (5x - 2)$
- b) $7a - (3a + 4)$
- c) $5y - (2y - 7) + 3y$
- d) $8x - (4x - 3) - (2x + 5)$

4. Simplify:
   - a) $2(x + 3) + 5(x - 1)$
   - b) $4(2a - 3) - 3(a + 2)$
   - c) $5(3y + 2) - 2(4y - 5)$
   - d) $3(x - 4) + 2(3x + 5) - 4(x - 1)$

Section C: Multiplication (15 marks)

1. Multiply:
   - a) $5 \times 3x$
   - b) $4a \times 7$
   - c) $3x \times 2y$
   - d) $5a \times 4b$
   - e) $(-3x) \times 4y$
2. Multiply (with powers):
   - a) $x \times x$
   - b) $x^2 \times x^3$
   - c) $3x \times 4x^2$
   - d) $2a^2 \times 5a^3$
   - e) $4y^3 \times 3y^2$
3. Expand:
   - a) $3(x + 5)$
   - b) $5(2a - 3)$
   - c) $-2(3y + 7)$
   - d) $x(x + 4)$

- e) $2a(3a - 5)$

4. Expand and simplify:

    - a) $2(x + 5) + 3(x - 2)$
    - b) $4(2a - 1) - 2(a + 3)$
    - c) $5(y + 3) - 3(2y - 4)$

Section D: Substitution (15 marks)

1. If $x = 4$, find:

    - a) $3x + 7$
    - b) $2x - 5$
    - c) $5x + 12$

2. If $a = -3$, find:

    - a) $2a + 8$
    - b) $4a - 10$
    - c) $a^2 + 5a + 6$

3. If $x = 2$ and $y = 5$, find:

    - a) $3x + 2y$
    - b) $4x - y + 8$
    - c) $xy + 2x - 3y$

4. If $a = 4$ and $b = -2$, find:

    - a) $2a + 3b$
    - b) $a^2 - b^2$
    - c) $3a^2 + 2ab - b^2$

5. If $p = -1$ and $q = 3$, find:

    - a) $p^2 + q^2$
    - b) $2p^3 - 3q$
    - c) $p^2 + 2pq - q^2$

Section E: Factorization (20 marks)

1. Find HCF:
   - a) 12 and 18
   - b) 15x and 20y
   - c) $6a^2$ and 9a
   - d) 8xy and $12x^2$
2. Factorize (common factor):
   - a) 8x + 12
   - b) 15a - 20
   - c) 9y + 15
   - d) 6p - 18q
3. Factorize:
   - a) $3x^2 + 6x$
   - b) $8a^2 - 12a$
   - c) $10y^2 + 15y$
   - d) $4x^2 - 6x$
4. Factorize:
   - a) 6ab + 9ac
   - b) 12xy - 18xz
   - c) 15pq + 20pr
   - d) $8a^2b - 12ab^2$
5. Factorize by grouping:
   - a) ax + ay + bx + by
   - b) 2m + 2n + am + an
   - c) xy + 2x + 3y + 6
   - d) pq - 3p + 4q - 12

Section F: Mixed Problems (15 marks)

1. Simplify: $3x - [2x - (x + 5)]$
2. Expand: $(x + 3)(x + 2)$ [Hint: $x(x + 2) + 3(x + 2)$]
3. Factorize completely: $12x^2 + 18x - 6$
4. If $x = -2$, show that $x^2 - 3x + 2 = 12$
5. Simplify then substitute $x = 3$: $5(x + 2) - 2(x - 4)$

WEEK 11: ALGEBRAIC EXPRESSIONS II

TOPIC: Expansion and Factorization of Quadratic Expressions; Algebraic Fractions

CONTENT:

A. REVISION OF EXPANSION

Distributive Property Review: $a(b + c) = ab + ac$

Example 1: Quick review:

a) $3(x + 4) = 3x + 12$

b) $5(2a - 3) = 10a - 15$

c) $x(x + 5) = x^2 + 5x$

d) $2y(3y - 4) = 6y^2 - 8y$

B. EXPANDING TWO BRACKETS

Method: Each term in the first bracket multiplies each term in the second bracket.

FOIL Method:

- **F**irst terms
- **O**uter terms
- **I**nner terms
- **L**ast terms

Example 2: Expand $(x + 3)(x + 2)$

Solution:

F: $x \times x = x^2$

O: $x \times 2 = 2x$

I: $3 \times x = 3x$

L: $3 \times 2 = 6$

$(x + 3)(x + 2) = x^2 + 2x + 3x + 6$

$= x^2 + 5x + 6$

Example 3: Expand $(x + 5)(x + 4)$

Solution:

$= x^2 + 4x + 5x + 20$

$= x^2 + 9x + 20$

Example 4: Expand $(x + 7)(x - 3)$

Solution:

$= x^2 - 3x + 7x - 21$

$= x^2 + 4x - 21$

Example 5: Expand $(x - 5)(x + 2)$

Solution:

$= x^2 + 2x - 5x - 10$

$= x^2 - 3x - 10$

Example 6: Expand $(x - 4)(x - 6)$

Solution:

$= x^2 - 6x - 4x + 24$

$= x^2 - 10x + 24$

Example 7: Expand $(2x + 3)(x + 4)$

Solution:

$= 2x^2 + 8x + 3x + 12$

$= 2x^2 + 11x + 12$

Example 8: Expand $(3x - 2)(2x + 5)$

Solution:

$= 6x^2 + 15x - 4x - 10$

$= 6x^2 + 11x - 10$

Example 9: Expand $(2x - 3)(3x - 4)$

Solution:

$= 6x^2 - 8x - 9x + 12$

$= 6x^2 - 17x + 12$

Example 10: Expand $(5x + 2)(x - 3)$

Solution:

$= 5x^2 - 15x + 2x - 6$

$= 5x^2 - 13x - 6$

C. SPECIAL PRODUCTS

1. Perfect Square: $(a + b)^2 = a^2 + 2ab + b^2$

Example 11:

$(x + 3)^2 = x^2 + 2(x)(3) + 3^2$

$= x^2 + 6x + 9$

Example 12:

$(x + 5)^2 = x^2 + 2(x)(5) + 25$

$= x^2 + 10x + 25$

Example 13:

$(2x + 3)^2 = (2x)^2 + 2(2x)(3) + 3^2$

$= 4x^2 + 12x + 9$

---

2. Perfect Square: $(a - b)^2 = a^2 - 2ab + b^2$

Example 14:

$(x - 4)^2 = x^2 - 2(x)(4) + 4^2$
$= x^2 - 8x + 16$

Example 15:

$(x - 7)^2 = x^2 - 2(x)(7) + 49$
$= x^2 - 14x + 49$

Example 16:

$(3x - 2)^2 = (3x)^2 - 2(3x)(2) + 2^2$
$= 9x^2 - 12x + 4$

---

3. Difference of Two Squares: $(a + b)(a - b) = a^2 - b^2$

Example 17:

$(x + 5)(x - 5) = x^2 - 5^2$
$= x^2 - 25$

Example 18:

$(x + 7)(x - 7) = x^2 - 49$

Example 19:

$(2x + 3)(2x - 3) = (2x)^2 - 3^2$
$= 4x^2 - 9$

Example 20:

$(5x + 4)(5x - 4) = 25x^2 - 16$

D. QUADRATIC EXPRESSIONS

Definition: A quadratic expression is an algebraic expression of the form $ax^2 + bx + c$, where $a \neq 0$.

General Form: $ax^2 + bx + c$

Where:

- **a** = coefficient of x^2 (leading coefficient)
- **b** = coefficient of x
- **c** = constant term

Examples:

a) $x^2 + 5x + 6$ (a=1, b=5, c=6)
b) $2x^2 - 7x + 3$ (a=2, b=-7, c=3)
c) $x^2 - 9$ (a=1, b=0, c=-9)
d) $3x^2 + 4x$ (a=3, b=4, c=0)

E. FACTORIZING QUADRATIC EXPRESSIONS (SIMPLE CASES)

Type 1: $x^2 + bx + c$ (where a = 1)

Method: Find two numbers that:

- **Multiply** to give c
- **Add** to give b

Example 21: Factorize $x^2 + 7x + 12$

Solution:
Need two numbers that multiply to 12 and add to 7
Factors of 12: 1×12, 2×6, 3×4
3 and 4: $3\times4=12$, $3+4=7$ ✓

$x^2 + 7x + 12 = (x + 3)(x + 4)$

Check: $(x + 3)(x + 4) = x^2 + 4x + 3x + 12 = x^2 + 7x + 12$ ✓

Example 22: Factorize $x^2 + 9x + 20$

Solution:

Factors of 20: 1×20, 2×10, 4×5

4 and 5: 4×5=20, 4+5=9 ✓

$x^2 + 9x + 20 = (x + 4)(x + 5)$

Example 23: Factorize $x^2 + 8x + 15$

Solution:

Factors of 15: 1×15, 3×5

3 and 5: 3×5=15, 3+5=8 ✓

$x^2 + 8x + 15 = (x + 3)(x + 5)$

Example 24: Factorize $x^2 + 11x + 24$

Solution:

Factors of 24: 1×24, 2×12, 3×8, 4×6

3 and 8: 3×8=24, 3+8=11 ✓

$x^2 + 11x + 24 = (x + 3)(x + 8)$

Type 2: $x^2 - bx + c$ (middle term negative)

Example 25: Factorize $x^2 - 8x + 15$

Solution:

Need two negative numbers that multiply to +15 and add to -8

-3 and -5: $(-3)\times(-5)=15$, $(-3)+(-5)=-8$ ✓

$x^2 - 8x + 15 = (x - 3)(x - 5)$

Example 26: Factorize $x^2 - 10x + 21$

Solution:

-3 and -7: $(-3)\times(-7)=21$, $(-3)+(-7)=-10$ ✓

$x^2 - 10x + 21 = (x - 3)(x - 7)$

Example 27: Factorize $x^2 - 9x + 20$

Solution:

-4 and -5: $(-4)\times(-5)=20$, $(-4)+(-5)=-9$ ✓

$x^2 - 9x + 20 = (x - 4)(x - 5)$

Type 3: $x^2 + bx - c$ (constant term negative)

Example 28: Factorize $x^2 + 3x - 10$

Solution:

Need numbers that multiply to -10 and add to +3

5 and -2: $5\times(-2)=-10$, $5+(-2)=3$ ✓

$x^2 + 3x - 10 = (x + 5)(x - 2)$

Example 29: Factorize $x^2 + 2x - 15$

Solution:

5 and -3: $5\times(-3)=-15$, $5+(-3)=2$ ✓

$x^2 + 2x - 15 = (x + 5)(x - 3)$

Example 30: Factorize $x^2 + 5x - 24$

Solution:

8 and -3: $8\times(-3)=-24$, $8+(-3)=5$ ✓

$x^2 + 5x - 24 = (x + 8)(x - 3)$

Type 4: $x^2 - bx - c$ (both middle and constant negative)

Example 31: Factorize $x^2 - 2x - 15$

Solution:

-5 and 3: $(-5)\times3=-15$, $(-5)+3=-2$ ✓

$x^2 - 2x - 15 = (x - 5)(x + 3)$

Example 32: Factorize $x^2 - 4x - 21$

Solution:

-7 and 3: $(-7)\times3=-21$, $(-7)+3=-4$ ✓

$x^2 - 4x - 21 = (x - 7)(x + 3)$

Example 33: Factorize $x^2 - 6x - 16$

Solution:

-8 and 2: $(-8)\times2=-16$, $(-8)+2=-6$ ✓

$x^2 - 6x - 16 = (x - 8)(x + 2)$

Type 5: Difference of Two Squares: $a^2 - b^2$

Formula: $a^2 - b^2 = (a + b)(a - b)$

Example 34: Factorize $x^2 - 25$

Solution:

$x^2 - 25 = x^2 - 5^2$

$= (x + 5)(x - 5)$

Example 35: Factorize $x^2 - 49$

Solution:

$= x^2 - 7^2$

$= (x + 7)(x - 7)$

Example 36: Factorize $4x^2 - 9$

Solution:

$= (2x)^2 - 3^2$

$= (2x + 3)(2x - 3)$

Example 37: Factorize $9x^2 - 16$

Solution:

$= (3x)^2 - 4^2$

$= (3x + 4)(3x - 4)$

Type 6: Perfect Square Trinomials

Pattern 1: $a^2 + 2ab + b^2 = (a + b)^2$ **Pattern 2:** $a^2 - 2ab + b^2 = (a - b)^2$

Example 38: Factorize $x^2 + 6x + 9$

Solution:

Check: Is it a perfect square?

First term: $x^2 \rightarrow a = x$

Last term: $9 = 3^2 \rightarrow b = 3$

Middle term: $6x = 2(x)(3)$ ✓

$x^2 + 6x + 9 = (x + 3)^2$

Example 39: Factorize $x^2 - 10x + 25$

Solution:

$x^2 \rightarrow a = x$

$25 = 5^2 \rightarrow b = 5$

$10x = 2(x)(5)$ ✓

$x^2 - 10x + 25 = (x - 5)^2$

Example 40: Factorize $4x^2 + 12x + 9$

Solution:

$4x^2 = (2x)^2 \rightarrow a = 2x$

$9 = 3^2 \rightarrow b = 3$

$12x = 2(2x)(3)$ ✓

$4x^2 + 12x + 9 = (2x + 3)^2$

F. ALGEBRAIC FRACTIONS - INTRODUCTION

Definition: An algebraic fraction is a fraction where the numerator and/or denominator contains algebraic expressions.

Examples:

a) $x/5$

b) $(3x + 2)/7$

c) $(x^2 - 4)/(x + 2)$

d) $(2a + 3)/(5a - 1)$

G. SIMPLIFYING ALGEBRAIC FRACTIONS

Method: Cancel common factors from numerator and denominator.

Example 41: Simplify: $(6x)/(9)$

Solution:

HCF = 3

$(6x)/(9) = (3 \times 2x)/(3 \times 3)$

$= 2x/3$

Example 42: Simplify: $(12a)/(18)$

Solution:

HCF = 6

$(12a)/(18) = 2a/3$

Example 43: Simplify: $(4x^2)/(8x)$

Solution:

HCF = 4x

$(4x^2)/(8x) = (4x \times x)/(4x \times 2)$

$= x/2$

Example 44: Simplify: $(15x^2y)/(25xy^2)$

Solution:

HCF = 5xy

$(15x^2y)/(25xy^2) = (5xy \times 3x)/(5xy \times 5y)$

$= 3x/5y$

Example 45: Simplify: $(x + 2)/(2x + 4)$

Solution:

Factorize denominator:

$2x + 4 = 2(x + 2)$

$(x + 2)/(2x + 4) = (x + 2)/(2(x + 2))$

$= 1/2$

Example 46: Simplify: $(x^2 - 9)/(x + 3)$

Solution:

Factorize numerator:

$x^2 - 9 = (x + 3)(x - 3)$

$(x^2 - 9)/(x + 3) = (x + 3)(x - 3)/(x + 3)$

$= x - 3$

H. ADDITION AND SUBTRACTION OF ALGEBRAIC FRACTIONS

With Same Denominator:

Example 47: Simplify: $(2x)/5 + (3x)/5$

Solution:

$= (2x + 3x)/5$

$= 5x/5$

$= x$

Example 48: Simplify: $(4a)/7 - (a)/7$

Solution:

$= (4a - a)/7$

$= 3a/7$

Example 49: Simplify: $(x + 2)/4 + (x - 3)/4$

Solution:

$= (x + 2 + x - 3)/4$

$= (2x - 1)/4$

With Different Denominators:

Method: Find LCM of denominators, convert to equivalent fractions, then add/subtract.

Example 50: Simplify: $x/2 + x/3$

Solution:

LCM of 2 and 3 = 6

$x/2 = 3x/6$

$x/3 = 2x/6$

$x/2 + x/3 = 3x/6 + 2x/6$

$= 5x/6$

Example 51: Simplify: $(2a)/3 + a/4$

Solution:

LCM = 12

$(2a)/3 = 8a/12$

$a/4 = 3a/12$

$(2a)/3 + a/4 = 8a/12 + 3a/12$

$= 11a/12$

Example 52: Simplify: $(3x)/4 - x/6$

Solution:

LCM = 12

$(3x)/4 = 9x/12$

$x/6 = 2x/12$

$(3x)/4 - x/6 = 9x/12 - 2x/12$

$= 7x/12$

Example 53: Simplify: $(x + 1)/2 - (x - 3)/4$

Solution:

LCM = 4

$(x + 1)/2 = (2(x + 1))/4 = (2x + 2)/4$

$(x + 1)/2 - (x - 3)/4 = (2x + 2)/4 - (x - 3)/4$

$= (2x + 2 - x + 3)/4$

$= (x + 5)/4$

EVALUATION:

1. Expand: a) $(x + 4)(x + 5)$ b) $(x - 3)(x + 7)$
2. Expand: $(x + 3)^2$
3. Expand: $(2x - 1)(x + 4)$
4. Factorize: $x^2 + 9x + 20$
5. Factorize: $x^2 - 7x + 12$
6. Factorize: $x^2 + 4x - 12$
7. Factorize: $x^2 - 36$
8. Simplify: $(8x)/(12)$
9. Simplify: $(x + 3)/(2x + 6)$
10. Add: $(2x)/5 + x/5$

REVISION QUESTIONS:

1. What is a quadratic expression?
2. State the FOIL method for expanding brackets

3. What is $a^2 - b^2$ equal to?
4. How do you check if factorization is correct?
5. Expand and simplify: $(x + 5)(x - 5)$
6. What pattern does $x^2 + 8x + 16$ follow?
7. Factorize: $x^2 + 11x + 30$
8. What's the first step in simplifying algebraic fractions?
9. How do you add fractions with different denominators?
10. Simplify: $(x^2 - 4)/(x + 2)$
11. Expand: $(3x - 2)^2$
12. Factorize: $9x^2 - 25$
13. Add: $x/3 + x/4$
14. Factorize: $x^2 - 5x - 14$
15. Simplify: $(6x^2)/(9x)$

ASSIGNMENT:

Section A: Expansion (25 marks)

1. Expand:
   - a) $(x + 2)(x + 3)$
   - b) $(x + 5)(x + 7)$
   - c) $(x + 6)(x + 4)$
   - d) $(x + 1)(x + 9)$
2. Expand:
   - a) $(x - 3)(x + 5)$
   - b) $(x + 4)(x - 2)$
   - c) $(x - 6)(x + 8)$
   - d) $(x + 7)(x - 1)$
3. Expand:

- a) $(x - 2)(x - 5)$
- b) $(x - 4)(x - 3)$
- c) $(x - 7)(x - 2)$
- d) $(x - 6)(x - 1)$

4. Expand:

   - a) $(2x + 3)(x + 4)$
   - b) $(3x - 2)(x + 5)$
   - c) $(2x - 1)(3x + 2)$
   - d) $(4x + 3)(2x - 5)$

5. Expand (perfect squares):

   - a) $(x + 4)^2$
   - b) $(x - 5)^2$
   - c) $(2x + 3)^2$
   - d) $(3x - 2)^2$

6. Expand (difference of squares):

   - a) $(x + 6)(x - 6)$
   - b) $(x + 8)(x - 8)$
   - c) $(3x + 4)(3x - 4)$
   - d) $(5x + 2)(5x - 2)$

Section B: Factorization (30 marks)

1. Factorize ($x^2 + bx + c$, both positive):

   - a) $x^2 + 8x + 12$
   - b) $x^2 + 10x + 21$
   - c) $x^2 + 7x + 10$
   - d) $x^2 + 13x + 36$

2. Factorize ($x^2 - bx + c$):

- a) $x^2 - 9x + 18$
- b) $x^2 - 11x + 24$
- c) $x^2 - 8x + 15$
- d) $x^2 - 13x + 42$

3. Factorize ($x^2 + bx - c$):
   - a) $x^2 + 4x - 12$
   - b) $x^2 + 7x - 18$
   - c) $x^2 + 2x - 24$
   - d) $x^2 + 6x - 27$
4. Factorize ($x^2 - bx - c$):
   - a) $x^2 - 3x - 18$
   - b) $x^2 - 5x - 24$
   - c) $x^2 - 7x - 30$
   - d) $x^2 - 2x - 35$
5. Factorize (difference of squares):
   - a) $x^2 - 16$
   - b) $x^2 - 64$
   - c) $4x^2 - 25$
   - d) $9x^2 - 49$
   - e) $16x^2 - 1$
6. Factorize (perfect squares):
   - a) $x^2 + 10x + 25$
   - b) $x^2 - 12x + 36$
   - c) $x^2 + 14x + 49$
   - d) $4x^2 + 12x + 9$
   - e) $9x^2 - 30x + 25$

Section C: Algebraic Fractions - Simplification (20 marks)

1. Simplify:
    - a) $(8x)/12$
    - b) $(15a)/20$
    - c) $(18y)/24$
    - d) $(21x)/35$
2. Simplify:
    - a) $(6x^2)/(9x)$
    - b) $(10a^3)/(15a^2)$
    - c) $(12y^2)/(18y)$
    - d) $(20x^3)/(25x^2)$
3. Simplify:
    - a) $(12xy)/(18x)$
    - b) $(15a^2b)/(20ab)$
    - c) $(24x^2y^2)/(36xy^3)$
    - d) $(30a^3b^2)/(45a^2b^3)$
4. Simplify by factoring:
    - a) $(x + 3)/(3x + 9)$
    - b) $(2x + 6)/(x + 3)$
    - c) $(x^2 - 4)/(x + 2)$
    - d) $(x^2 - 9)/(x - 3)$

Section D: Addition and Subtraction of Algebraic Fractions (15 marks)

1. Same denominator:
    - a) $(3x)/7 + (2x)/7$
    - b) $(5a)/9 - (2a)/9$
    - c) $(x + 2)/5 + (x - 1)/5$
    - d) $(2y + 3)/6 - (y - 2)/6$

2. Different denominators:

    - a) $x/2 + x/4$
    - b) $(2a)/3 + a/6$
    - c) $(3y)/4 - y/8$
    - d) $(2x)/5 + (3x)/10$

3. More complex:

    - a) $(x + 1)/3 + (x - 2)/6$
    - b) $(2a - 1)/4 - (a + 3)/8$
    - c) $(3y + 2)/5 + (y - 1)/10$
    - d) $(x + 4)/6 - (x - 5)/9$

Section E: Mixed Problems (10 marks)

4. Expand then factorize your answer: $(x + 5)(x - 5)$
5. Factorize then expand to check: $x^2 + 9x + 18$
6. Simplify: $(x^2 - 16)/(x - 4)$, then find value when $x = 5$
7. Add then simplify: $x/2 + x/3 + x/4$
8. If $x = 3$, find value of $(x^2 - 9)/(x + 3)$

JSS2 MATHEMATICS COMPREHENSIVE LESSON NOTES

SECOND TERM

WEEK 1: REVIEW OF FIRST TERM'S WORK

TOPIC: Emphasis on Algebraic Expressions, Solving Quadratic Equations, and Transactions at Home and Offices

CONTENT:

A. REVIEW OF ALGEBRAIC EXPRESSIONS

Key Concepts:

1. Terms and Coefficients

- Term: A single number, variable, or product
- Coefficient: The numerical part of a variable term
- Like terms: Terms with same variables and powers
- Unlike terms: Terms with different variables or powers

Example 1: Identify in $5x^2 - 3xy + 7y^2 - 4x + 9$

Terms: $5x^2$, $-3xy$, $7y^2$, $-4x$, 9
Coefficients: 5 (of x^2), -3 (of xy), 7 (of y^2), -4 (of x)
Constant: 9

2. Simplification by Collecting Like Terms

Example 2: Simplify: $7a + 4b - 3a + 9b - 5a$

Solution:
a terms: $7a - 3a - 5a = -a$
b terms: $4b + 9b = 13b$
Answer: $-a + 13b$ or $13b - a$

Example 3: Simplify: $4x^2 + 5x - 2x^2 + 7x + 3 - 8$

Solution:

x^2 terms: $4x^2 - 2x^2 = 2x^2$

x terms: $5x + 7x = 12x$

Constants: $3 - 8 = -5$

Answer: $2x^2 + 12x - 5$

3. Removing Brackets

Example 4: Simplify: $5x - (3x - 2y) + 4y$

Solution:

$= 5x - 3x + 2y + 4y$

$= 2x + 6y$

Example 5: Expand: $3(2a - 5) + 2(3a + 4)$

Solution:

$= 6a - 15 + 6a + 8$

$= 12a - 7$

4. Multiplication

Example 6: Multiply:

a) $3x \times 4x^2 = 12x^3$

b) $5a^2 \times 2a^3 = 10a^5$

c) $(-3y^2) \times 4y = -12y^3$

5. Substitution

Example 7: If $x = 3$ and $y = -2$, find: $2x^2 - 3xy + y^2$

Solution:

$= 2(3)^2 - 3(3)(-2) + (-2)^2$

$= 2(9) - 3(-6) + 4$

$= 18 + 18 + 4$

$= 40$

B. EXPANSION OF BRACKETS

1. Single Bracket

Example 8: Expand: $5(3x - 7)$

$= 15x - 35$

Example 9: Expand: $-3(2a + 5b - 4)$

$= -6a - 15b + 12$

2. Two Brackets (FOIL Method)

Example 10: Expand: $(x + 5)(x + 3)$

F: $x \times x = x^2$
O: $x \times 3 = 3x$
I: $5 \times x = 5x$
L: $5 \times 3 = 15$

Answer: $x^2 + 8x + 15$

Example 11: Expand: $(2x - 3)(x + 4)$

$= 2x^2 + 8x - 3x - 12$
$= 2x^2 + 5x - 12$

Example 12: Expand: $(3a - 2)(2a - 5)$

$= 6a^2 - 15a - 4a + 10$
$= 6a^2 - 19a + 10$

3. Perfect Squares

Example 13: Expand:

a) $(x + 4)^2 = x^2 + 8x + 16$

b) $(x - 5)^2 = x^2 - 10x + 25$

c) $(2x + 3)^2 = 4x^2 + 12x + 9$

4. Difference of Two Squares

Example 14: Expand:

a) $(x + 7)(x - 7) = x^2 - 49$

b) $(3x + 5)(3x - 5) = 9x^2 - 25$

c) $(2a + 7)(2a - 7) = 4a^2 - 49$

C. FACTORIZATION

1. Common Factor

Example 15: Factorize:

a) $12x + 18 = 6(2x + 3)$

b) $8a^2 - 12ab = 4a(2a - 3b)$

c) $15x^2y - 10xy^2 = 5xy(3x - 2y)$

2. Grouping

Example 16: Factorize: $ax + ay + bx + by$

$= a(x + y) + b(x + y)$

$= (x + y)(a + b)$

Example 17: Factorize: $3m + 3n + am + an$

$= 3(m + n) + a(m + n)$

$= (m + n)(3 + a)$

3. Quadratic Expressions

Example 18: Factorize: $x^2 + 9x + 20$

Need: Two numbers that multiply to 20 and add to 9

4 and 5: $4 \times 5 = 20$, $4 + 5 = 9$ ✓

Answer: $(x + 4)(x + 5)$

Example 19: Factorize: $x^2 - 7x + 12$

Need: Two negative numbers that multiply to 12 and add to -7

-3 and -4: $(-3) \times (-4) = 12$, $(-3) + (-4) = -7$ ✓

Answer: $(x - 3)(x - 4)$

Example 20: Factorize: $x^2 + 3x - 18$

Need: Numbers that multiply to -18 and add to 3

6 and -3: $6 \times (-3) = -18$, $6 + (-3) = 3$ ✓

Answer: $(x + 6)(x - 3)$

Example 21: Factorize: $x^2 - 5x - 24$

-8 and 3: $(-8) \times 3 = -24$, $(-8) + 3 = -5$ ✓

Answer: $(x - 8)(x + 3)$

4. Difference of Two Squares

Example 22: Factorize:

a) $x^2 - 36 = (x + 6)(x - 6)$

b) $4x^2 - 25 = (2x + 5)(2x - 5)$

c) $9a^2 - 16 = (3a + 4)(3a - 4)$

5. Perfect Square Trinomials

Example 23: Factorize:

a) $x^2 + 10x + 25 = (x + 5)^2$

b) $x^2 - 14x + 49 = (x - 7)^2$

c) $4x^2 + 12x + 9 = (2x + 3)^2$

D. SOLVING QUADRATIC EQUATIONS BY FACTORIZATION

Quadratic Equation: An equation of the form $ax^2 + bx + c = 0$

Method:

11. Arrange in standard form ($ax^2 + bx + c = 0$)
12. Factorize the left side
13. Set each factor equal to zero
14. Solve for x

Example 24: Solve: $x^2 + 5x + 6 = 0$

Solution:

Step 1: Factorize

$x^2 + 5x + 6 = (x + 2)(x + 3)$

Step 2: Set equal to zero

$(x + 2)(x + 3) = 0$

Step 3: Apply zero product property

Either $(x + 2) = 0$ or $(x + 3) = 0$

Step 4: Solve

$x + 2 = 0 \rightarrow x = -2$

$x + 3 = 0 \rightarrow x = -3$

Answer: $x = -2$ or $x = -3$

Example 25: Solve: $x^2 - 7x + 12 = 0$

Solution:

Factorize: $(x - 3)(x - 4) = 0$

$x - 3 = 0 \rightarrow x = 3$

$x - 4 = 0 \rightarrow x = 4$

Answer: $x = 3$ or $x = 4$

Example 26: Solve: $x^2 + 2x - 15 = 0$

Solution:

Factorize: $(x + 5)(x - 3) = 0$

$x + 5 = 0 \rightarrow x = -5$

$x - 3 = 0 \rightarrow x = 3$

Answer: $x = -5$ or $x = 3$

Example 27: Solve: $x^2 - 4x - 21 = 0$

Solution:

Factorize: $(x - 7)(x + 3) = 0$

$x - 7 = 0 \rightarrow x = 7$

$x + 3 = 0 \rightarrow x = -3$

Answer: $x = 7$ or $x = -3$

Example 28: Solve: $x^2 - 9 = 0$

Solution:

Factorize: $(x + 3)(x - 3) = 0$

$x + 3 = 0 \rightarrow x = -3$

$x - 3 = 0 \rightarrow x = 3$

Answer: $x = -3$ or $x = 3$

Example 29: Solve: $2x^2 - 8x = 0$

Solution:

Factorize: $2x(x - 4) = 0$

$2x = 0 \rightarrow x = 0$

$x - 4 = 0 \rightarrow x = 4$

Answer: $x = 0$ or $x = 4$

Example 30: Solve: $x^2 = 5x + 6$

Solution:

Rearrange: $x^2 - 5x - 6 = 0$

Factorize: $(x - 6)(x + 1) = 0$

$x - 6 = 0 \rightarrow x = 6$

$x + 1 = 0 \rightarrow x = -1$

Answer: $x = 6$ or $x = -1$

E. ALGEBRAIC FRACTIONS

1. Simplification

Example 31: Simplify:

a) $(8x)/12 = 2x/3$

b) $(6a^2)/(9a) = 2a/3$

c) $(x + 3)/(2x + 6) = (x + 3)/(2(x + 3)) = 1/2$

d) $(x^2 - 9)/(x + 3) = (x + 3)(x - 3)/(x + 3) = x - 3$

2. Addition and Subtraction

Example 32: Simplify: $(3x)/5 + x/5$

$= (3x + x)/5 = 4x/5$

Example 33: Simplify: $x/2 + x/3$

LCM = 6

$= 3x/6 + 2x/6$

$= 5x/6$

Example 34: Simplify: (2a)/3 - a/4

LCM = 12

= 8a/12 - 3a/12

= 5a/12

F. TRANSACTIONS AT HOME AND OFFICES

1. Percentage Applications

Example 35: A dress costs ₦6,500. After 20% discount, find: a) Discount amount b) Selling price

Solution:

a) Discount = 20% of 6,500

= 0.20 × 6,500

= ₦1,300

b) Selling price = 6,500 - 1,300

= ₦5,200

OR: 80% of 6,500 = ₦5,200

Example 36: A trader buys goods for ₦12,000 and sells at 30% profit. Find: a) Profit b) Selling price

Solution:

a) Profit = 30% of 12,000

= ₦3,600

b) SP = CP + Profit

= 12,000 + 3,600

= ₦15,600

OR: 130% of 12,000 = ₦15,600

2. Simple Interest

Formula: I = (PRT)/100

Example 37: Find simple interest on ₦20,000 at 6% per annum for 3 years.

Solution:

I = (20,000 × 6 × 3)/100

= 360,000/100

= ₦3,600

Amount = Principal + Interest

= 20,000 + 3,600

= ₦23,600

Example 38: A man borrowed ₦50,000 and repaid ₦59,500. If the rate was 7% per annum, for how long did he borrow?

Solution:

Interest = 59,500 - 50,000 = ₦9,500

I = (PRT)/100

9,500 = (50,000 × 7 × T)/100

9,500 = 3,500T

T = 9,500 ÷ 3,500

T = 2.71 years ≈ 2 years 8½ months

3. Sharing in Ratios

Example 39: Share ₦180,000 among three partners in ratio 2:3:4.

Solution:

Total parts = 2 + 3 + 4 = 9

First partner = 2/9 × 180,000 = ₦40,000

Second partner = 3/9 × 180,000 = ₦60,000

Third partner = 4/9 × 180,000 = ₦80,000

Verification: 40,000 + 60,000 + 80,000 = 180,000 ✓

4. Commission and Salary

Example 40: A salesperson earns ₦25,000 basic salary plus 4% commission on sales. If monthly sales are ₦350,000, find total earnings.

Solution:
Commission = 4% of 350,000
= 0.04 × 350,000
= ₦14,000

Total earnings = Basic + Commission
= 25,000 + 14,000
= ₦39,000

5. Value Added Tax (VAT)

Example 41: An item costs ₦8,000 before VAT. If VAT is 7.5%, find total cost.

Solution:
VAT = 7.5% of 8,000
= 0.075 × 8,000
= ₦600

Total cost = 8,000 + 600
= ₦8,600
OR: 107.5% of 8,000 = ₦8,600

6. Hire Purchase

Example 42: A TV costs ₦45,000 cash or ₦15,000 deposit and 6 monthly installments of ₦6,000. Find: a) Hire purchase price b) Extra amount paid

Solution:

a) HP price = Deposit + Total installments

= 15,000 + (6 × 6,000)

= 15,000 + 36,000

= ₦51,000

b) Extra = HP price - Cash price

= 51,000 - 45,000

= ₦6,000

7. Budgeting

Example 43: A family's monthly income is ₦120,000. They budget:

- Food: 35%
- Rent: 30%
- Transport: 15%
- Utilities: 10%
- Savings: 10%

Calculate each amount.

Solution:

Food = 35% of 120,000 = ₦42,000

Rent = 30% of 120,000 = ₦36,000

Transport = 15% of 120,000 = ₦18,000

Utilities = 10% of 120,000 = ₦12,000

Savings = 10% of 120,000 = ₦12,000

Total = 42,000 + 36,000 + 18,000 + 12,000 + 12,000 = ₦120,000 ✓

EVALUATION:

1. Simplify: $8x + 5y - 3x + 9y - 2x$
2. Expand: $3(2a - 5) + 2(a + 4)$
3. Factorize: a) $12x + 18$ b) $x^2 + 8x + 15$
4. Solve: $x^2 - 9x + 20 = 0$
5. If $x = -2$, find: $3x^2 - 4x + 7$
6. Simplify: $(6a^2)/9a$
7. Add: $x/3 + x/4$
8. A dress costs ₦8,000. After 25% discount, what do you pay?
9. Find simple interest on ₦15,000 at 8% for 2 years
10. Share ₦90,000 in ratio 2:3:5

REVISION QUESTIONS:

1. What is the difference between an expression and an equation?
2. When factorizing $x^2 + bx + c$, what do the two numbers multiply and add to?
3. In solving quadratic equations, what property allows us to set each factor to zero?
4. How do you find discount amount?
5. State the formula for simple interest
6. Expand: $(x + 4)(x - 4)$
7. Factorize: $x^2 - 16$
8. Solve: $x^2 + x - 12 = 0$
9. How do you calculate commission?
10. What is hire purchase?
11. Simplify: $4(x + 3) - 2(x - 5)$
12. If cost price is ₦10,000 and profit is 40%, find selling price
13. Add: $2x/5 + x/3$
14. Factorize: $2x^2 - 8x$
15. What does VAT stand for and how is it calculated?

ASSIGNMENT:

Section A: Simplification and Expansion (20 marks)

1. Simplify by collecting like terms:
   - a) $9a - 4b + 3a + 7b - 2a$
   - b) $5x^2 + 3x - 2x^2 + 7x - x^2 + 4$
   - c) $8xy - 3x + 5xy + 2y - 2xy - x$
2. Remove brackets and simplify:
   - a) $7x - (4x - 3y) + 5y$
   - b) $3(2a + 5) - 2(a - 3)$
   - c) $5(x - 2) + 3(2x + 4) - 2(x - 1)$
3. Expand:
   - a) $(x + 6)(x + 4)$
   - b) $(x - 5)(x + 3)$
   - c) $(2x + 3)(x - 4)$
   - d) $(x + 7)^2$
   - e) $(x - 6)(x + 6)$

Section B: Factorization (20 marks)

1. Factorize by common factor:
   - a) $15x + 20$
   - b) $12a^2 - 18a$
   - c) $8xy - 12xz$
2. Factorize by grouping:
   - a) $xy + 3x + 2y + 6$
   - b) $ab - 4a + 3b - 12$

3. Factorize quadratics:

   - a) $x^2 + 10x + 21$
   - b) $x^2 - 11x + 28$
   - c) $x^2 + 6x - 27$
   - d) $x^2 - 5x - 36$
   - e) $x^2 - 64$
   - f) $x^2 + 16x + 64$

Section C: Solving Quadratic Equations (15 marks)

Solve:

1. $x^2 + 7x + 10 = 0$
2. $x^2 - 9x + 18 = 0$
3. $x^2 + 4x - 21 = 0$
4. $x^2 - 6x - 16 = 0$
5. $x^2 - 25 = 0$
6. $x^2 + 8x = 0$
7. $x^2 = 7x + 18$
8. $2x^2 - 10x = 0$
9. $x^2 - 12x + 36 = 0$
10. $x^2 + x = 20$

Section D: Algebraic Fractions (10 marks)

1. Simplify:

   - a) $(10x)/15$
   - b) $(8a^2)/(12a)$
   - c) $(x + 4)/(2x + 8)$

2. Add or subtract:

   - a) $(4x)/7 + (3x)/7$

- b) $x/3 + x/6$
- c) $(3a)/4 - a/8$
- d) $(x + 2)/5 + (x - 1)/5$

Section E: Substitution (10 marks)

3. If $x = 4$, find:
    - a) $5x - 12$
    - b) $2x^2 + 3x - 7$
4. If $a = -3$ and $b = 2$, find:
    - a) $3a + 4b$
    - b) $a^2 - 2ab + b^2$
5. If $p = 5$ and $q = -2$, find:
    - a) $2p^2 - 3pq + q^2$
    - b) $(p + q)(p - q)$

Section F: Transactions (25 marks)

1. A shirt costs ₦5,400 after 25% discount. Find original price.
2. A trader buys goods for ₦18,000 and sells at 35% profit. Find:
    - a) Profit amount
    - b) Selling price
3. Find simple interest on:
    - a) ₦25,000 at 7% for 4 years
    - b) ₦40,000 at 5.5% for 3 years
4. A man borrowed ₦60,000 at 9% per annum. If he repaid ₦73,200, for how long did he borrow?
5. Share ₦240,000 among A, B, C in ratio 3:5:8. Find each share.

6. A salesperson earns ₦30,000 plus 5% commission on sales. If sales are ₦420,000, find total earnings.
7. An item costs ₦12,500 before 7.5% VAT. Find total cost.
8. A laptop costs ₦80,000 cash or ₦25,000 deposit and 8 monthly installments of ₦8,500. Find:
   - a) Hire purchase price
   - b) Extra amount paid
9. A woman's salary is ₦95,000. If 10% is pension and 5% is tax, find:
   - a) Total deductions
   - b) Net salary
10. A family budgets their ₦150,000 income as:
    - Food: 30%
    - Rent: 25%
    - Transport: 20%
    - Others: 15%
    - Savings: 10% Calculate each amount and verify total.

WEEK 2: SIMPLE EQUATIONS

TOPIC: Meaning of Algebraic Equations; Differences Between Expressions and Equations; Solving Simple Algebraic Equations

CONTENT:

A. ALGEBRAIC EQUATIONS

Definition: An algebraic equation is a mathematical statement showing that two algebraic expressions are equal, connected by an equals sign (=).

Parts of an Equation:

- **Left-Hand Side (LHS):** Expression before the = sign
- **Right-Hand Side (RHS):** Expression after the = sign
- **Equals Sign (=):** Shows both sides have the same value

Example 1: In the equation $3x + 5 = 11$

LHS = $3x + 5$
RHS = 11
The equation states that $3x + 5$ equals 11

B. DIFFERENCES BETWEEN EXPRESSIONS AND EQUATIONS

Algebraic Expression	Algebraic Equation
No equals sign (=)	Has equals sign (=)
Cannot be solved	Can be solved
Can only be simplified	Has a solution/answer
Examples: $3x + 5$, $2a - 7$, $x^2 + 3x$	Examples: $3x + 5 = 11$, $2a - 7 = 15$, $x^2 = 9$

Example 2: Identify as expression or equation:

a) $5x + 7$ → Expression (no = sign)
b) $5x + 7 = 22$ → Equation (has = sign)

c) $3a - 2b$ → Expression

d) $3a - 2b = 12$ → Equation

e) $x^2 + 4x - 5$ → Expression

f) $x^2 + 4x = 5$ → Equation

C. MEANING OF SOLVING AN EQUATION

To solve an equation means to find the value of the unknown variable that makes the equation true.

Solution/Root: The value that satisfies the equation.

Example 3: Is $x = 3$ a solution to $2x + 5 = 11$?

Check by substitution:
LHS $= 2(3) + 5 = 6 + 5 = 11$
RHS $= 11$
LHS = RHS ✓

Yes, $x = 3$ is the solution

Example 4: Is $a = 5$ a solution to $3a - 7 = 10$?

Check:
LHS $= 3(5) - 7 = 15 - 7 = 8$
RHS $= 10$
LHS ≠ RHS ✗

No, $a = 5$ is not the solution

D. SOLVING SIMPLE LINEAR EQUATIONS

Linear Equation: An equation where the highest power of the variable is 1. General form: ax + b = c

Rules for Solving:

11. **Addition Property:** Add the same number to both sides
12. **Subtraction Property:** Subtract the same number from both sides
13. **Multiplication Property:** Multiply both sides by the same number
14. **Division Property:** Divide both sides by the same number (not zero)

Goal: Isolate the variable on one side of the equation.

Type 1: x + a = b

Example 5: Solve: x + 7 = 15

Solution:

x + 7 = 15

Subtract 7 from both sides:

x + 7 - 7 = 15 - 7

x = 8

Check: 8 + 7 = 15 ✓

Example 6: Solve: y + 12 = 20

Solution:

y + 12 = 20

y = 20 - 12

y = 8

Example 7: Solve: a + 3.5 = 10

Solution:

a = 10 - 3.5

a = 6.5

Type 2: x - a = b

Example 8: Solve: x - 5 = 12

Solution:

x - 5 = 12

Add 5 to both sides:

x - 5 + 5 = 12 + 5

x = 17

Check: 17 - 5 = 12 ✓

Example 9: Solve: m - 8 = 15

Solution:

m = 15 + 8

m = 23

Example 10: Solve: y - 4.2 = 6.8

Solution:

y = 6.8 + 4.2

y = 11

Type 3: ax = b

Example 11: Solve: 5x = 30

Solution:

5x = 30

Divide both sides by 5:

5x/5 = 30/5

x = 6

Check: 5(6) = 30 ✓

Example 12: Solve: 7y = 42

Solution:

y = 42/7

y = 6

Example 13: Solve: 3a = 18

Solution:

a = 18/3

a = 6

Type 4: x/a = b

Example 14: Solve: x/4 = 7

Solution:

x/4 = 7

Multiply both sides by 4:

x/4 × 4 = 7 × 4

x = 28

Check: 28/4 = 7 ✓

Example 15: Solve: y/5 = 9

Solution:

$y = 9 \times 5$

$y = 45$

Example 16: Solve: $a/3 = 12$

Solution:

$a = 12 \times 3$

$a = 36$

Type 5: ax + b = c

Example 17: Solve: $3x + 5 = 20$

Solution:

$3x + 5 = 20$

Subtract 5 from both sides:

$3x = 20 - 5$

$3x = 15$

Divide by 3:

$x = 15/3$

$x = 5$

Check: $3(5) + 5 = 15 + 5 = 20$ ✓

Example 18: Solve: $5y - 7 = 18$

Solution:

$5y = 18 + 7$

$5y = 25$

$y = 25/5$

$y = 5$

Example 19: Solve: $2a + 9 = 25$

Solution:

2a = 25 - 9

2a = 16

a = 8

Example 20: Solve: 4x - 11 = 17

Solution:

4x = 17 + 11

4x = 28

x = 7

Type 6: ax - b = c

Example 21: Solve: 6x - 15 = 33

Solution:

6x = 33 + 15

6x = 48

x = 8

Example 22: Solve: 7y - 12 = 30

Solution:

7y = 30 + 12

7y = 42

y = 6

Type 7: Equations with x on both sides

Example 23: Solve: 5x + 3 = 2x + 18

Solution:

Collect x terms on left, constants on right:

$5x - 2x = 18 - 3$

$3x = 15$

$x = 5$

Check:

LHS = $5(5) + 3 = 28$

RHS = $2(5) + 18 = 28$ ✓

Example 24: Solve: $7y - 4 = 3y + 20$

Solution:

$7y - 3y = 20 + 4$

$4y = 24$

$y = 6$

Example 25: Solve: $8a + 7 = 5a + 25$

Solution:

$8a - 5a = 25 - 7$

$3a = 18$

$a = 6$

Example 26: Solve: $4x - 9 = x + 12$

Solution:

$4x - x = 12 + 9$

$3x = 21$

$x = 7$

Type 8: Equations with brackets

Example 27: Solve: $3(x + 4) = 24$

Solution:

Expand bracket:

$3x + 12 = 24$

$3x = 24 - 12$

$3x = 12$

$x = 4$

Check: $3(4 + 4) = 3(8) = 24$ ✓

Example 28: Solve: $2(y - 5) = 18$

Solution:

$2y - 10 = 18$

$2y = 18 + 10$

$2y = 28$

$y = 14$

Example 29: Solve: $5(2a + 3) = 55$

Solution:

$10a + 15 = 55$

$10a = 55 - 15$

$10a = 40$

$a = 4$

Example 30: Solve: $4(x - 3) + 5 = 21$

Solution:

$4x - 12 + 5 = 21$

$4x - 7 = 21$

$4x = 21 + 7$

$4x = 28$

$x = 7$

Type 9: Equations with fractions

Example 31: Solve: $x/3 + 5 = 11$

Solution:

$x/3 = 11 - 5$

$x/3 = 6$

$x = 6 \times 3$

$x = 18$

Example 32: Solve: $(2x + 1)/3 = 7$

Solution:

Multiply both sides by 3:

$2x + 1 = 21$

$2x = 21 - 1$

$2x = 20$

$x = 10$

Example 33: Solve: $(y - 4)/5 = 3$

Solution:

$y - 4 = 3 \times 5$

$y - 4 = 15$

$y = 19$

Type 10: Equations with negative coefficients

Example 34: Solve: $-3x = 15$

Solution:

$x = 15/(-3)$

$x = -5$

Check: -3(-5) = 15 ✓

Example 35: Solve: -2y + 7 = 15

Solution:

-2y = 15 - 7

-2y = 8

y = 8/(-2)

y = -4

Example 36: Solve: 5 - 3x = 20

Solution:

-3x = 20 - 5

-3x = 15

x = -5

E. CHECKING SOLUTIONS

Always verify your solution by substituting back into the original equation.

Example 37: Solve and check: 4x - 7 = 13

Solution:

4x = 13 + 7

4x = 20

x = 5

Check:

LHS = 4(5) - 7 = 20 - 7 = 13 ✓

RHS = 13 ✓

F. WORD PROBLEMS LEADING TO SIMPLE EQUATIONS

Example 38: When 7 is added to a number, the result is 23. Find the number.

Solution:

Let the number = x

$x + 7 = 23$

$x = 23 - 7$

$x = 16$

The number is 16.

Example 39: A number multiplied by 5 gives 45. Find the number.

Solution:

Let the number = x

$5x = 45$

$x = 9$

The number is 9.

Example 40: Three times a number, decreased by 8, equals 19. Find the number.

Solution:

Let the number = x

$3x - 8 = 19$

$3x = 27$

$x = 9$

The number is 9.

Example 41: The sum of two consecutive numbers is 27. Find the numbers.

Solution:

Let first number = x

Second number = $x + 1$

$x + (x + 1) = 27$

$2x + 1 = 27$

$2x = 26$

$x = 13$

First number = 13

Second number = 14

Example 42: Mary is 5 years older than John. The sum of their ages is 35. Find their ages.

Solution:

Let John's age = x

Mary's age = x + 5

$x + (x + 5) = 35$

$2x + 5 = 35$

$2x = 30$

$x = 15$

John = 15 years

Mary = 20 years

Example 43: A rectangle's length is 3 times its width. If the perimeter is 48 cm, find the dimensions.

Solution:

Let width = x

Length = 3x

Perimeter = 2(length + width)

$48 = 2(3x + x)$

$48 = 2(4x)$

$48 = 8x$

$x = 6$

Width = 6 cm

Length = 18 cm

Example 44: A father is 4 times as old as his son. In 5 years, he will be 3 times as old. Find their current ages.

Solution:

Let son's current age = x

Father's current age = 4x

In 5 years:

Son's age = x + 5

Father's age = 4x + 5

Equation: $4x + 5 = 3(x + 5)$

$4x + 5 = 3x + 15$

$4x - 3x = 15 - 5$

$x = 10$

Son = 10 years

Father = 40 years

Example 45: The cost of 5 pens and 3 books is ₦1,700. If each book costs ₦200, find the cost of each pen.

Solution:

Cost of 3 books = 3 × 200 = ₦600

Cost of 5 pens = 1,700 - 600 = ₦1,100

Let cost of 1 pen = x

$5x = 1,100$

$x = 220$

Each pen costs ₦220.

EVALUATION:

1. State the difference between an expression and an equation
2. Is $x = 4$ a solution to $3x + 5 = 17$? Verify
3. Solve: a) $x + 12 = 25$ b) $3y = 21$ c) $a/4 = 8$
4. Solve: $5x - 7 = 23$
5. Solve: $7y + 3 = 4y + 21$
6. Solve: $2(x + 5) = 24$
7. Solve: $(3x - 1)/4 = 5$
8. A number increased by 15 equals 42. Find the number
9. Five times a number minus 12 is 38. Find the number
10. Two consecutive numbers sum to 51. Find the numbers

REVISION QUESTIONS:

1. What is an algebraic equation?
2. What are the parts of an equation?
3. What does it mean to solve an equation?
4. State four properties used in solving equations
5. How do you check if your solution is correct?
6. Solve: $8x = 56$
7. Solve: $x - 9 = 14$
8. What's the first step in solving $4x + 7 = 23$?
9. How do you solve equations with x on both sides?
10. Solve: $3x + 8 = 2x + 15$

11. What do you do first when solving: $5(x - 3) = 20$?
12. Solve: $x/5 + 2 = 7$
13. A number multiplied by 6 gives 48. What's the equation?
14. Solve: $12 - 2x = 4$
15. If length = 2w and perimeter = 36, find w

ASSIGNMENT:

Section A: Basic Concepts (10 marks)

1. Distinguish between algebraic expression and equation (give 3 examples each)
2. State whether expression or equation:
   - a) $5x + 7$
   - b) $3y - 2 = 11$
   - c) $2a + 3b - 5$
   - d) $x^2 = 16$
3. Check if the given value is a solution:
   - a) $x = 5$ for $2x + 3 = 13$
   - b) $y = 3$ for $4y - 5 = 9$
   - c) $a = 6$ for $3a + 2 = 23$

Section B: Simple Equations (30 marks)

Solve:

1. $x + 15 = 28$
2. $y - 12 = 19$
3. $6x = 42$
4. $a/5 = 9$
5. $3x + 7 = 25$
6. $5y - 9 = 31$

7. 2a + 13 = 29
8. 4x - 15 = 17
9. 7y + 5 = 40
10. 3x - 8 = 19

Section C: Equations with Variables on Both Sides (20 marks)

Solve:

1. 5x + 4 = 2x + 19
2. 7y - 3 = 4y + 18
3. 8a + 5 = 3a + 35
4. 6x - 7 = 2x + 13
5. 9y + 2 = 5y + 26
6. 4x + 11 = x + 29
7. 10a - 8 = 6a + 20
8. 3x + 15 = x + 33

Section D: Equations with Brackets (15 marks)

Solve:

1. 2(x + 5) = 22
2. 3(y - 4) = 21
3. 5(a + 3) = 40
4. 4(x - 2) + 7 = 23
5. 3(2y + 1) = 27
6. 2(3x - 5) = 16
7. 5(x + 2) - 3 = 27

Section E: Equations with Fractions (10 marks)

Solve:

1. x/4 + 3 = 8

2. $(y + 5)/3 = 6$
3. $(2x - 1)/5 = 3$
4. $(3a + 2)/4 = 5$

Section F: Word Problems (15 marks)

1. A number increased by 18 equals 45. Find the number.
2. When a number is multiplied by 7, the result is 91. Find the number.
3. Four times a number, decreased by 9, equals 31. Find the number.
4. The sum of three consecutive numbers is 48. Find the numbers.
5. Peter is 7 years older than James. The sum of their ages is 43. Find their ages.
6. A rectangle's length is twice its width. If the perimeter is 54 cm, find the dimensions.
7. The cost of 4 oranges and 5 mangoes is ₦450. If each mango costs ₦60, find the cost of each orange.
8. A number divided by 5, then increased by 8, equals 15. Find the number.
9. Twice a number plus 13 equals three times the number minus 5. Find the number.
10. A father is 5 times as old as his daughter. The sum of their ages is 48 years. Find their ages.

WEEK 3: LINEAR INEQUALITIES I

TOPIC: Definition of Linear Inequalities; Introduction to Symbols and Basic Properties

CONTENT:

A. INTRODUCTION TO INEQUALITIES

Definition: An inequality is a mathematical statement that compares two expressions using inequality symbols instead of an equals sign.

Equality vs Inequality:

- **Equation:** $3x + 5 = 11$ (both sides are equal)
- **Inequality:** $3x + 5 > 11$ (one side is greater than the other)

B. INEQUALITY SYMBOLS

Symbol	Meaning	Example	Reading
$>$	Greater than	$x > 5$	x is greater than 5
$<$	Less than	$x < 3$	x is less than 3
$\geq$	Greater than or equal to	$x \geq 7$	x is greater than or equal to 7
$\leq$	Less than or equal to	$x \leq 4$	x is less than or equal to 4
$\neq$	Not equal to	$x \neq 2$	x is not equal to 2

C. UNDERSTANDING INEQUALITY SYMBOLS

Example 1: $x > 5$

This means x can be any value greater than 5
Examples: 6, 7, 10, 5.1, 100, etc.
NOT: 5 (x must be greater than, not equal to 5)

Example 2: $y < 3$

This means y can be any value less than 3

Examples: 2, 1, 0, -5, 2.9, etc.

NOT: 3 (y must be less than, not equal to 3)

Example 3: $a \geq 7$

This means a can be 7 or any value greater than 7

Examples: 7, 8, 9, 10, 7.5, 100, etc.

Example 4: $b \leq 4$

This means b can be 4 or any value less than 4

Examples: 4, 3, 2, 1, 0, -10, 3.9, etc.

D. GRAPHICAL REPRESENTATION ON NUMBER LINE

Open Circle (○): Used for > and < (value not included) **Closed/Filled Circle (●):** Used for ≥ and ≤ (value included)

Example 5: Represent $x > 3$ on a number line

○————————————→

0 1 2 3 4 5 6

Open circle at 3, arrow to the right

Example 6: Represent $x < 2$ on a number line

←————————————○

-2 -1 0 1 2 3 4

Open circle at 2, arrow to the left

Example 7: Represent $x \geq 4$ on a number line

●────────────────→

0 1 2 3 4 5 6

Closed circle at 4, arrow to the right

Example 8: Represent $x \leq 1$ on a number line

←────────────────●

-2 -1 0 1 2 3 4

Closed circle at 1, arrow to the left

Example 9: Represent $-2 < x \leq 3$

○────────────────●

-3 -2 -1 0 1 2 3 4

Open circle at -2, closed circle at 3
x is between -2 and 3 (not including -2, but including 3)

E. SOLUTION SETS OF INEQUALITIES

Solution Set: All values that satisfy the inequality.

Example 10: List three values that satisfy $x > 5$

Examples: 6, 7, 10
(Any value greater than 5)

Example 11: List three values that satisfy $y \leq 2$

Examples: 2, 1, 0
(Any value less than or equal to 2)

Example 12: Which of the following satisfy $x \geq 3$?

Values: 2, 3, 4, 5, 1

Check:

2: $2 \geq 3$? No ✗

3: $3 \geq 3$? Yes ✓

4: $4 \geq 3$? Yes ✓

5: $5 \geq 3$? Yes ✓

1: $1 \geq 3$? No ✗

Answer: 3, 4, 5

F. PROPERTIES OF INEQUALITIES

Property 1: Addition Property If $a > b$, then $a + c > b + c$

Example 13:

If $x > 5$, then $x + 3 > 5 + 3$
So $x + 3 > 8$

Property 2: Subtraction Property If $a > b$, then $a - c > b - c$

Example 14:

If $y < 10$, then $y - 4 < 10 - 4$
So $y - 4 < 6$

Property 3: Multiplication Property (Positive Number) If $a > b$ and $c > 0$, then $ac > bc$

Example 15:

If $x > 3$, then $2x > 2(3)$
So $2x > 6$

Property 4: Division Property (Positive Number) If $a > b$ and $c > 0$, then $a/c > b/c$

Example 16:

If $3y < 12$, then $3y/3 < 12/3$

So $y < 4$

Property 5: Multiplication/Division by Negative Number CRITICAL: When multiplying or dividing by a negative number, **reverse** the inequality sign.

Example 17:

If $x > 5$, then $-2x < -2(5)$ [Note: sign reversed!]

So $-2x < -10$

Example 18:

If $-3y < 15$, then $-3y/(-3) > 15/(-3)$ [Sign reversed!]

So $y > -5$

Example 19:

If $-x > 7$, multiply by -1:

$-x \times (-1) < 7 \times (-1)$ [Sign reversed!]

$x < -7$

G. COMPARING INEQUALITY STATEMENTS

Example 20: Which is stronger: $x > 5$ or $x \geq 5$?

$x > 5$ is stronger (more restrictive)

Because $x \geq 5$ includes more values (includes 5 itself)

Example 21: True or False?

a) If $x > 3$, then $x > 2$ True ✓ (anything greater than 3 is also greater than 2)

b) If $x < 5$, then $x < 10$ True ✓

c) If $x \leq 4$, then $x < 4$ False ✗ (≤ 4 includes 4, but < 4 doesn't)

d) If $x \geq 7$, then $x = 7$ False ✗ (≥ 7 includes many values, not just 7)

H. DOUBLE INEQUALITIES

Definition: An inequality with two inequality signs.

Example 22: $2 < x < 7$

This means: $x > 2$ AND $x < 7$
x is between 2 and 7 (not including 2 and 7)
Examples: 3, 4, 5, 6, 2.1, 6.9

Example 23: $-3 \leq y \leq 5$

This means: $y \geq -3$ AND $y \leq 5$
y is between -3 and 5 (including both -3 and 5)
Examples: -3, -2, 0, 3, 5

Example 24: Represent $-1 < x \leq 4$ on a number line

○────────────────●

-2 -1 0 1 2 3 4 5

Open circle at -1, closed circle at 4

I. TRANSLATING WORDS TO INEQUALITIES

Example 25: Write as inequalities:

a) x is greater than 10 → $x > 10$
b) y is at most 5 → $y \leq 5$
c) a is at least 8 → $a \geq 8$
d) b is less than 12 → $b < 12$

e) c is no more than 15 → $c \leq 15$

f) d is no less than 20 → $d \geq 20$

g) e is between 3 and 7 → $3 < e < 7$

Example 26: Write in words:

a) $x > 15$ → x is greater than 15

b) $y \leq 20$ → y is less than or equal to 20 OR y is at most 20

c) $a \geq 30$ → a is greater than or equal to 30 OR a is at least 30

d) $5 < b < 10$ → b is between 5 and 10

J. REAL-LIFE SITUATIONS WITH INEQUALITIES

Example 27: Age Restrictions

"You must be at least 18 years old to vote"

If age = a, then: $a \geq 18$

Example 28: Speed Limits

"Speed must not exceed 100 km/h"

If speed = s, then: $s \leq 100$

Example 29: Temperature

"Temperature should be kept below 5°C"

If temperature = t, then: $t < 5$

Example 30: Weight Limits

"Elevator capacity is at most 1000 kg"

If weight = w, then: $w \leq 1000$

Example 31: Height Requirements

"Children taller than 1.2m can ride"

If height = h, then: $h > 1.2$

Example 32: Price Range

"Budget is between ₦5,000 and ₦10,000"

If price = p, then: $5{,}000 \leq p \leq 10{,}000$

K. SIMPLE SOLVING (Introduction)

Example 33: Solve: $x + 3 > 8$

Solution:

$x + 3 > 8$

Subtract 3 from both sides:

$x > 8 - 3$

$x > 5$

Number line:

```
   ○————————————————→
 0  1  2  3  4  5  6  7
```

Example 34: Solve: $y - 5 < 12$

Solution:

$y - 5 < 12$

Add 5 to both sides:

$y < 12 + 5$

$y < 17$

Example 35: Solve: $2x \geq 14$

Solution:

$2x \geq 14$

Divide both sides by 2:

$x \geq 14/2$

$x \geq 7$

Example 36: Solve: $x/3 \leq 5$

Solution:

$x/3 \leq 5$

Multiply both sides by 3:

$x \leq 5 \times 3$

$x \leq 15$

Example 37: Solve: $-x > 4$

Solution:

$-x > 4$

Multiply by -1 (reverse sign!):

$x < -4$

EVALUATION:

1. Define inequality and give 3 examples
2. Explain the difference between $>$ and $\geq$
3. Represent on a number line: a) $x < 4$ b) $x \geq 2$
4. List 3 values that satisfy: a) $x > 7$ b) $y \leq 3$
5. What happens to the inequality sign when you multiply or divide by a negative number?
6. Write as inequality: "a is at least 15"
7. Solve: $x + 7 > 12$
8. True or False: If $x \geq 5$, then x can be 5
9. Represent: $-2 \leq x < 3$ on a number line
10. Write in words: $y < 20$

REVISION QUESTIONS:

1. What is the difference between an equation and an inequality?
2. State the five inequality symbols and their meanings

3. When do we use an open circle on a number line?
4. When do we use a closed circle?
5. Give 3 real-life examples of inequalities
6. What does "at most" mean in inequalities?
7. What does "at least" mean?
8. State the critical rule about multiplying/dividing by negative numbers
9. Can $x = 5$ satisfy $x \geq 5$? Explain
10. Can $x = 5$ satisfy $x > 5$? Explain
11. Solve: $3x > 15$
12. What values satisfy: $2 < x \leq 7$?
13. Represent $x \leq -1$ on a number line
14. Write as inequality: "Temperature is below 0°C"
15. If $x > 3$, is $x > 2$ also true? Why?

ASSIGNMENT:

Section A: Symbols and Meanings (10 marks)

1. State the meaning of:
    - a) $>$
    - b)
    - c) $\geq$
    - d) $\leq$
    - e) $\neq$
2. Write in words:
    - a) $x > 12$
    - b) $y \leq 25$
    - c) $a \geq 30$
    - d) $b < 8$

- e) $5 < c < 15$

Section B: Number Line Representation (15 marks)

Draw number lines to represent:

1. $x > 4$
2. $x < 2$
3. $x \geq 6$
4. $x \leq 0$
5. $x > -3$
6. $-2 < x < 5$
7. $-1 \leq x \leq 4$
8. $0 < x \leq 3$

Section C: Solution Sets (15 marks)

1. List 5 values that satisfy:
   - a) $x > 7$
   - b) $y < 5$
   - c) $a \geq 10$
   - d) $b \leq 2$
2. Which of the following satisfy $x \geq 4$?

   Values: 2, 3, 4, 5, 6, 1
3. Which of the following satisfy $y < 8$?

   Values: 7, 8, 9, 6, 10, 5
4. True or False (explain):
   - a) If $x > 5$, then $x = 6$ is a solution
   - b) If $y \leq 3$, then $y = 3$ is a solution
   - c) If $a < 10$, then $a = 10$ is a solution

- d) If $b \geq 7$, then $b = 6$ is a solution

Section D: Word to Symbols (15 marks)

Write as inequalities:

1. x is greater than 20
2. y is less than 15
3. a is at most 25
4. b is at least 30
5. c is no more than 18
6. d is no less than 12
7. e is between 5 and 10
8. f is greater than or equal to 8
9. Age (a) must be at least 21
10. Speed (s) should not exceed 80 km/h
11. Temperature (t) is below freezing (0°C)
12. Weight (w) can be at most 50 kg
13. Height (h) must exceed 1.5 m
14. Price (p) ranges from ₦2,000 to ₦5,000
15. Score (s) must be above 40

Section E: Properties (15 marks)

1. If $x > 8$, what is:
   - a) $x + 5$
   - b) $x - 3$
   - c) $2x$
2. If $y < 12$, what is:
   - a) $y + 4$
   - b) $y - 7$

3. Complete:

    - a) If $a > 6$, then $a + 3$ __ 9
    - b) If $b < 10$, then $b - 2$ __ 8
    - c) If $c \geq 5$, then $3c$ __ 15
    - d) If $d \leq 20$, then $d/4$ __ 5

4. What happens to the inequality sign when:

    - a) You multiply both sides by -2
    - b) You divide both sides by -5
    - c) You add 10 to both sides
    - d) You subtract 7 from both sides

Section F: Simple Solving (20 marks)

Solve and represent on a number line:

1. $x + 5 > 12$
2. $y - 3 < 10$
3. $4a \geq 20$
4. $b/2 \leq 6$
5. $x + 8 < 15$
6. $3y > 18$
7. $a - 7 \geq 2$
8. $x/5 < 3$
9. $2x + 4 > 10$
10. $3y - 5 < 7$

Section G: Real-Life Applications (10 marks)

Write an inequality for each situation:

1. A bus can carry at most 50 passengers. (Let p = number of passengers)

2. You must be at least 16 years old to drive. (Let a = age)

3. The temperature in a freezer should be below -5°C. (Let t = temperature)
4. A bag can hold no more than 20 kg. (Let w = weight)
5. Children over 1.2 m tall can ride the roller coaster. (Let h = height)
6. The movie is suitable for viewers aged 13 and above. (Let a = age)
7. Workers must complete between 30 and 50 units per day. (Let u = units)
8. The budget should not exceed ₦50,000. (Let b = budget)
9. Students need at least 50% to pass. (Let s = score as percentage)
10. Room temperature should be kept between 20°C and 25°C. (Let t = temperature)

WEEK 4: LINEAR INEQUALITIES II

TOPIC: Word Problems Leading to Simple Inequalities in One Variable; Solving and Interpreting Results

CONTENT:

A. SOLVING LINEAR INEQUALITIES

Linear Inequality: An inequality where the highest power of the variable is 1.

General Forms:

- ax + b > c
- ax + b < c
- ax + b ≥ c
- ax + b ≤ c

Steps to Solve:

1. Simplify both sides if necessary
2. Collect variable terms on one side, constants on the other
3. Isolate the variable
4. Remember: Reverse inequality sign when multiplying/dividing by negative number
5. Check solution
6. Represent on number line

Type 1: x + a > b

Example 1: Solve: x + 7 > 15

Solution:

x + 7 > 15

x > 15 - 7

x > 8

Number line:

○——————————→

5 6 7 8 9 10 11

Solution set: $\{x: x > 8\}$

or $x \in (8, \infty)$

Example 2: Solve: $y + 5 \leq 12$

Solution:

$y \leq 12 - 5$

$y \leq 7$

Number line:

←——————————●

3 4 5 6 7 8 9

Type 2: x - a < b

Example 3: Solve: $x - 9 < 15$

Solution:

$x < 15 + 9$

$x < 24$

Example 4: Solve: $y - 4 \geq 11$

Solution:

$y \geq 11 + 4$

$y \geq 15$

Type 3: ax > b

Example 5: Solve: $5x > 30$

Solution:

$x > 30/5$

$x > 6$

Check: Try $x = 7$

$5(7) = 35 > 30$ ✓

Example 6: Solve: $3y \leq 21$

Solution:

$y \leq 21/3$

$y \leq 7$

Type 4: x/a < b

Example 7: Solve: $x/4 < 6$

Solution:

$x < 6 \times 4$

$x < 24$

Example 8: Solve: $y/3 \geq 5$

Solution:

$y \geq 5 \times 3$

$y \geq 15$

Type 5: ax + b > c

Example 9: Solve: $3x + 5 > 20$

Solution:

$3x > 20 - 5$

$3x > 15$

$x > 15/3$

$x > 5$

Check: Try $x = 6$

$3(6) + 5 = 23 > 20$ ✓

Example 10: Solve: $2y - 7 \leq 13$

Solution:

$2y \leq 13 + 7$

$2y \leq 20$

$y \leq 10$

Example 11: Solve: $5a + 8 < 33$

Solution:

$5a < 33 - 8$

$5a < 25$

$a < 5$

Example 12: Solve: $4x - 9 \geq 15$

Solution:

$4x \geq 15 + 9$

$4x \geq 24$

$x \geq 6$

Type 6: Inequalities with Variable on Both Sides

Example 13: Solve: $5x + 3 > 2x + 18$

Solution:

$5x - 2x > 18 - 3$

$3x > 15$

$x > 5$

Check: Try $x = 6$

LHS: $5(6) + 3 = 33$

RHS: $2(6) + 18 = 30$

$33 > 30$ ✓

Example 14: Solve: $7y - 4 \leq 3y + 20$

Solution:

$7y - 3y \leq 20 + 4$

$4y \leq 24$

$y \leq 6$

Example 15: Solve: $8a + 7 < 5a + 28$

Solution:

$8a - 5a < 28 - 7$

$3a < 21$

$a < 7$

Type 7: Inequalities with Brackets

Example 16: Solve: $3(x + 4) > 24$

Solution:

$3x + 12 > 24$

$3x > 24 - 12$

$3x > 12$

$x > 4$

Example 17: Solve: $2(y - 5) \leq 18$

Solution:

$2y - 10 \leq 18$

$2y \leq 18 + 10$

$2y \leq 28$

$y \leq 14$

Example 18: Solve: $5(2a + 3) \geq 65$

Solution:

$10a + 15 \geq 65$

$10a \geq 50$

$a \geq 5$

Type 8: Negative Coefficients (CRITICAL)

Example 19: Solve: $-3x > 12$

Solution:

Divide by -3 (REVERSE THE SIGN!):

$x < 12/(-3)$

$x < -4$

Number line:

←————————————○

-7 -6 -5 -4 -3 -2 -1

Example 20: Solve: $-2y \leq 10$

Solution:

Divide by -2 (reverse sign):

$y \geq 10/(-2)$

$y \geq -5$

Example 21: Solve: -5a + 7 > 22

Solution:

-5a > 22 - 7

-5a > 15

Divide by -5 (reverse sign):

a < 15/(-5)

a < -3

Example 22: Solve: 8 - 4x < 20

Solution:

-4x < 20 - 8

-4x < 12

Divide by -4 (reverse sign):

x > 12/(-4)

x > -3

Type 9: Fractions

Example 23: Solve: (x + 5)/3 > 4

Solution:

Multiply both sides by 3:

x + 5 > 12

x > 12 - 5

x > 7

Example 24: Solve: (2y - 3)/5 ≤ 3

Solution:

2y - 3 ≤ 15

2y ≤ 18

y ≤ 9

Type 10: Double Inequalities

Example 25: Solve: $5 < 2x + 1 < 13$

Solution:

Split into two inequalities:

Part 1: $5 < 2x + 1$

$4 < 2x$

$2 < x$ or $x > 2$

Part 2: $2x + 1 < 13$

$2x < 12$

$x < 6$

Combined: $2 < x < 6$

Number line:

○————————○

0 1 2 3 4 5 6 7

Solution: $\{x: 2 < x < 6\}$

Example 26: Solve: $-1 \leq 3x - 4 \leq 8$

Solution:

Part 1: $-1 \leq 3x - 4$

$3 \leq 3x$

$1 \leq x$ or $x \geq 1$

Part 2: $3x - 4 \leq 8$

$3x \leq 12$

$x \leq 4$

Combined: $1 \leq x \leq 4$

Solution: $\{x: 1 \leq x \leq 4\}$

B. WORD PROBLEMS WITH INEQUALITIES

Example 27: A number increased by 15 is greater than 42. Find the possible values.

Solution:

Let the number = x

$x + 15 > 42$

$x > 42 - 15$

$x > 27$

Answer: The number must be greater than 27.

Example 28: Five times a number is at most 45. Find the possible values.

Solution:

Let the number = x

$5x \leq 45$

$x \leq 9$

Answer: The number can be at most 9.

Example 29: Three times a number, decreased by 8, is less than 19. Find possible values.

Solution:

Let the number = x

$3x - 8 < 19$

$3x < 27$

$x < 9$

Answer: The number must be less than 9.

Example 30: Twice a number plus 7 is at least 23. What are the possible values?

Solution:

Let the number = x

$2x + 7 \geq 23$

$2x \geq 16$

$x \geq 8$

Answer: The number must be at least 8.

Age Problems:

Example 31: John's age plus 5 is more than 20. How old can John be?

Solution:

Let John's age = x

$x + 5 > 20$

$x > 15$

Answer: John must be older than 15 years.

Example 32: Mary is at least 3 years older than twice her brother's age. If her brother is 7, how old can Mary be?

Solution:

Brother's age = 7

Twice brother's age = $2 \times 7 = 14$

Let Mary's age = x

$x \geq 14 + 3$

$x \geq 17$

Answer: Mary must be at least 17 years old.

Money Problems:

Example 33: You have ₦5,000. After buying books that cost ₦350 each, you want to have at least ₦2,000 left. How many books can you buy?

Solution:

Let number of books = x

Cost of books = 350x

Money left = 5,000 - 350x

$5{,}000 - 350x \geq 2{,}000$

$-350x \geq 2{,}000 - 5{,}000$

$-350x \geq -3{,}000$

Divide by -350 (reverse sign):

$x \leq -3{,}000/(-350)$

$x \leq 8.57$

Answer: You can buy at most 8 books.

Example 34: A taxi charges ₦200 plus ₦50 per kilometer. If you have at most ₦1,500, how far can you travel?

Solution:

Let distance = x km

Cost = 200 + 50x

$200 + 50x \leq 1{,}500$

$50x \leq 1{,}300$

$x \leq 26$

Answer: You can travel at most 26 km.

Perimeter Problems:

Example 35: A rectangle's length is 3 times its width. If the perimeter must be less than 80 cm, find possible widths.

Solution:

Let width = x

Length = 3x

Perimeter = 2(length + width) = 2(3x + x) = 8x

$8x < 80$

$x < 10$

Answer: Width must be less than 10 cm.

Example 36: An isosceles triangle has equal sides of length x cm and base of 8 cm. If perimeter is at least 26 cm, find possible values of x.

Solution:

Perimeter = x + x + 8 = 2x + 8

$2x + 8 \geq 26$

$2x \geq 18$

$x \geq 9$

Answer: Each equal side must be at least 9 cm.

Grade/Score Problems:

Example 37: To pass a test, you need an average of at least 60% on 5 tests. If your first 4 scores are 55, 62, 58, and 65, what must you score on the 5th test?

Solution:

Let 5th score = x

Average = (55 + 62 + 58 + 65 + x)/5

$(55 + 62 + 58 + 65 + x)/5 \geq 60$

$(240 + x)/5 \geq 60$

$240 + x \geq 300$

$x \geq 60$

Answer: You must score at least 60 on the 5th test.

Temperature Problems:

Example 38: For a science experiment, temperature must be kept between 15°C and 25°C. Currently it's 18°C. If temperature rises by 2°C every hour, for how long can the experiment continue?

Solution:

Let hours = x

Final temperature = 18 + 2x

Must satisfy: $15 \leq 18 + 2x \leq 25$

Upper limit: $18 + 2x \leq 25$

$2x \leq 7$

$x \leq 3.5$

Answer: Experiment can continue for at most 3.5 hours (or 3 hours 30 minutes).

Weight/Capacity Problems:

Example 39: An elevator can carry at most 800 kg. There are already 5 people averaging 65 kg each. What's the maximum weight of additional person(s) that can enter?

Solution:
Current weight = $5 \times 65 = 325$ kg
Let additional weight = x

$325 + x \leq 800$
$x \leq 475$

Answer: Additional weight can be at most 475 kg.

Profit/Loss Problems:

Example 40: A trader buys items for ₦250 each. If he must make at least ₦50 profit on each, what's the minimum selling price?

Solution:
Cost price = ₦250
Let selling price = x
Profit = x - 250

$x - 250 \geq 50$
$x \geq 300$

Answer: Minimum selling price is ₦300.

Example 41: A company's monthly expenses are ₦150,000. If they sell products at ₦800 each, how many must they sell to make profit?

Solution:

Let number of products = x

Revenue = 800x

For profit: Revenue > Expenses

800x > 150,000

x > 187.5

Answer: They must sell at least 188 products.

C. INTERPRETING SOLUTIONS

Example 42: After solving x > 5, interpret:

Interpretation:

- x can be any value greater than 5
- x = 5 is NOT included
- Examples: 5.1, 6, 7, 10, 100, etc.
- Graphically: Open circle at 5, arrow to right
- In real context: "More than 5"

Example 43: After solving y ≤ 12, interpret:

Interpretation:

- y can be 12 or any value less than 12
- y = 12 IS included
- Examples: 12, 11, 10, 0, -5, etc.
- Graphically: Closed circle at 12, arrow to left
- In real context: "At most 12" or "12 or fewer"

Example 44: After solving $3 < x \leq 8$, interpret:

Interpretation:

- x is between 3 and 8
- x = 3 is NOT included (open circle)
- x = 8 IS included (closed circle)
- Examples: 4, 5, 6, 7, 8, 3.1, 7.9
- In real context: "More than 3 but not more than 8"

EVALUATION:

1. Solve: $3x + 7 > 22$
2. Solve: $5y - 8 \leq 17$
3. Solve: $2x + 5 > x + 12$
4. Solve: $-4a < 16$
5. Solve: $(x + 3)/2 \geq 5$
6. A number increased by 12 is less than 30. Find possible values
7. Solve and represent on number line: $4x - 3 \leq 13$
8. Twice a number minus 7 is at least 15. Find possible values
9. Solve: $3 < 2x + 1 < 11$
10. You have ₦8,000. Books cost ₦650 each. You want at least ₦2,500 left. How many books can you buy?

REVISION QUESTIONS:

1. State the steps to solve a linear inequality
2. What's the critical rule about negative coefficients?
3. How do you check if your solution is correct?
4. What does "at most" mean in word problems?
5. What does "at least" mean?
6. Solve: $7x + 4 > 25$

7. Solve: $6y - 9 \leq 27$
8. How do you solve double inequalities?
9. Interpret: $x \geq 10$ in real-life context
10. Solve: $-3x + 5 > 14$
11. A perimeter is at most 60 cm. Write an inequality
12. Solve: $4(x - 3) < 20$
13. What does an open circle mean on a number line?
14. Give 3 values that satisfy: $2 < x \leq 7$
15. Solve: $5x - 2 > 3x + 8$

ASSIGNMENT:

Section A: Basic Solving (25 marks)

Solve and represent on a number line:

1. $x + 8 > 15$
2. $y - 5 \leq 12$
3. $3a > 18$
4. $b/4 \leq 5$
5. $2x + 7 > 19$
6. $5y - 3 \leq 22$
7. $4a + 9 < 33$
8. $3x - 8 \geq 13$
9. $7y + 5 > 40$
10. $2a - 11 \leq 15$

Section B: Variables on Both Sides (15 marks)

Solve:

1. $6x + 4 > 3x + 19$

2. $8y - 5 \leq 5y + 16$
3. $7a + 3 < 4a + 24$
4. $9x - 7 \geq 5x + 17$
5. $5y + 12 > 2y + 30$
6. $10a - 8 \leq 6a + 20$

Section C: With Brackets (10 marks)

Solve:

1. $3(x + 5) > 27$
2. $4(y - 2) \leq 24$
3. $2(3a + 4) \geq 26$
4. $5(x - 3) + 7 < 32$
5. $3(2y + 1) - 4 > 17$

Section D: Negative Coefficients (10 marks)

Solve:

1. $-2x > 14$
2. $-5y \leq 30$
3. $-3a + 7 > 19$
4. $6 - 4x < 18$
5. $12 - 3y \geq 0$

Section E: Fractions (10 marks)

Solve:

1. $(x + 4)/3 > 5$
2. $(2y - 5)/4 \leq 3$
3. $(3a + 1)/2 \geq 7$
4. $(x - 2)/5 < 4$

Section F: Double Inequalities (10 marks)

Solve:

1. $3 < x + 5 < 12$
2. $7 \leq 2y + 1 \leq 15$
3. $-2 < 3x - 5 < 10$
4. $4 \leq 5a - 6 \leq 19$
5. $-5 < 2x + 3 \leq 9$

Section G: Word Problems (20 marks)

1. A number increased by 18 is greater than 45. Find possible values.
2. Four times a number is at most 52. Find possible values.
3. Three times a number, decreased by 7, is less than 26. Find possible values.
4. Twice a number plus 11 is at least 35. Find possible values.
5. Peter's age minus 4 is more than 18. How old can Peter be?
6. The sum of Mary's age and 7 is at most 30. How old can Mary be?
7. You have ₦12,000. After buying shirts that cost ₦850 each, you want at least ₦3,500 left. How many shirts can you buy?
8. A taxi charges ₦300 plus ₦80 per kilometer. If you have at most ₦2,100, how far can you travel?
9. A rectangle's length is twice its width. If perimeter must be less than 72 cm, find possible widths.
10. An isosceles triangle has equal sides x cm and base 12 cm. If perimeter is at least 38 cm, find possible values of x.
11. To get grade A, average must be at least 80%. If your first 4 scores are 75, 82, 78, 85, what must you score on the 5th test?

12. An elevator can carry at most 900 kg. There are 6 people averaging 70 kg each. What's maximum weight of additional person(s)?
13. A trader buys items for ₦320 each. If he must make at least ₦80 profit on each, what's minimum selling price?
14. A company's monthly expenses are ₦200,000. If they sell products at ₦650 each, how many must they sell to make profit?
15. For an experiment, temperature must be between 18°C and 28°C. Currently it's 20°C. If temperature rises 1.5°C per hour, for how long can experiment continue?
16. A bus can carry at most 48 passengers. There are already 32 passengers. How many more can board?
17. Your phone has 35% battery. It drains 5% per hour. For how many more hours can you use it before it reaches 10%?
18. A garden's length is 4 m more than twice its width. If perimeter is at most 56 m, find possible widths.
19. The sum of three consecutive even numbers is less than 90. What's the largest possible value of the first number?
20. A worker earns ₦3,500 daily plus ₦200 per unit produced. If he wants to earn at least ₦8,500 daily, how many units must he produce?

WEEK 6: GRAPHS I

TOPIC: Graphical Representation; Graphs of Cartesian Plane; Introduction to Graphs of Linear Equations in Two Variables

CONTENT:

A. INTRODUCTION TO GRAPHS

Definition: A graph is a visual representation of data or mathematical relationships using points, lines, or curves on a coordinate system.

Why Use Graphs?

- Visual representation of data
- Easy to understand patterns and trends
- Compare information quickly
- Show relationships between variables
- Make predictions

Types of Graphs:

- Bar graphs
- Line graphs
- Pie charts
- Pictographs
- Scatter plots
- Linear graphs (focus of this topic)

B. THE CARTESIAN PLANE (COORDINATE SYSTEM)

Definition: The Cartesian plane (also called coordinate plane or xy-plane) is a two-dimensional surface formed by two perpendicular number lines.

Components:

1. The x-axis (Horizontal axis)

```
←————————————————————————→
   x-axis
```

- Runs left to right
- Positive direction: right
- Negative direction: left

2. The y-axis (Vertical axis)

```
    ↑
    |
  y-axis
    |
    ↓
```

- Runs up and down
- Positive direction: up
- Negative direction: down

3. Origin (O)

- Point where x-axis and y-axis intersect
- Coordinates: (0, 0)
- The reference point

Complete Cartesian Plane:

```
     y-axis
       ↑
       |
  II   |   I
       |
←——————————————O——————————————→ x-axis
```

|

III | IV

|

↓

C. THE FOUR QUADRANTS

Quadrant	Position	x-value	y-value	Example Point
I (First)	Top right	Positive (+)	Positive (+)	(3, 4)
II (Second)	Top left	Negative (-)	Positive (+)	(-2, 5)
III (Third)	Bottom left	Negative (-)	Negative (-)	(-4, -3)
IV (Fourth)	Bottom right	Positive (+)	Negative (-)	(5, -2)

D. COORDINATES (ORDERED PAIRS)

Definition: A coordinate is an ordered pair (x, y) that locates a point on the Cartesian plane.

Format: (x, y)

- **x-coordinate (abscissa):** Distance from y-axis (horizontal)
- **y-coordinate (ordinate):** Distance from x-axis (vertical)

Important: Order matters! (3, 4) ≠ (4, 3)

E. PLOTTING POINTS

Steps to Plot a Point (x, y):

15. Start at the origin (0, 0)
16. Move x units along x-axis (right if +, left if -)
17. From there, move y units parallel to y-axis (up if +, down if -)
18. Mark the point

Example 1: Plot point A(3, 4)

```
      y
      |
    5 |
    4 |        • A(3,4)
    3 |
    2 |
    1 |
      |
——————————O—————————— x
    1  2  3  4  5
```

Steps:

- From O, move 3 units right (x = 3)
- Then move 4 units up (y = 4)
- Mark point A

Example 2: Plot point B(-2, 3)

```
      y
      |
    4 |
    3 |  • B(-2,3)
    2 |
    1 |
      |
——————————O—————————— x
 -3 -2 -1  1  2  3
```

Steps:

- From O, move 2 units left (x = -2)

- Then move 3 units up (y = 3)
- Mark point B

Example 3: Plot point C(4, -2)

```
     y
     |
   2 |
   1 |
     |
——————————O—————————— x
     |  1  2  3  4  5
  -1 |
  -2 |       • C(4,-2)
     |
```

Steps:
- From O, move 4 units right (x = 4)
- Then move 2 units down (y = -2)
- Mark point C

Example 4: Plot point D(-3, -4)

```
     y
     |
   1 |
     |
——————————O—————————— x
 -4 -3 -2 -1  1  2
  -1 |
  -2 |
  -3 |
  -4 | • D(-3,-4)
```

|

Steps:

- From O, move 3 units left (x = -3)
- Then move 4 units down (y = -4)
- Mark point D

F. SPECIAL POINTS

Points on the Axes:

On x-axis: y-coordinate = 0

Examples: (3, 0), (-2, 0), (5, 0)

```
      y
      |
      |
 ———•———O———•— x
 (-2,0)   (3,0)
      |
```

On y-axis: x-coordinate = 0

Examples: (0, 4), (0, -3), (0, 2)

```
      y
      |
    4 |• (0,4)
      |
    2 |• (0,2)
      |
 ——————O—————— x
```

```
     |
  -3 |• (0,-3)
     |
```

At Origin: Both coordinates = 0

Point: (0, 0)

G. READING COORDINATES

Example 5: State coordinates of points shown:

```
      y
      |
    4 | B •
    3 |
    2 |    • C
    1 |
      |
 ————————————O———————————— x
 -3 -2 -1  1  2  3  4
   -1 |
   -2 |   • A
      |
```

Point A: x = 1, y = -2 → A(1, -2)

Point B: x = -2, y = 4 → B(-2, 4)

Point C: x = 3, y = 2 → C(3, 2)

H. LINEAR EQUATIONS IN TWO VARIABLES

Definition: A linear equation in two variables is an equation that can be written in the form: **$ax + by = c$**

Where a, b, c are constants, and x, y are variables.

Examples:

a) $x + y = 5$
b) $2x + 3y = 12$
c) $y = 2x + 1$
d) $x - y = 4$
e) $y = 3x$

Characteristics:

- Highest power of variables is 1
- Graph is always a straight line
- Has infinitely many solutions (ordered pairs)

I. FINDING SOLUTIONS TO LINEAR EQUATIONS

Solution: Any ordered pair (x, y) that satisfies the equation.

Example 6: Is (2, 3) a solution to $x + y = 5$?

Check:
$x + y = 5$
$2 + 3 = 5$ ✓

Yes, (2, 3) is a solution.

Example 7: Is (4, 1) a solution to $x + y = 5$?

Check:

x + y = 5

4 + 1 = 5 ✓

Yes, (4, 1) is a solution.

Example 8: Is (3, 1) a solution to x + y = 5?

Check:

x + y = 5

3 + 1 = 4 ✗

No, (3, 1) is not a solution.

J. FINDING MULTIPLE SOLUTIONS

Method: Choose values for one variable, solve for the other.

Example 9: Find 5 solutions to x + y = 6

Create a table:

x	y = 6 - x	(x, y)
0	6 - 0 = 6	(0, 6)
1	6 - 1 = 5	(1, 5)
2	6 - 2 = 4	(2, 4)
3	6 - 3 = 3	(3, 3)
4	6 - 4 = 2	(4, 2)

Solutions: (0, 6), (1, 5), (2, 4), (3, 3), (4, 2)

Example 10: Find 4 solutions to y = 2x

x	y = 2x	(x, y)
0	2(0) = 0	(0, 0)
1	2(1) = 2	(1, 2)
2	2(2) = 4	(2, 4)
3	2(3) = 6	(3, 6)

Solutions: (0, 0), (1, 2), (2, 4), (3, 6)

Example 11: Find 5 solutions to $2x + y = 8$

x	y = 8 - 2x	(x, y)
0	8 - 2(0) = 8	(0, 8)
1	8 - 2(1) = 6	(1, 6)
2	8 - 2(2) = 4	(2, 4)
3	8 - 2(3) = 2	(3, 2)
4	8 - 2(4) = 0	(4, 0)

Solutions: (0, 8), (1, 6), (2, 4), (3, 2), (4, 0)

K. INTRODUCTION TO GRAPHING LINEAR EQUATIONS

Key Concept: All solutions to a linear equation lie on a straight line.

Example 12: Graph $x + y = 5$

Step 1: Find at least 3 solutions (2 minimum, 3 for accuracy)

x	y	(x, y)
0	5	(0, 5)
2	3	(2, 3)
5	0	(5, 0)

Step 2: Plot the points on Cartesian plane

```
      y
      |
    6 |
    5 |• (0,5)
    4 |
    3 |    • (2,3)
    2 |
    1 |
      |
——————————O————•——— x
      |    (5,0)
        1  2  3  4  5
```

Step 3: Draw a straight line through the points

Step 4: Label the line with the equation

L. INTERCEPTS

x-intercept: Point where line crosses x-axis (y = 0) **y-intercept:** Point where line crosses y-axis (x = 0)

Example 13: Find intercepts of x + y = 6

x-intercept (let y = 0):

x + 0 = 6
x = 6
x-intercept: (6, 0)

y-intercept (let x = 0):

$0 + y = 6$

$y = 6$

y-intercept: (0, 6)

Example 14: Find intercepts of $2x + 3y = 12$

x-intercept:

$2x + 3(0) = 12$

$2x = 12$

$x = 6$

x-intercept: (6, 0)

y-intercept:

$2(0) + 3y = 12$

$3y = 12$

$y = 4$

y-intercept: (0, 4)

M. SLOPE (GRADIENT)

Definition: The slope/gradient of a line measures its steepness.

Formula:

Slope (m) = Change in y / Change in x

$= (y_2 - y_1) / (x_2 - x_1)$

= Rise / Run

Example 15: Find slope of line through (1, 2) and (3, 6)

$m = (y_2 - y_1) / (x_2 - x_1)$

$= (6 - 2) / (3 - 1)$

$= 4 / 2$

$= 2$

Slope = 2

Types of Slopes:

Positive slope: / (rises left to right)
Negative slope: \ (falls left to right)
Zero slope: — (horizontal line)
Undefined slope: | (vertical line)

N. SPECIAL LINEAR GRAPHS

1. Horizontal Lines: y = k (constant)

Example 16: Graph y = 3

All points have y = 3
Points: (-2, 3), (0, 3), (2, 3), (4, 3)

```
   y
   |
 4 |
 3 |•———————•———————•———————• y = 3
 2 |
 1 |
   |
————————————O———————————— x
-2 -1  0  1  2  3  4
```

Horizontal line through y = 3
Slope = 0

2. Vertical Lines: x = k (constant)

Example 17: Graph x = 2

All points have x = 2

Points: (2, -1), (2, 0), (2, 2), (2, 4)

```
    y
    |
  4 |   •
  3 |   |
  2 |   • x = 2
  1 |   |
    |  •
——————————O—————————•—— x
    | 2
 -1 |   •
    |
```

Vertical line through x = 2

Slope = undefined

3. Lines through Origin

Example 18: Graph y = x

Points: (-2, -2), (0, 0), (2, 2), (4, 4)

```
    y
    |
  4 |       •
  3 |     /
  2 |   •
  1 | /
    |•
```

——————O—————— x

-1 /

-2•

Line through origin at 45°

Slope = 1

EVALUATION:

1. What is a Cartesian plane?
2. Name the four quadrants and state signs of coordinates in each
3. Plot points: A(3, 2), B(-2, 4), C(-1, -3), D(4, -2)
4. What are the coordinates of the origin?
5. Is (3, 5) a solution to x + y = 8? Verify
6. Find 4 solutions to y = 3x
7. What is the x-intercept of a line?
8. Find intercepts of x + y = 7
9. Draw a graph of y = 2
10. State the coordinates of a point on the y-axis

REVISION QUESTIONS:

1. What is the purpose of using graphs?
2. How do you plot a point (x, y)?
3. What is an ordered pair?
4. In which quadrant is point (-3, 5)?
5. What is special about points on the x-axis?
6. Define a linear equation in two variables
7. How many solutions does a linear equation have?
8. What shape is the graph of a linear equation?

9. How do you find the y-intercept?
10. What is the slope of a horizontal line?
11. Find three solutions to $x + y = 10$
12. Plot point (0, -4). On which axis does it lie?
13. What happens when you reverse coordinates (3, 4) to (4, 3)?
14. Find slope of line through (0, 0) and (4, 8)
15. What is the equation of a vertical line through $x = 5$?

ASSIGNMENT:

Section A: Cartesian Plane (15 marks)

1. Define:
    - a) Cartesian plane
    - b) x-axis
    - c) y-axis
    - d) Origin
    - e) Ordered pair
2. State the quadrant and signs of coordinates:
    - a) (5, 7)
    - b) (-3, 4)
    - c) (-2, -6)
    - d) (4, -3)
3. On which axis do these points lie?
    - a) (0, 5)
    - b) (-3, 0)
    - c) (0, -2)
    - d) (7, 0)

Section B: Plotting Points (20 marks)

Draw a Cartesian plane and plot these points:

1. A(2, 3)
2. B(-4, 2)
3. C(-2, -5)
4. D(5, -3)
5. E(0, 4)
6. F(-3, 0)
7. G(4, 4)
8. H(-1, -1)

Label each point clearly.

Section C: Reading Coordinates (10 marks)

[Draw a Cartesian plane with plotted points]

State the coordinates of points P, Q, R, S, T shown on the plane.

Section D: Solutions to Equations (20 marks)

1. Check if these are solutions:
   - a) (2, 5) for x + y = 7
   - b) (3, 4) for 2x + y = 10
   - c) (1, 6) for y = 5x
   - d) (4, 2) for x - y = 2
2. Complete the table for x + y = 9:

x	0	1	3	5	9
y					

3. Complete the table for y = 3x:

x	0	1	2	4

x	0	1	2	4
y				

4. Complete the table for $2x + y = 10$:

x	0	2	3	5
y				

Section E: Intercepts (15 marks)

Find x-intercept and y-intercept:

1. $x + y = 8$
2. $2x + y = 6$
3. $x + 3y = 12$
4. $3x + 2y = 18$
5. $x - y = 5$

Section F: Slope (10 marks)

Find the slope of the line through:

1. (1, 3) and (4, 9)
2. (0, 2) and (3, 8)
3. (-1, 4) and (2, 10)
4. (2, 5) and (6, 13)
5. (0, 0) and (5, 15)

Section G: Graphing (10 marks)

1. Draw a Cartesian plane and graph:
   - a) $y = 4$ (horizontal line)
   - b) $x = -2$ (vertical line)
2. For equation $x + y = 6$:
   - a) Find 4 solutions

- b) Plot the points
- c) Draw the line
- d) State intercepts

WEEK 8: GRAPHS II

TOPIC: Plotting Linear Graphs in Two Variables from Real-Life Situations; Solving Quantitative Reasoning Problems on Graphs

CONTENT:

A. REVIEW OF GRAPHING LINEAR EQUATIONS

Steps to Graph a Linear Equation:

3. Find at least 3 solutions (ordered pairs)
4. Create a table of values
5. Plot the points on Cartesian plane
6. Draw a straight line through points
7. Label the line with equation
8. Mark intercepts if required

B. GRAPHING LINEAR EQUATIONS IN STANDARD FORM

Example 1: Graph 2x + y = 6

Step 1: Find solutions (choose x-values, solve for y)

x	y = 6 - 2x	(x, y)
0	6 - 2(0) = 6	(0, 6)
1	6 - 2(1) = 4	(1, 4)
2	6 - 2(2) = 2	(2, 2)
3	6 - 2(3) = 0	(3, 0)

Step 2: Plot and draw line

```
 y
 |
7 |
```

6 |• (0,6)
5 | \
4 | • (1,4)
3 | \
2 | • (2,2)
1 | \
  | • (3,0)
——————————O——————————————————— x
1 2 3 4

Intercepts:

- y-intercept: (0, 6)
- x-intercept: (3, 0)

Example 2: Graph x + 2y = 8

Find solutions:

x	y = (8 - x)/2	(x, y)
0	(8 - 0)/2 = 4	(0, 4)
2	(8 - 2)/2 = 3	(2, 3)
4	(8 - 4)/2 = 2	(4, 2)
8	(8 - 8)/2 = 0	(8, 0)

Graph:

y
|
5 |
4 |• (0,4)
3 | \• (2,3)
2 | \• (4,2)

```
  1 |     \
    |       \• (8,0)
————————O———————————————— x
     2  4  6  8
```

Example 3: Graph 3x + 2y = 12

Using intercepts method (faster):

x-intercept (y = 0):

$3x + 2(0) = 12$

$3x = 12$

$x = 4$

Point: (4, 0)

y-intercept (x = 0):

$3(0) + 2y = 12$

$2y = 12$

$y = 6$

Point: (0, 6)

Third point (check): Let x = 2

$3(2) + 2y = 12$

$6 + 2y = 12$

$2y = 6$

$y = 3$

Point: (2, 3)

Plot: (0, 6), (2, 3), (4, 0)

C. GRAPHING IN SLOPE-INTERCEPT FORM (y = mx + c)

Where:

- m = slope (gradient)
- c = y-intercept

Example 4: Graph y = 2x + 1

Analysis:

- Slope (m) = 2 (rise 2, run 1)
- y-intercept (c) = 1, so starts at (0, 1)

Table:

x	y = 2x + 1	(x, y)
-1	2(-1) + 1 = -1	(-1, -1)
0	2(0) + 1 = 1	(0, 1)
1	2(1) + 1 = 3	(1, 3)
2	2(2) + 1 = 5	(2, 5)

Graph:

```
    y
    |
  6 |
  5 |     • (2,5)
  4 |    /
  3 |  • (1,3)
  2 |/
  1 |• (0,1)
    |
————————————O———————————— x
-2 -1  0  1  2  3
```

-1 • (-1,-1)
|

Example 5: Graph y = -x + 4

Analysis:

- Slope = -1 (negative slope, line falls)
- y-intercept = 4

Table:

x	y = -x + 4	(x, y)
0	-0 + 4 = 4	(0, 4)
1	-1 + 4 = 3	(1, 3)
2	-2 + 4 = 2	(2, 2)
4	-4 + 4 = 0	(4, 0)

Example 6: Graph y = 3x - 2

Table:

x	y = 3x - 2	(x, y)
0	3(0) - 2 = -2	(0, -2)
1	3(1) - 2 = 1	(1, 1)
2	3(2) - 2 = 4	(2, 4)
3	3(3) - 2 = 7	(3, 7)

D. REAL-LIFE APPLICATIONS OF LINEAR GRAPHS

Application 1: Cost and Pricing

Example 7: A taxi charges ₦200 flat rate plus ₦50 per kilometer.

a) Write equation for total cost (C) based on distance (d)

C = 50d + 200

b) Create table:

Distance (km)	Cost (₦)	Point
0	200	(0, 200)
2	300	(2, 300)
4	400	(4, 400)
6	500	(6, 500)
10	700	(10, 700)

c) Graph:

```
Cost (₦)
   |
800 |
700 |            • (10, 700)
600 |
500 |        • (6, 500)
400 |    • (4, 400)
300 |  • (2, 300)
200 |• (0, 200)
100 |
   |
————————————O———————————————————— Distance (km)
     2  4  6  8  10
```

d) Use graph to answer:

- Cost for 5 km? ₦450 (from graph)

- Distance for ₦600? 8 km

Application 2: Temperature Conversion

Example 8: Convert Celsius to Fahrenheit: F = (9/5)C + 32

Simplified: F = 1.8C + 32

Table:

Celsius (°C)	Fahrenheit (°F)	Point
0	32	(0, 32)
10	50	(10, 50)
20	68	(20, 68)
30	86	(30, 86)
40	104	(40, 104)

Use graph to:

- Convert 25°C to °F
- Convert 77°F to °C

Application 3: Simple Interest

Example 9: Interest on ₦10,000 at 5% per annum

Formula: I = (10,000 × 5 × T)/100 = 500T

Table:

Time (years)	Interest (₦)	Point
0	0	(0, 0)
1	500	(1, 500)

Time (years)	Interest (₦)	Point
2	1,000	(2, 1000)
3	1,500	(3, 1500)
5	2,500	(5, 2500)

From graph, find:

- Interest after 4 years
- Time to earn ₦2,000 interest

Application 4: Distance-Time Graphs

Example 10: A car travels at constant speed of 60 km/h

Equation: Distance = 60 × Time **D = 60T**

Table:

Time (hours)	Distance (km)	Point
0	0	(0, 0)
1	60	(1, 60)
2	120	(2, 120)
3	180	(3, 180)
4	240	(4, 240)

Graph:

```
Distance (km)
   |
250 |          •
200 |        /
150 |      /
100 |    /
```

```
 50 |  /
    |/
————————O—————————————— Time (hours)
      1  2  3  4
```

Slope = Speed = 60 km/h

Application 5: Mobile Phone Plans

Example 11:

- Plan A: ₦500/month + ₦10 per minute
- Plan B: ₦800/month + ₦5 per minute

Equations:

- Plan A: C = 10m + 500
- Plan B: C = 5m + 800

Where: m = minutes used, C = total cost

Tables:

Plan A:

Minutes	Cost (₦)
0	500
20	700
40	900
60	1100

Plan B:

Minutes	Cost (₦)

Minutes	Cost (₦)
0	800
20	900
40	1000
60	1100

Graph both on same axes:

Questions:

- Which plan is cheaper for 30 minutes?
- At how many minutes do plans cost the same?
- Which plan for heavy users?

E. READING AND INTERPRETING GRAPHS

Example 12: Given graph of C = 3d + 50

```
Cost (₦)
   |
200 |        •
150 |      /
100 |    /
 50 |  /•
    |/
————————————O——————————————————— d
    10 20 30 40 50
```

Interpret:

- What does 50 represent? **Initial cost**
- What does 3 represent? **Cost per unit of d**
- Find cost when d = 25: **₦125**

- Find d when cost = ₦170: **d = 40**

Example 13: Distance-Time Graph Interpretation

```
   Distance (km)
      |
   120 |    ————————
    80 |   /      \
    40 |  /        \
      | /         \
——————————O———————————————————— Time (hours)
        1  2  3  4  5
```

Interpret:

- 0-1 hour: Traveling (rising line)
- 1-3 hours: Stationary (flat line)
- 3-5 hours: Returning (falling line)
- Average speed 0-1 hour: 80 km/h
- Total distance: 80 km one way

F. SIMULTANEOUS EQUATIONS (GRAPHICAL SOLUTION)

Definition: Finding values of x and y that satisfy two equations simultaneously.

Graphical Method:

- Graph both equations
- Point of intersection is the solution

Example 14: Solve graphically:

$x + y = 5$

$x - y = 1$

Equation 1: x + y = 5

x	y	Point
0	5	(0, 5)
2	3	(2, 3)
5	0	(5, 0)

Equation 2: x - y = 1

x	y	Point
0	-1	(0, -1)
1	0	(1, 0)
3	2	(3, 2)

Graph both lines:

```
   y
   |
 5 |•
 4 | \
 3 |  ×• (3,2) [Intersection]
 2 |  /\
 1 | /  \
   |/    \
——————————O——————————• x
-1 |•  1  2  3  4  5
   |
```

Solution: x = 3, y = 2

Check:

x + y = 5: 3 + 2 = 5 ✓

x - y = 1: 3 - 2 = 1 ✓

Example 15: Solve graphically:

y = 2x + 1

y = -x + 7

Graph both and find intersection point

Solution will be at the point where both lines meet

G. USING GRAPHS FOR PREDICTIONS

Example 16: Population Growth

Given graph showing town population over years:

```
   Population
      |
   5000|          •
   4000|        /
   3000|      /
   2000|  /
   1000|/
      |
 ————————————O——————————————————————— Year
     2020 2021 2022 2023 2024
```

Predict:

- Population in 2025: **Extend line**
- When will population reach 6000?: **Extend line to y = 6000**

H. QUANTITATIVE REASONING WITH GRAPHS

Example 17: A shop sells notebooks at ₦150 each. Fixed costs are ₦3,000.

a) Revenue equation: R = 150n (where n = number of notebooks) **b) Cost equation:** C = 3,000 (fixed) **c) Profit equation:** P = 150n - 3,000

Graph:

```
Amount (₦)
   |
6000|        /
5000|      / Revenue
4000|    /
3000|———————————————— Cost
2000|  /
1000|/
   |
——————————O————————————————— Notebooks
     10 20 30 40
```

Break-even point: Where Revenue = Cost

150n = 3,000
n = 20 notebooks

Example 18: Water Tank Problem

Tank has 100 liters. Water drains at 5 liters/minute.

Equation: V = 100 - 5t

Time (min)	Volume (L)
0	100

Time (min)	Volume (L)
5	75
10	50
15	25
20	0

Questions from graph:

- How long to half-empty?
- Volume after 8 minutes?
- When will tank be empty?

EVALUATION:

1. Graph y = 2x + 3 (use x = -1, 0, 1, 2)
2. Find intercepts of 3x + y = 9
3. A phone call costs ₦20 plus ₦5 per minute. Write equation and graph
4. From a distance-time graph, how do you find speed?
5. Graph x + y = 7 and x - y = 1 on same axes. Find intersection point
6. What does the slope represent in a cost graph?
7. Interpret: A flat horizontal line on distance-time graph
8. A tank drains from 80L to 0L in 10 minutes. Write equation
9. Graph y = -2x + 6
10. At what point do lines y = x + 2 and y = -x + 6 intersect?

REVISION QUESTIONS:

1. What are the steps to graph a linear equation?
2. How do you find x-intercept?
3. What is the slope-intercept form?

4. In $y = mx + c$, what does m represent?
5. How do you solve simultaneous equations graphically?
6. What does a steep slope indicate?
7. What does a negative slope look like?
8. How do you use a graph for prediction?
9. What is a break-even point?
10. Describe distance-time graph when stationary
11. In cost equation $C = 5n + 100$, what does 100 represent?
12. How do you graph $x = 3$?
13. What does intersection point of two lines represent?
14. How do you read values from a graph?
15. What's the advantage of using graphs over tables?

ASSIGNMENT:

Section A: Graphing Equations (30 marks)

Graph the following on separate Cartesian planes:

1. $x + y = 8$ (find 4 points)
2. $2x + y = 10$ (use intercepts method)
3. $y = 3x - 2$ (find 4 points)
4. $x - y = 4$ (find 4 points)
5. $y = -2x + 6$ (find 4 points)
6. $3x + 2y = 12$ (use intercepts)

Section B: Real-Life Applications (30 marks)

1. **Taxi Problem:** A taxi charges ₦300 plus ₦60 per km.
    - a) Write equation for cost (C) vs distance (d)
    - b) Complete table for d = 0, 2, 4, 6, 8

- c) Draw graph
- d) Use graph to find cost for 5 km
- e) Use graph to find distance for ₦720

2. **Mobile Phone:** Monthly cost = ₦600 + ₦8 per minute
   - a) Write equation
   - b) Create table (0, 20, 40, 60, 80 minutes)
   - c) Graph the relationship
   - d) Find cost for 50 minutes
   - e) Find minutes for ₦1,000
3. **Water Tank:** Tank starts with 120 liters, drains at 6 L/min
   - a) Write equation for volume (V) vs time (t)
   - b) Table for t = 0, 5, 10, 15, 20
   - c) Graph
   - d) When is tank half-empty?
   - e) When is tank empty?

Section C: Intercepts and Slope (15 marks)

Find x-intercept, y-intercept, and slope:

1. $x + y = 10$
2. $2x + 3y = 12$
3. $y = 4x - 8$
4. $3x - y = 9$
5. $y = -3x + 12$

Section D: Simultaneous Equations (15 marks)

Solve graphically (graph both on same axes):

1. $x + y = 6$ $x - y = 2$
2. $y = 2x + 1$ $y = -x + 7$

3. $2x + y = 8$ $x + y = 5$

Section E: Interpretation (10 marks)

[Provide distance-time graph showing journey with stops]

From the graph, answer:

1. Total distance traveled
2. Total time taken
3. When was the person stationary?
4. Average speed for first hour
5. Describe the complete journey

WEEK 9: PLANE FIGURES/SHAPES

TOPIC: Identifying Plane Shapes in the Environment; Properties of Plane Shapes (Square, Rectangle, Parallelogram, Rhombus, Kite)

CONTENT:

A. INTRODUCTION TO PLANE FIGURES

Definition: A plane figure (or plane shape) is a flat, two-dimensional shape that lies entirely in one plane. It has length and width but no thickness.

Characteristics:

- Two-dimensional (2D)
- Flat surface
- Has area and perimeter
- Cannot be held or have volume

Examples of Plane Figures:

- Triangles
- Quadrilaterals (4-sided shapes)
- Circles
- Polygons

Focus: Quadrilaterals (4-sided polygons)

B. QUADRILATERALS

Definition: A quadrilateral is a polygon with four sides, four angles, and four vertices.

Properties of ALL Quadrilaterals:

1. Four sides
2. Four angles

3. Four vertices (corners)
4. Sum of interior angles = 360°
5. Two diagonals

Types of Quadrilaterals:

1. Square
2. Rectangle
3. Parallelogram
4. Rhombus
5. Trapezium
6. Kite

C. SQUARE

Definition: A square is a quadrilateral with all sides equal and all angles equal to 90°.

Properties:

1. **All sides are equal** (AB = BC = CD = DA)
2. **All angles are 90°** (right angles)
3. **Opposite sides are parallel**
4. **Diagonals are equal** in length
5. **Diagonals bisect each other at 90°**
6. **Diagonals bisect the angles** (divide into 45° each)
7. Has **4 lines of symmetry**
8. Has **rotational symmetry** of order 4

Diagram:

```
D————————————C
|     |
|     |
```

| |
A————————B

All sides equal: AB = BC = CD = DA

All angles = 90°

Formulas:

- **Perimeter:** P = 4s (where s = side length)
- **Area:** $A = s^2$
- **Diagonal:** $d = s\sqrt{2}$

Example 1: Square with side 5 cm

Perimeter = 4 × 5 = 20 cm

Area = $5^2 = 25\ cm^2$

Diagonal = $5\sqrt{2} \approx 7.07$ cm

Real-Life Examples:

- Chess board squares
- Floor tiles
- Window panes
- Checkerboard
- Book covers (sometimes)

D. RECTANGLE

Definition: A rectangle is a quadrilateral with opposite sides equal and all angles equal to 90°.

Properties:

9. **Opposite sides are equal** (AB = CD, BC = DA)
10. **All angles are 90°**
11. **Opposite sides are parallel**

12. **Diagonals are equal** in length
13. **Diagonals bisect each other** (but not at 90°)
14. Has **2 lines of symmetry** (vertical and horizontal)
15. Has **rotational symmetry** of order 2

Diagram:

```
D————————————————————C
|           |
|           |
A————————————————————B
```

Length: AB = DC (longer sides)
Width: BC = AD (shorter sides)
All angles = 90°

Formulas:

- **Perimeter:** $P = 2(l + w)$ or $P = 2l + 2w$
- **Area:** $A = l \times w$
- **Diagonal:** $d = \sqrt{(l^2 + w^2)}$

Example 2: Rectangle with length 8 cm and width 5 cm

Perimeter = $2(8 + 5) = 2(13) = 26$ cm
Area = $8 \times 5 = 40$ cm²
Diagonal = $\sqrt{(8^2 + 5^2)} = \sqrt{(64 + 25)} = \sqrt{89} \approx 9.43$ cm

Real-Life Examples:

- Books
- Doors
- Tables
- Paper (A4, etc.)

- Computer screens
- Football pitch

Difference from Square:

- Rectangle: Only opposite sides equal
- Square: All sides equal (special type of rectangle)

E. PARALLELOGRAM

Definition: A parallelogram is a quadrilateral with opposite sides parallel and equal.

Properties:

16. **Opposite sides are parallel** (AB ∥ DC, AD ∥ BC)
17. **Opposite sides are equal** (AB = DC, AD = BC)
18. **Opposite angles are equal** (∠A = ∠C, ∠B = ∠D)
19. **Adjacent angles are supplementary** (sum to 180°)
20. **Diagonals bisect each other** (but not equal, not at 90°)
21. **No lines of symmetry** (unless it's a rectangle)
22. Has **rotational symmetry** of order 2

Diagram:

```
   D————————————C
  /        /
 /        /
/        /
A————————————B
```

AB ∥ DC (parallel)

AD ∥ BC (parallel)

AB = DC, AD = BC

Formulas:

- **Perimeter:** P = 2(a + b) where a and b are adjacent sides
- **Area:** A = base × height (perpendicular height)
- **Area:** A = b × h

Example 3: Parallelogram with base 10 cm, side 6 cm, height 4 cm

Perimeter = 2(10 + 6) = 32 cm
Area = base × height = 10 × 4 = 40 cm^2

Real-Life Examples:

- Slanted roof supports
- Conveyor belt systems
- Some bridge designs
- Erasers (some shapes)
- Tiles (some patterns)

Key Point: Height is PERPENDICULAR distance between parallel sides, not the slant side length.

F. RHOMBUS

Definition: A rhombus is a parallelogram with all sides equal.

Properties:

1. **All sides are equal** (AB = BC = CD = DA)
2. **Opposite sides are parallel**
3. **Opposite angles are equal**
4. **Diagonals bisect each other at 90°** (right angles)
5. **Diagonals bisect the angles**
6. Has **2 lines of symmetry** (along diagonals)

7. Has **rotational symmetry** of order 2
8. Angles are NOT necessarily 90° (if they are, it's a square)

Diagram:

```
     D
    /|\
   / | \
  /  |  \
 A———+———C
  \  |  /
   \ | /
    \|/
     B
```

All sides equal: AB = BC = CD = DA
Diagonals AC and BD bisect at 90°

Formulas:

- **Perimeter:** $P = 4s$ (where s = side length)
- **Area:** $A = (d_1 \times d_2)/2$ (where d_1, d_2 are diagonals)
- **Area:** $A = \text{base} \times \text{height}$

Example 4: Rhombus with side 6 cm, diagonals 8 cm and 10 cm

Perimeter = 4 × 6 = 24 cm
Area = (8 × 10)/2 = 80/2 = 40 cm²

Real-Life Examples:

- Diamond shape (hence name "diamond")
- Kite shapes (some)
- Road signs (some)
- Jewelry designs

- Pattern designs

Difference from Square:

- Rhombus: Angles not necessarily 90°
- Square: All angles are 90° (special type of rhombus)

Difference from Parallelogram:

- Rhombus: All sides equal
- Parallelogram: Only opposite sides equal

G. KITE

Definition: A kite is a quadrilateral with two pairs of adjacent sides equal.

Properties:

9. **Two pairs of adjacent sides are equal** (AB = AD, CB = CD)
10. **One pair of opposite angles are equal** (the angles between unequal sides)
11. **Diagonals intersect at 90°** (perpendicular)
12. **One diagonal bisects the other**
13. **One diagonal is the axis of symmetry**
14. Has **1 line of symmetry**
15. **No rotational symmetry** (order 1)

Diagram:

```
    D
   /|\
  / | \
 /  |  \
A   |   C
 \  |  /
  \ | /
```

\|/
B

AD = AB (one pair)
CD = CB (another pair)
Diagonal DB bisects AC at 90°

Formulas:

- **Perimeter:** P = 2(a + b) where a and b are the unequal adjacent sides
- **Area:** $A = (d_1 \times d_2)/2$ (where d_1, d_2 are diagonals)

Example 5: Kite with sides 5 cm and 7 cm (each pair), diagonals 8 cm and 6 cm

Perimeter = 2(5 + 7) = 2(12) = 24 cm
Area = (8 × 6)/2 = 48/2 = 24 cm²

Real-Life Examples:

- Flying kites
- Arrowheads
- Some gemstone cuts
- Decorative designs
- Some logos

H. COMPARISON TABLE

Property	Square	Rectangle	Parallelogram	Rhombus	Kite
All sides equal	Yes	No	No	Yes	No
Opposite sides equal	Yes	Yes	Yes	Yes	No
Adjacent sides equal	Yes	No	No	Yes	Yes (2 pairs)
All angles 90°	Yes	Yes	No	No	No

Property	Square	Rectangle	Parallelogram	Rhombus	Kite
Opposite angles equal	Yes	Yes	Yes	Yes	One pair
Opposite sides parallel	Yes	Yes	Yes	Yes	No
Diagonals equal	Yes	Yes	No	No	No
Diagonals bisect each other	Yes	Yes	Yes	Yes	One bisects other
Diagonals perpendicular	Yes	No	No	Yes	Yes
Lines of symmetry	4	2	0	2	1
Rotational symmetry order	4	2	2	2	1

I. IDENTIFYING SHAPES IN THE ENVIRONMENT

Classroom Objects:

1. Floor tiles - Square/Rectangle
2. Whiteboard - Rectangle
3. Window panes - Square/Rectangle
4. Book covers - Rectangle
5. Desk tops - Rectangle

Home Objects:

1. Doors - Rectangle
2. Tiles - Square
3. Mirrors - Square/Rectangle
4. Beds - Rectangle
5. TVs/Monitors - Rectangle

Outdoor Objects:

1. Football pitch - Rectangle
2. Building facades - Rectangle
3. Road signs - Various shapes

4. Kites - Kite shape
5. Windows - Square/Rectangle

J. CALCULATING PERIMETER AND AREA

Example 6: Square courtyard with side 15 m

a) Perimeter = $4 \times 15 = 60$ m

b) Area = $15^2 = 225$ m²

c) Cost to fence at ₦500/m:
= 60×500 = ₦30,000

d) Cost to tile at ₦2,000/m²:
= $225 \times 2{,}000$ = ₦450,000

Example 7: Rectangular field 50 m by 30 m

a) Perimeter = $2(50 + 30) = 160$ m

b) Area = $50 \times 30 = 1{,}500$ m²

c) Diagonal = $\sqrt{(50^2 + 30^2)}$
$= \sqrt{(2{,}500 + 900)}$
$= \sqrt{3{,}400}$
≈ 58.3 m

Example 8: Rhombus-shaped garden with diagonals 12 m and 16 m

a) Area = $(12 \times 16)/2$
$= 192/2$
$= 96$ m²

b) If each side is 10 m:
Perimeter = 4 × 10 = 40 m

Example 9: Parallelogram-shaped plot with base 20 m, side 15 m, height 12 m

a) Perimeter = 2(20 + 15) = 70 m

b) Area = base × height
= 20 × 12
= 240 m²

Example 10: Kite-shaped logo with sides 8 cm and 10 cm, diagonals 12 cm and 14 cm

a) Perimeter = 2(8 + 10) = 36 cm

b) Area = (12 × 14)/2 = 84 cm²

K. PROBLEM SOLVING WITH SHAPES

Example 11: A rectangular garden is 24 m long and 18 m wide. A path 2 m wide runs around the inside. Find the area of the path.

Solution:
Outer rectangle: 24 m × 18 m
Area = 24 × 18 = 432 m²

Inner rectangle (minus path):
Length = 24 - 2(2) = 20 m
Width = 18 - 2(2) = 14 m
Area = 20 × 14 = 280 m²

Path area = 432 - 280 = 152 m²

Example 12: A square field has perimeter 120 m. Find: a) Side length b) Area c) Diagonal

Solution:

a) Perimeter = 4s

$120 = 4s$

$s = 30$ m

b) Area = s^2

$= 30^2$

$= 900\ m^2$

c) Diagonal = $s\sqrt{2}$

$= 30\sqrt{2}$

≈ 42.43 m

EVALUATION:

1. Define a quadrilateral
2. List 5 properties of a square
3. What's the difference between a square and a rhombus?
4. A rectangle has length 12 cm and width 8 cm. Find perimeter and area
5. State 3 properties of a parallelogram
6. How many lines of symmetry does a kite have?
7. A square has area 64 cm^2. Find its perimeter
8. What's the formula for area of a rhombus using diagonals?
9. Give 3 real-life examples of rectangles
10. All angles in a rectangle are __°

REVISION QUESTIONS:

1. What is a plane figure?
2. What is the sum of angles in any quadrilateral?

3. How many diagonals does a quadrilateral have?
4. Is every square a rectangle? Explain
5. Is every rectangle a square? Explain
6. Is every rhombus a parallelogram? Explain
7. What's unique about diagonals of a rhombus?
8. Which quadrilaterals have all sides equal?
9. Which quadrilaterals have perpendicular diagonals?
10. A square has perimeter 48 cm. Find area
11. What's the difference between a parallelogram and a rectangle?
12. How many lines of symmetry does a parallelogram have?
13. In which quadrilaterals are opposite sides parallel?
14. What shape is a square that has been "pushed over"?
15. Can a kite be a rhombus? Explain

ASSIGNMENT:

Section A: Definitions and Properties (20 marks)

1. Define and sketch:
   - a) Square
   - b) Rectangle
   - c) Parallelogram
   - d) Rhombus
   - e) Kite
2. List 5 properties of:
   - a) Square
   - b) Rectangle
   - c) Rhombus
3. State the number of lines of symmetry for each shape

4. State the order of rotational symmetry for each shape

Section B: Calculations - Square (15 marks)

5. Square with side 9 cm. Find:
   - a) Perimeter
   - b) Area
   - c) Diagonal
6. Square with perimeter 52 cm. Find:
   - a) Side length
   - b) Area
7. Square with area 144 cm². Find:
   - a) Side length
   - b) Perimeter
8. Square field has diagonal $20\sqrt{2}$ m. Find:
   - a) Side length
   - b) Area

Section B: Calculations - Rectangle (15 marks)

1. Rectangle 15 cm by 10 cm. Find:
   - a) Perimeter
   - b) Area
   - c) Diagonal
2. Rectangle has perimeter 46 cm and length 14 cm. Find:
   - a) Width
   - b) Area
3. Rectangle has area 120 cm² and width 8 cm. Find:
   - a) Length

- b) Perimeter

4. Rectangular field 50 m by 40 m. Find:
    - a) Cost to fence at ₦450/m
    - b) Cost to grass at ₦1,500/m²

Section D: Calculations - Parallelogram (10 marks)

1. Parallelogram with base 12 cm, side 8 cm, height 6 cm. Find:
    - a) Perimeter
    - b) Area
2. Parallelogram has area 96 cm² and base 12 cm. Find height
3. Parallelogram sides 10 cm and 14 cm. Find perimeter

Section E: Calculations - Rhombus (15 marks)

1. Rhombus with side 7 cm, diagonals 10 cm and 12 cm. Find:
    - a) Perimeter
    - b) Area
2. Rhombus has diagonals 16 cm and 12 cm. Find area
3. Rhombus has perimeter 36 cm. Find side length
4. Rhombus has area 60 cm² and one diagonal 10 cm. Find other diagonal

Section F: Calculations - Kite (10 marks)

1. Kite with sides 6 cm and 9 cm, diagonals 8 cm and 12 cm. Find:
    - a) Perimeter
    - b) Area
2. Kite has diagonals 14 cm and 10 cm. Find area
3. Kite has sides 5 cm and 8 cm. Find perimeter

Section G: Word Problems (15 marks)

1. A square carpet has perimeter 32 m. Find its area.
2. A rectangular room is 6 m long and 4 m wide. Find:
    - a) Area
    - b) Cost to tile at ₦3,500/m²
3. A rhombus-shaped signboard has diagonals 80 cm and 60 cm. Find its area.
4. A rectangular garden 25 m by 20 m has a path 2 m wide around the inside. Find path area.
5. A square field has area 625 m². Find its perimeter.
6. A parallelogram-shaped plot has base 30 m and height 18 m. Find area.
7. A rectangular field is twice as long as it is wide. If width is 15 m, find:
    - a) Length
    - b) Perimeter
    - c) Area

WEEK 10: SCALE DRAWING OF LENGTH AND DISTANCES

TOPIC: Meaning and Purpose of Scale Drawing; Measurement and Representation on Paper

CONTENT:

A. INTRODUCTION TO SCALE DRAWING

Definition: A scale drawing is a reduced or enlarged representation of an actual object or distance that maintains accurate proportions.

Why Use Scale Drawings?

- To represent large objects/distances on paper (buildings, maps, etc.)
- To represent small objects in larger form (cells, insects, etc.)
- To maintain accurate proportions
- To communicate designs and plans
- To measure actual distances from drawings

Real-Life Applications:

- Maps and atlases
- Architectural plans
- Engineering blueprints
- Model cars, planes, buildings
- Garden designs
- Room layouts

B. UNDERSTANDING SCALE

Definition of Scale: A scale is the ratio of the length on a drawing to the actual length of the object.

Format: Scale is written as:

- **Ratio form:** 1:100 (read as "1 to 100")

- **Fraction form:** 1/100
- **Statement form:** "1 cm represents 100 cm" or "1 cm = 1 m"

Meaning:

- Scale 1:100 means:
    - 1 cm on drawing = 100 cm in reality
    - 1 unit on drawing = 100 units in reality

C. TYPES OF SCALES

1. Reduction Scale (Most common)

- Drawing is smaller than actual object
- Scale ratio: 1:n where n > 1
- Examples: 1:10, 1:50, 1:100, 1:1000

Example: 1:50 scale

- 1 cm on paper = 50 cm in reality

2. Enlargement Scale (Less common)

- Drawing is larger than actual object
- Scale ratio: n:1 where n > 1
- Examples: 10:1, 100:1, 1000:1
- Used for: microscopic objects, small insects, etc.

Example: 10:1 scale

- 10 cm on paper = 1 cm in reality

3. Full Scale (1:1)

- Drawing is same size as object
- Rarely used except for small objects

D. COMMON SCALES USED

Building Plans:

- 1:50 (1 cm = 50 cm = 0.5 m)
- 1:100 (1 cm = 100 cm = 1 m)
- 1:200 (1 cm = 200 cm = 2 m)

Maps:

- 1:10,000 (1 cm = 10,000 cm = 100 m)
- 1:50,000 (1 cm = 50,000 cm = 500 m = 0.5 km)
- 1:100,000 (1 cm = 100,000 cm = 1,000 m = 1 km)
- 1:1,000,000 (1 cm = 10 km)

Models:

- Model cars: 1:24, 1:32, 1:64
- Model planes: 1:72, 1:48
- Architectural models: 1:200, 1:500

E. CALCULATING ACTUAL LENGTH FROM SCALE DRAWING

Formula: Actual Length = Drawing Length × Scale Factor

Example 1: Scale 1:100 A line measures 5 cm on drawing. Find actual length.

Scale 1:100 means 1 cm = 100 cm
Actual length = 5 × 100 = 500 cm = 5 m

Example 2: Scale 1:50 Drawing shows 8 cm. Find actual length.

1 cm = 50 cm
Actual length = 8 × 50 = 400 cm = 4 m

Example 3: Scale 1:200 On a plan, a room measures 3 cm by 2.5 cm. Find actual dimensions.

Length: 3 × 200 = 600 cm = 6 m

Width: 2.5 × 200 = 500 cm = 5 m

Actual room: 6 m by 5 m

Example 4: Map scale 1:10,000 Distance on map is 4.5 cm. Find actual distance.

Actual distance = 4.5 × 10,000 cm

= 45,000 cm

= 450 m

= 0.45 km

Example 5: Map scale 1:50,000 Two towns are 7 cm apart on map. Find actual distance.

Actual distance = 7 × 50,000 cm

= 350,000 cm

= 3,500 m

= 3.5 km

F. CALCULATING DRAWING LENGTH FROM ACTUAL LENGTH

Formula: Drawing Length = Actual Length ÷ Scale Factor

Example 6: Scale 1:100 Actual length is 12 m. Find drawing length.

Convert to same units: 12 m = 1,200 cm

Drawing length = 1,200 ÷ 100 = 12 cm

Example 7: Scale 1:50 A wall is 8 m long. How long on drawing?

8 m = 800 cm

Drawing length = 800 ÷ 50 = 16 cm

Example 8: Scale 1:200 Garden is 30 m by 20 m. Find dimensions on plan.

Length: 3,000 cm ÷ 200 = 15 cm

Width: 2,000 cm ÷ 200 = 10 cm

On plan: 15 cm by 10 cm

Example 9: Map scale 1:25,000 Actual distance is 5 km. Find map distance.

5 km = 5,000 m = 500,000 cm

Map distance = 500,000 ÷ 25,000 = 20 cm

Example 10: Scale 1:1000 Distance between two points is 750 m. Find drawing distance.

750 m = 75,000 cm

Drawing distance = 75,000 ÷ 1,000 = 75 cm

G. FINDING THE SCALE

Formula: Scale = Drawing Length : Actual Length (Both must be in same units)

Example 11: Drawing length is 6 cm, actual length is 300 cm. Find scale.

Scale = 6 : 300

= 1 : 50 (divide both by 6)

Scale is 1:50

Example 12: On a plan, 4 cm represents 8 m. Find scale.

Convert to same units: 8 m = 800 cm

Scale = 4 : 800

= 1 : 200 (divide both by 4)

Scale is 1:200

Example 13: A map shows 5 cm for actual distance of 2.5 km. Find scale.

2.5 km = 2,500 m = 250,000 cm

Scale = 5 : 250,000

= 1 : 50,000 (divide both by 5)

Scale is 1:50,000

Example 14: Drawing: 8 cm, Actual: 4 m. Find scale.

4 m = 400 cm

Scale = 8 : 400

= 1 : 50

Scale is 1:50

H. SCALE DRAWINGS - PRACTICAL EXAMPLES

Example 15: Room Plan (Scale 1:50)

Given: Actual room 7 m long, 5 m wide

Step 1: Calculate drawing dimensions

Length: 700 cm ÷ 50 = 14 cm

Width: 500 cm ÷ 50 = 10 cm

Step 2: Draw rectangle 14 cm × 10 cm

Step 3: Add features (door 1 m wide, window 1.5 m)

Door: 100 cm ÷ 50 = 2 cm

Window: 150 cm ÷ 50 = 3 cm

Step 4: Label: "Scale 1:50" and dimensions

Example 16: Garden Design (Scale 1:100)

Actual garden: 20 m by 15 m with:

- Path 2 m wide along one side
- Flower bed 4 m × 3 m

- Lawn area (remaining)

Drawing dimensions:

Garden: 2,000 cm ÷ 100 = 20 cm by 15 cm

Path: 200 cm ÷ 100 = 2 cm wide

Flower bed: 4 cm × 3 cm

Draw and label all sections

Example 17: Classroom Floor Plan (Scale 1:50)

Actual dimensions:

- Room: 10 m × 8 m
- Teacher's desk: 1.5 m × 1 m
- Student desks: 1.2 m × 0.6 m (6 desks)
- Door: 1 m wide
- Windows: 2 m wide (2 windows)

Calculate drawing dimensions:

Room: 20 cm × 16 cm

Teacher's desk: 3 cm × 2 cm

Student desks: 2.4 cm × 1.2 cm

Door: 2 cm

Windows: 4 cm each

I. AREA CALCULATIONS FROM SCALE DRAWINGS

Key Concept: Area scale is the square of length scale.

If scale is 1:n, then area scale is 1:n^2

Example 18: Scale 1:100 Rectangle on drawing: 5 cm × 4 cm Find actual area.

Method 1: Find actual dimensions first

Actual length = $5 \times 100 = 500$ cm = 5 m

Actual width = $4 \times 100 = 400$ cm = 4 m

Actual area = $5 \times 4 = 20\ m^2$

Method 2: Use area scale

Drawing area = $5 \times 4 = 20\ cm^2$

Area scale = $1:100^2 = 1:10{,}000$

Actual area = $20 \times 10{,}000\ cm^2 = 200{,}000\ cm^2 = 20\ m^2$

Example 19: Scale 1:50 Drawing shows room with area 12 cm^2. Find actual area.

Area scale = $1:50^2 = 1:2{,}500$

Actual area = $12 \times 2{,}500 = 30{,}000\ cm^2 = 3\ m^2$

Example 20: Scale 1:200 Field on plan measures 8 cm × 6 cm. Find actual area.

Drawing area = 48 cm^2

Area scale = 1:40,000

Actual area = $48 \times 40{,}000\ cm^2 = 1{,}920{,}000\ cm^2 = 192\ m^2$

OR:

Actual length = $8 \times 200 = 1{,}600$ cm = 16 m

Actual width = $6 \times 200 = 1{,}200$ cm = 12 m

Actual area = $16 \times 12 = 192\ m^2$

J. MAP READING AND DISTANCE CALCULATIONS

Example 21: Map scale 1:50,000

Two villages are 8 cm apart on map.

a) Actual distance:

$8 \times 50{,}000 = 400{,}000$ cm = 4 km

b) If traveling at 60 km/h, time taken:

Time = $4 \div 60 = 0.067$ hours = 4 minutes

Example 22: Map scale 1:25,000

A road measures 12 cm on map.

a) Actual length:

$12 \times 25{,}000 = 300{,}000$ cm = 3 km

b) If walking at 5 km/h:

Time = $3 \div 5 = 0.6$ hours = 36 minutes

Example 23: Map scale 1:100,000

Town A to Town B is 15 km. How far on map?

15 km = 1,500,000 cm

Map distance = $1{,}500{,}000 \div 100{,}000 = 15$ cm

K. CREATING SCALE DRAWINGS

Steps to Create a Scale Drawing:

1. **Choose appropriate scale**
   - Consider paper size
   - Consider actual dimensions
   - Common scales: 1:50, 1:100, 1:200
2. **Calculate drawing dimensions**
   - Convert actual to same units

   - Divide by scale factor
3. **Draw using ruler and pencil**
   - Use sharp pencil
   - Draw lightly first
   - Check measurements
4. **Add details and labels**
   - Mark dimensions
   - Show scale
   - Add title
5. **Make neat and clear**
   - Use straight lines
   - Label clearly
   - Show orientation (N, S, E, W if needed)

Example 24: Draw a scale plan of rectangular plot 40 m × 30 m at scale 1:500

Step 1: Choose scale (given): 1:500

Step 2: Calculate dimensions

Length: 4,000 cm ÷ 500 = 8 cm
Width: 3,000 cm ÷ 500 = 6 cm

Step 3: Draw rectangle 8 cm × 6 cm

Step 4: Label

- Mark "40 m" and "30 m"
- Write "Scale 1:500"
- Add title "Plot Plan"

L. COMMON MISTAKES TO AVOID

1. **Not converting to same units**

   Wrong: Scale 1:100, actual 5 m

      Drawing = 5 ÷ 100 = 0.05 m ✗

   Right: Scale 1:100, actual 5 m = 500 cm

      Drawing = 500 ÷ 100 = 5 cm ✓

2. **Confusing which to divide/multiply**

   - Actual to drawing: DIVIDE by scale
   - Drawing to actual: MULTIPLY by scale

3. **Forgetting to simplify scale**

   Scale 6:300 should be simplified to 1:50

4. **Wrong area scale**

   If length scale is 1:100

   Area scale is 1:10,000 (not 1:100)

M. PRACTICAL APPLICATIONS

Application 1: Building Plans

An architect draws a house plan at 1:100.

- Living room: 6 m × 5 m → 6 cm × 5 cm on plan
- Bedroom: 4 m × 3.5 m → 4 cm × 3.5 cm on plan
- Kitchen: 3 m × 2.5 m → 3 cm × 2.5 cm on plan

Application 2: Maps

A town map at 1:10,000 shows:

- School to hospital: 5 cm on map = 500 m actual
- Hospital to market: 3.5 cm = 350 m actual
- Total journey: 850 m

Application 3: Models

Model car at scale 1:32

- Actual car length: 4.8 m
- Model length: 480 cm ÷ 32 = 15 cm

EVALUATION:

1. Define scale drawing
2. What is the purpose of using scales?
3. Scale 1:50 means 1 cm represents ___
4. Drawing shows 8 cm, scale 1:100. Find actual length
5. Actual length 12 m, scale 1:200. Find drawing length
6. If scale is 1:100, what is the area scale?
7. On a plan, 3 cm represents 6 m. Find the scale
8. Map scale 1:50,000. Towns are 6 cm apart. Find actual distance
9. Room is 8 m × 6 m. At scale 1:50, find drawing dimensions
10. Why must both lengths be in same units when finding scale?

REVISION QUESTIONS:

1. What is a scale?
2. Give 3 examples of where scale drawings are used
3. What is a reduction scale?
4. What is an enlargement scale?
5. Convert 1:100 scale to statement form
6. Scale 1:200, drawing 5 cm. Find actual length

7. Actual 10 m, scale 1:50. Find drawing length
8. How do you find the scale if given both lengths?
9. If length scale is 1:100, what is area scale?
10. Scale 1:1000, drawing area 6 cm². Find actual area
11. Why is scale 1:50 more detailed than 1:100?
12. Map scale 1:25,000. What does 1 cm represent?
13. Drawing: 4 cm, Actual: 2 m. Find scale
14. What are common scales for building plans?
15. What are common scales for maps?

ASSIGNMENT:

Section A: Understanding Scale (15 marks)

1. Explain what scale 1:100 means
2. Write in statement form:
    - a) 1:50
    - b) 1:200
    - c) 1:1000
    - d) 1:25,000
3. Which shows more detail: 1:50 or 1:100? Explain
4. Give 5 real-life examples where scale drawings are used
5. What units must be the same when:
    - a) Finding actual length from drawing
    - b) Finding scale

Section B: Actual Length from Drawing (20 marks)

Scale 1:100. Find actual length:

1. Drawing: 5 cm
2. Drawing: 12 cm
3. Drawing: 8.5 cm
4. Drawing: 15 cm

Scale 1:50. Find actual length: 5. Drawing: 6 cm 6. Drawing: 10 cm 7. Drawing: 3.5 cm

Scale 1:200. Find actual dimensions: 8. Rectangle 4 cm × 3 cm 9. Square 7 cm 10. Room 6 cm × 4.5 cm

Section C: Drawing Length from Actual (20 marks)

Scale 1:100. Find drawing length:

1. Actual: 8 m
2. Actual: 15 m
3. Actual: 6.5 m
4. Actual: 12 m

Scale 1:50. Find drawing length: 5. Actual: 5 m 6. Actual: 8 m 7. Actual: 3.5 m

Scale 1:200. Find drawing dimensions: 8. Room 10 m × 8 m 9. Garden 24 m × 18 m 10. Field 50 m × 40 m

Section D: Finding Scale (15 marks)

Find the scale:

1. Drawing 4 cm, Actual 200 cm
2. Drawing 5 cm, Actual 10 m
3. Drawing 6 cm, Actual 3 m
4. Drawing 8 cm, Actual 16 m
5. Drawing 3 cm, Actual 600 cm
6. Map 10 cm, Actual 5 km
7. Plan 7 cm, Actual 14 m

Section E: Area Calculations (15 marks)

1. Scale 1:100
   - Drawing area: 8 cm^2
   - Find actual area
2. Scale 1:50
   - Drawing rectangle: 6 cm × 4 cm
   - Find actual area
3. Scale 1:200
   - Drawing square: 5 cm side
   - Find actual area
4. Scale 1:100
   - Actual area: 20 m^2
   - Find drawing area
5. What is the area scale if length scale is:
   - a) 1:50
   - b) 1:200
   - c) 1:1000

Section F: Map Problems (20 marks)

6. Map scale 1:50,000
   - Two points 8 cm apart on map
   - Find actual distance in km
7. Map scale 1:25,000
   - Actual distance 5 km
   - Find map distance
8. Map scale 1:10,000

- Road measures 12 cm on map
- Find actual length in meters

9. Map scale 1:100,000
    - Towns 15 km apart
    - How far on map?
10. Map scale 1:50,000
    - Points A to B: 6 cm on map
    - Points B to C: 8 cm on map
    - Total actual distance A to C?
11. Map scale 1:25,000
    - Distance on map: 10 cm
    - Walking speed: 5 km/h
    - Time to walk?

Section G: Practical Scale Drawing (15 marks)

1. Draw a scale plan of:
    - Rectangular room 8 m × 6 m
    - Scale 1:100
    - Show door 1 m wide
    - Show window 1.5 m wide
    - Label all dimensions
    - Show scale clearly
2. Draw a scale plan of:
    - Square plot 20 m side
    - Scale 1:200
    - Show house 12 m × 8 m in center
    - Label dimensions

3. Draw to scale 1:50:
    - Rectangle 5 m × 3 m
    - Calculate and mark drawing dimensions

WEEK 11: QUANTITATIVE APTITUDE

TOPIC: Meaning of Quantitative Aptitude; Reasons for Studying It; Solving Problems Related to Plane Shapes/Figures and Scale

CONTENT:

A. INTRODUCTION TO QUANTITATIVE APTITUDE

Definition: Quantitative aptitude is the ability to handle numerical and mathematical problems efficiently and accurately. It involves reasoning with numbers, solving mathematical problems, and applying mathematical concepts to real-life situations.

Components:

4. **Numerical ability** - working with numbers
5. **Problem-solving skills** - finding solutions
6. **Logical reasoning** - thinking systematically
7. **Speed and accuracy** - solving quickly and correctly
8. **Application skills** - using math in real situations

B. IMPORTANCE OF QUANTITATIVE APTITUDE

Why Study Quantitative Aptitude?

1. **Academic Success**
    - Essential for mathematics exams
    - Needed for science subjects
    - Important for competitive exams

- Foundation for higher studies

2. **Career Development**
   - Required in many professions
   - Important for job interviews
   - Needed for aptitude tests
   - Essential for business
3. **Daily Life Applications**
   - Shopping and budgeting
   - Time management
   - Measurement and estimation
   - Financial planning
4. **Mental Development**
   - Improves logical thinking
   - Enhances problem-solving skills
   - Develops analytical ability
   - Sharpens mental calculation
5. **Competitive Advantage**
   - Better exam performance
   - Success in entrance tests
   - Improved job prospects
   - Business decision-making

C. TYPES OF QUANTITATIVE REASONING PROBLEMS

1. Number Problems 2. Ratio and Proportion 3. Percentage Applications 4. Profit and Loss 5. Simple and Compound Interest 6. Time and Work 7. Time, Speed, and Distance 8. Averages 9. Mensuration (Areas and Volumes) 10. Data Interpretation

D. QUANTITATIVE REASONING WITH PLANE SHAPES

Type 1: Perimeter Problems

Example 1: A rectangular field has length 50 m and perimeter 180 m. Find: a) Width b) Area c) Cost to fence at ₦350 per meter

Solution:

a) Perimeter = $2(l + w)$

$180 = 2(50 + w)$

$180 = 100 + 2w$

$80 = 2w$

$w = 40$ m

b) Area = $l \times w$

$= 50 \times 40$

$= 2{,}000 \text{ m}^2$

c) Cost = Perimeter × Rate

$= 180 \times 350$

= ₦63,000

Example 2: A square field has perimeter 120 m. Find: a) Side length b) Area c) Diagonal length

Solution:

a) Perimeter = $4s$

$120 = 4s$

$s = 30$ m

b) Area = s^2

$= 30^2$

$= 900 \text{ m}^2$

c) Diagonal = $s\sqrt{2}$

$= 30\sqrt{2}$

≈ 42.43 m

Type 2: Area Problems

Example 3: A rectangle has area 240 m² and length 20 m. Find: a) Width b) Perimeter c) Cost to tile at ₦2,500/m²

Solution:

a) Area = $l \times w$

$240 = 20 \times w$

$w = 12$ m

b) Perimeter = $2(20 + 12)$

$= 64$ m

c) Cost = $240 \times 2{,}500$

= ₦600,000

Example 4: A rhombus has diagonals 24 cm and 18 cm. Find: a) Area b) If each side is 15 cm, find perimeter

Solution:

a) Area = $(d_1 \times d_2)/2$

$= (24 \times 18)/2$

$= 216$ cm²

b) Perimeter = 4×15

$= 60$ cm

Type 3: Comparison Problems

Example 5: Square A has side 10 cm. Square B has side 15 cm. a) Find area of each b) How much larger is B than A? c) Find ratio of areas

Solution:

a) Area A = $10^2 = 100$ cm²

Area B = $15^2 = 225$ cm²

b) Difference = 225 - 100 = 125 cm²

B is 125 cm² larger

c) Ratio = 100:225 = 4:9

Example 6: Rectangle P: 12 cm × 8 cm Rectangle Q: 15 cm × 10 cm Which has greater perimeter? By how much?

Solution:

Perimeter P = 2(12 + 8) = 40 cm

Perimeter Q = 2(15 + 10) = 50 cm

Q has greater perimeter by 50 - 40 = 10 cm

Type 4: Cost Calculation Problems

Example 7: A rectangular plot 40 m × 30 m needs:

- Fencing at ₦450 per meter
- Tiling at ₦3,000 per m²

Find total cost.

Solution:

Perimeter = 2(40 + 30) = 140 m

Fencing cost = 140 × 450 = ₦63,000

Area = 40 × 30 = 1,200 m²

Tiling cost = 1,200 × 3,000 = ₦3,600,000

Total cost = 63,000 + 3,600,000 = ₦3,663,000

Example 8: A square garden of side 20 m has:

- Path 2 m wide around inside
- Grass costs ₦1,500/m² for inner area
- Paving costs ₦2,500/m² for path

Find total cost.

Solution:

Outer area = 20^2 = 400 m²

Inner garden side = 20 - 2(2) = 16 m

Inner area = 16^2 = 256 m²

Path area = 400 - 256 = 144 m²

Grass cost = 256 × 1,500 = ₦384,000

Path cost = 144 × 2,500 = ₦360,000

Total = ₦744,000

E. QUANTITATIVE REASONING WITH SCALE

Type 1: Scale Conversion Problems

Example 9: A building plan uses scale 1:100.

- Drawing dimensions: 15 cm × 12 cm
- Find: a) Actual dimensions b) Actual area c) Cost to build at ₦50,000/m²

Solution:

a) Length = 15 × 100 = 1,500 cm = 15 m

Width = 12 × 100 = 1,200 cm = 12 m

b) Area = 15 × 12 = 180 m²

c) Cost = 180 × 50,000 = ₦9,000,000

Example 10: Map scale 1:50,000

- School to library: 8 cm on map
- Library to hospital: 6 cm on map
- Find total actual distance in km

Solution:

School to library = 8 × 50,000 = 400,000 cm = 4 km

Library to hospital = 6 × 50,000 = 300,000 cm = 3 km

Total distance = 4 + 3 = 7 km

Type 2: Time and Distance with Scale

Example 11: Map scale 1:25,000

- Two towns are 10 cm apart on map
- Traveling at 50 km/h, find time taken

Solution:

Actual distance = 10 × 25,000 = 250,000 cm = 2.5 km

Time = Distance ÷ Speed

= 2.5 ÷ 50

= 0.05 hours

= 3 minutes

Example 12: Scale 1:100,000

- Runner completes route in 30 minutes at 12 km/h
- How long is route on map?

Solution:

Time = 30 min = 0.5 hours

Actual distance = Speed × Time

= 12 × 0.5

= 6 km = 600,000 cm

Map distance = 600,000 ÷ 100,000 = 6 cm

F. COMBINED REASONING PROBLEMS

Example 13: A rectangular field 60 m × 40 m has:

- Square pond of side 10 m in one corner
- Path 3 m wide along all sides inside
- Remaining area for crops

Find: a) Area for crops b) If yield is 5 kg/m², total yield c) Selling at ₦200/kg, total revenue

Solution:

a) Total area = 60 × 40 = 2,400 m²

Pond area = 10² = 100 m²

Inner dimensions (minus path):

54 m × 34 m = 1,836 m²

Crop area = 1,836 - 100 = 1,736 m²

b) Yield = 1,736 × 5 = 8,680 kg

c) Revenue = 8,680 × 200 = ₦1,736,000

Example 14: A town map at scale 1:20,000 shows:

- Park: rectangle 6 cm × 4 cm
- School: square 3 cm side
- Road connecting them: 5 cm

Find: a) Actual areas b) Actual road length c) Walking time at 4 km/h

Solution:

a) Park:

Actual: 120 m × 80 m = 9,600 m²

School:

Actual: 60 m × 60 m = 3,600 m²

b) Road = 5 × 20,000 = 100,000 cm = 1 km

c) Time = 1 ÷ 4 = 0.25 hours = 15 minutes

G. PERCENTAGE PROBLEMS WITH SHAPES

Example 15: A rectangular plot has area 1,500 m².

- 40% for building
- 25% for garden
- 20% for parking
- Remaining for lawn

Find area of each section.

Solution:

Building = 40% of 1,500 = 600 m²

Garden = 25% of 1,500 = 375 m^2

Parking = 20% of 1,500 = 300 m^2

Lawn = 1,500 - (600 + 375 + 300) = 225 m^2

OR: Lawn = 15% of 1,500 = 225 m^2

Example 16: A square field's side is increased by 20%. Find percentage increase in: a) Perimeter b) Area

Solution:

Let original side = s

New side = 1.2s

a) Original perimeter = 4s

New perimeter = 4(1.2s) = 4.8s

Increase = 4.8s - 4s = 0.8s

% increase = (0.8s/4s) × 100% = 20%

b) Original area = s^2

New area = $(1.2s)^2 = 1.44s^2$

Increase = $1.44s^2 - s^2 = 0.44s^2$

% increase = $(0.44s^2/s^2) \times 100\%$ = 44%

H. RATIO AND PROPORTION PROBLEMS

Example 17: Two rectangles have areas in ratio 3:5. If smaller has area 90 cm^2, find: a) Area of larger b) If both have same width 6 cm, find lengths

Solution:

a) Let areas be 3x and 5x

3x = 90

x = 30

Larger area = $5 \times 30 = 150$ cm²

b) Smaller: Area = $l \times w$

$90 = l \times 6$

$l = 15$ cm

Larger: $150 = l \times 6$

$l = 25$ cm

Example 18: Sides of a rectangle are in ratio 4:3. If perimeter is 140 cm, find: a) Dimensions b) Area

Solution:

a) Let sides be $4x$ and $3x$

Perimeter = $2(4x + 3x) = 14x$

$14x = 140$

$x = 10$

Length = $4 \times 10 = 40$ cm

Width = $3 \times 10 = 30$ cm

b) Area = $40 \times 30 = 1{,}200$ cm²

I. PRACTICAL REASONING STRATEGIES

Strategy 1: Understand the Problem

- Read carefully
- Identify what is given
- Identify what is asked
- Draw diagrams if helpful

Strategy 2: Choose the Right Formula

- Perimeter formulas
- Area formulas
- Scale formulas
- Cost = Quantity × Rate

Strategy 3: Work Systematically

- Show all steps
- Convert units if needed
- Check calculations
- Verify answer makes sense

Strategy 4: Use Estimation

- Round numbers for quick check
- Estimate before calculating
- Check if answer is reasonable

Strategy 5: Multiple Approaches

- Try different methods
- Use what works best
- Check answer different way

J. COMMON QUESTION PATTERNS

Pattern 1: "Find and Calculate"

- Find missing dimension
- Then calculate another value

Pattern 2: "Comparison"

- Compare two shapes

- Find difference or ratio

Pattern 3: "Cost Calculation"

- Find area/perimeter
- Multiply by rate

Pattern 4: "Percentage Change"

- Find new value after % increase/decrease
- Calculate effect

Pattern 5: "Word to Math"

- Translate word problem
- Set up equation
- Solve

EVALUATION:

1. Define quantitative aptitude
2. Give 5 reasons for studying quantitative aptitude
3. A rectangle 15 m × 10 m costs ₦400/m to fence. Find total cost
4. Square has area 144 cm². Find perimeter
5. Map scale 1:50,000. Distance 6 cm. Find actual distance in km
6. Rectangle area 180 m², length 15 m. Find width
7. A field's length increased 10%. Perimeter increases by?
8. Rhombus diagonals 16 cm and 12 cm. Find area
9. Scale 1:100, actual 8 m. Find drawing length
10. Plot 20 m × 15 m. If 30% for building, find building area

---

REVISION QUESTIONS:

1. What is quantitative aptitude?

2. Why is it important for careers?
3. List 5 types of quantitative reasoning problems
4. How do you calculate area of rectangle?
5. Formula for perimeter of square?
6. How to find actual length from scale drawing?
7. If side increases 20%, area increases by?
8. Rectangle vs square: main difference?
9. Scale 1:200 means?
10. Area of rhombus using diagonals?
11. How to check if answer is reasonable?
12. Cost formula for fencing?
13. If map scale is 1:25,000, what does 1 cm represent?
14. Parallelogram area formula?
15. Strategy for solving word problems?

ASSIGNMENT:

Section A: Plane Shapes Problems (25 marks)

1. Rectangle 25 m × 18 m. Find:
   - a) Perimeter
   - b) Area
   - c) Diagonal
   - d) Cost to fence at ₦380/m
   - e) Cost to tile at ₦2,800/m²
2. Square field has perimeter 96 m. Find:
   - a) Side length
   - b) Area
   - c) Cost to grass at ₦1,500/m²

3. Rhombus diagonals 20 cm and 16 cm, sides 13 cm. Find:
    - a) Area
    - b) Perimeter
4. Two squares A and B. A has side 12 cm, B has side 18 cm.
    - a) Find areas
    - b) Find ratio of areas
    - c) How much larger is B?
5. Rectangular garden 30 m × 20 m has path 2 m wide inside.
    - a) Outer area
    - b) Inner area (without path)
    - c) Path area

Section B: Scale Problems (25 marks)

1. Scale 1:100
    - Drawing: 8 cm × 6 cm
    - Find actual dimensions and area
2. Scale 1:50
    - Actual: 12 m × 9 m
    - Find drawing dimensions
3. Map scale 1:50,000
    - Towns 12 cm apart
    - Find actual distance in km
    - Time to drive at 60 km/h
4. Building plan scale 1:200
    - Room on plan: 4 cm × 3 cm
    - Find actual area

- Cost to build at ₦45,000/m²

5. Find scale:

   - Drawing 6 cm, Actual 12 m

Section C: Combined Problems (25 marks)

1. Rectangular plot 50 m × 40 m

   - Building occupies 35%
   - Garden 25%
   - Parking 20%
   - Lawn (remaining)

   Find area of each section

2. Map scale 1:25,000

   - Park shown as 8 cm × 6 cm rectangle
   - Swimming pool shown as 2 cm × 2 cm square

   Find:

   - a) Actual areas
   - b) Which is larger and by how much?

3. Square field side increased 25%

   - Original side: 20 m
   - Find % increase in: a) Perimeter b) Area

4. Rectangle length:width = 5:3

   - Perimeter = 96 cm
   - Find dimensions and area

5. Garden plan scale 1:100

   - Total: 20 m × 15 m
   - Path 1.5 m wide around inside

- Grass costs ₦1,200/m²
- Paving costs ₦2,000/m²

Find total cost

Section D: Application Problems (25 marks)

1. A school has rectangular field 80 m × 60 m
    - Football pitch in center: 70 m × 50 m
    - Running track around it

    Find:

    - a) Track area
    - b) Cost to grass pitch at ₦1,800/m²
    - c) Cost for track surface at ₦2,500/m²
2. Town map scale 1:100,000
    - Hospital to school: 5 cm
    - School to market: 8 cm
    - Market to hospital: 12 cm (different route)

    Find:

    - a) Actual distances in km
    - b) Which route hospital-market is shorter?
    - c) Time to walk hospital to school at 5 km/h
3. Rectangular room 9 m × 6 m
    - Door: 1 m wide
    - Windows: 1.5 m wide (2 windows)
    - Paint walls (excluding door/windows)
    - Wall height: 3 m

    Find:

    - a) Total wall area

- b) Door and window area
- c) Area to paint
- d) Paint cost at ₦350/m²

4. Two rectangles P and Q

    - P: 20 cm × 15 cm
    - Q has same perimeter as P but is square

    Find:

    - a) Perimeter of P
    - b) Side of Q
    - c) Areas of both
    - d) Which has larger area?

5. Plot 40 m × 30 m drawn at scale 1:500

    - Find: a) Drawing dimensions b) Actual area c) Drawing area d) Area scale ratio

---

WEEK 1: REVISION OF SECOND TERM'S EXAMINATION

TOPIC: Representation of Real-Life Situations on a Graph and Reasons

CONTENT:

A. REVIEW OF GRAPHING CONCEPTS

Key Points from Second Term:

- Cartesian plane and coordinates
- Linear equations in two variables
- Graphing linear equations
- Reading and interpreting graphs
- Real-life applications

B. WHY USE GRAPHS FOR REAL-LIFE SITUATIONS?

Advantages of Graphs:

1. **Visual Representation**
    - Easy to understand at a glance
    - Shows patterns and trends clearly
    - Better than long tables or text
2. **Quick Comparison**
    - Compare different data sets easily
    - See relationships between variables
    - Identify maximum and minimum values
3. **Prediction and Estimation**

- Extend lines to predict future values
- Interpolate between known points
- Make informed decisions

4. **Communication**

    - Universal language
    - Effective in presentations
    - Clear to wide audience

5. **Problem Solving**

    - Visualize solutions
    - Identify break-even points
    - Optimize decisions

C. TYPES OF REAL-LIFE GRAPHS

1. Distance-Time Graphs

Shows relationship between distance traveled and time taken.

Example 1: A car travels from home to market

```
 Distance (km)
    |
 50 |        ______________
 40 |       /       \
 30 |      /         \
 20 |     /           \
 10 |    /             \
    |____/               \___
_____________O_________________________________________________ Time (hours)
    0  1  2  3  4  5  6
```

Interpretation:
0-2 hrs: Traveling to market (30 km away)
2-4 hrs: At market (stationary - flat line)
4-6 hrs: Returning home

Key Features:

- **Slope = Speed** (steeper = faster)
- **Horizontal line = Stationary** (not moving)
- **Rising line = Going away**
- **Falling line = Returning**

Example 2: Journey Analysis

Distance from home vs Time:
- First hour: Travels 40 km (speed = 40 km/h)
- Next 2 hours: Stationary
- Last 2 hours: Returns 40 km (speed = 20 km/h)

---

2. Temperature-Time Graphs

Shows how temperature changes over time.

Example 3: Daily Temperature

```
Temperature (°C)
   |
30 |     ___/\___
25 |   _/        \_
20 |  _/           \_
15 | /               \
10 |/                 \
   |_______________________
————————O———————————————————————— Time
```

```
     6am 9am 12pm 3pm 6pm
```

Pattern:
- Coolest at dawn
- Rises through morning
- Peaks at midday
- Falls in evening

---

3. Cost-Quantity Graphs

Shows relationship between cost and quantity purchased.

Example 4: Mobile Phone Bills

Plan: ₦500 monthly + ₦5 per minute

Equation: $C = 5m + 500$

```
  Cost (₦)
      |
 1500 |          /
 1250 |        /
 1000 |      /
  750 |    /
  500 |___/
      |
 ―――――――――――――O――――――――――――――――――――――――――――― Minutes
     0  50 100 150 200
```

Interpretation:
- Fixed cost ₦500 (when m = 0)
- Each minute costs ₦5 more
- 100 minutes costs ₦1,000

4. Profit-Loss Graphs

Example 5: Business Analysis

Fixed costs: ₦200,000 per month **Selling price:** ₦500 per unit **Cost per unit:** ₦300

Equations:

- Revenue: R = 500n
- Cost: C = 300n + 200,000

```
  Amount (₦)
     |
 800k |       /Revenue
 600k |     /
 400k |   /  Cost
 200k |——/————
     | /
 ————————O——————————————————— Units (n)
     0  500 1000
```

Break-even point: Where Revenue = Cost

500n = 300n + 200,000

200n = 200,000

n = 1,000 units

---

5. Growth Graphs

Example 6: Population Growth

```
  Population
     |
 5000 |         •
```

```
4000 |          •
3000 |      •
2000 |  •
1000 |•
     |__________________
————————————O———————————————————————————————— Year
    2020 2021 2022 2023 2024
```

Shows steady growth over years

Can predict future population

---

6. Conversion Graphs

Example 7: Currency Conversion

If $1 = ₦450, graph shows relationship.

```
  Naira (₦)
       |
4500 |              •
3600 |          •
2700 |      •
1800 |   •
 900  |•
       |__________________
  ————————————O———————————————————————————————— Dollars ($)
       0  2  4  6  8  10
```

Straight line through origin

Slope = 450 (exchange rate)

D. READING AND INTERPRETING GRAPHS

Skills Needed:

1. **Identify Variables**
   - What's on x-axis?
   - What's on y-axis?
   - What do they represent?
2. **Read Values**
   - Find coordinates of points
   - Interpolate between points
   - Extrapolate beyond data
3. **Identify Patterns**
   - Increasing or decreasing?
   - Linear or curved?
   - Constant or varying?
4. **Make Conclusions**
   - What does it show?
   - What can we predict?
   - What decisions can be made?

Example 8: Water Tank Graph

```
Volume (L)
      |
200 |•
150 |  \
100 |    \
 50 |      \
```

| •

——————O——————————— Time (min)

0 5 10

Questions:

a) Initial volume? 200 L

b) Final volume? 0 L

c) Time to empty? 10 minutes

d) Rate of drainage? 200/10 = 20 L/min

e) Volume after 3 min? 200 - 3(20) = 140 L

Example 9: Taxi Fare Graph

Equation: C = 300 + 60d

C = Cost, d = distance in km

Cost (₦)

|

1500 | •

1200 | •

900 | •

600 |•

300 |•

|________________

——————O—————————————————— Distance (km)

0 2 4 6 8 10

From graph:

- Fixed charge: ₦300

- Rate per km: ₦60

- Cost for 5 km: ₦600
- Distance for ₦1,200: 15 km

E. CREATING GRAPHS FROM REAL-LIFE DATA

Steps:

1. **Collect Data**
    - Identify variables
    - Take measurements
    - Record in table
2. **Choose Scale**
    - Suit the data range
    - Make graph readable
    - Use convenient units
3. **Label Axes**
    - State variable names
    - Include units
    - Clear and neat
4. **Plot Points**
    - Mark accurately
    - Use appropriate symbols
    - Check each point
5. **Draw Line/Curve**
    - Use ruler for straight lines
    - Smooth curve for curved lines
    - Best fit if scattered data
6. **Title and Legend**

- Descriptive title
- Explain symbols if needed
- Date if relevant

Example 10: Student's Savings

Data collected:

Month	Amount Saved (₦)
Jan	2,000
Feb	2,500
Mar	3,000
Apr	3,500
May	4,000

Create graph showing savings growth

Analysis:

- Linear growth
- Saves ₦500 more each month
- Equation: $S = 500m + 1{,}500$
- Predict June: ₦4,500

F. PRACTICAL APPLICATIONS

Application 1: Business Decision

A company compares two pricing strategies by graphing revenue vs units sold. Graph helps choose better strategy.

Application 2: Health Monitoring

Patient's temperature graphed over days. Helps doctor see improvement or worsening.

Application 3: Academic Performance

Student's test scores graphed over terms. Shows progress, identifies weak areas.

Application 4: Budget Planning

Family expenses graphed monthly. Helps identify spending patterns, areas to save.

Application 5: Sports Performance

Athlete's running time graphed over training weeks. Shows improvement, motivates training.

G. COMMON GRAPH MISTAKES TO AVOID

1. **Wrong Scale**
   - Too cramped or too spread out
   - Inconsistent intervals
2. **Unlabeled Axes**
   - Readers can't understand graph
   - No units shown
3. **Inaccurate Plotting**
   - Points in wrong positions
   - Careless measurement
4. **Wrong Line Type**
   - Straight line through scattered data
   - Connecting unrelated points
5. **Missing Title**
   - Purpose unclear
   - Context missing

EVALUATION:

1. Give 3 advantages of using graphs
2. What does a horizontal line mean on distance-time graph?
3. How do you find speed from distance-time graph?
4. Draw a graph showing: Taxi charges ₦200 + ₦40/km
5. From a temperature graph, temperature was 25°C at 9am and 32°C at 2pm. Find rate of increase
6. What is a break-even point on business graph?
7. If graph passes through origin, what does this mean?
8. How do you predict future values from a graph?
9. What's the difference between interpolation and extrapolation?
10. Why should axes be labeled?

REVISION QUESTIONS:

1. List 5 types of real-life graphs
2. What does slope represent on distance-time graph?
3. Explain "break-even point" in business
4. How do you read values from a graph?
5. What pattern shows on population growth graph?
6. In cost equation $C = 5x + 100$, what does 100 represent?
7. How can graphs help in decision making?
8. What's the purpose of a conversion graph?
9. If a line is steeper, what does this indicate about rate?
10. Why use graphs instead of tables?
11. What does flat line mean on distance-time graph?
12. How do you choose appropriate scale?
13. What's the advantage of visual representation?

14. Can you predict exact future values from graphs? Explain
15. Give 3 real-life situations where graphs are useful

ASSIGNMENT:

Section A: Interpretation (25 marks)

The graph below shows John's journey from home to school and back:

[Provide distance-time graph with multiple stages]

Answer:

1. How far is school from home?
2. How long did John stay at school?
3. What was his speed going to school?
4. What was his speed returning home?
5. Total time for entire journey?

Section B: Temperature Graph (20 marks)

A patient's temperature was recorded:

Time	6am	9am	12pm	3pm	6pm
Temp (°C)	38	38.5	39	38.5	38

1. Draw the graph
2. What was highest temperature and when?
3. When was temperature normal (37°C)?
4. Describe the pattern
5. What can doctor conclude?

Section C: Cost Graphs (25 marks)

6. Mobile Plan: ₦600 + ₦8 per minute

- Draw graph (0 to 100 minutes)
- Find cost for 50 minutes
- Find minutes for ₦1,400

7. Two plans:

   - Plan A: ₦500 + ₦10/min
   - Plan B: ₦800 + ₦5/min
   - Graph both
   - Find break-even point
   - Which is cheaper for 40 minutes?

Section D: Business Application (15 marks)

Company data:

- Fixed costs: ₦150,000/month
- Selling price: ₦400 per item
- Production cost: ₦250 per item

1. Write equations for revenue and cost
2. Draw both on same graph
3. Find break-even point
4. Profit if 1,500 items sold?
5. How many items for ₦100,000 profit?

Section E: Create Your Own (15 marks)

1. Collect data about:

   - Your daily study hours for one week, OR
   - Prices of 5 items at different shops, OR
   - Temperature at different times one day

2. Present in table
3. Draw appropriate graph

4. Write 5 questions about your graph

5. Answer your questions

WEEK 2: ANGLES AND POLYGONS

TOPIC: Definition of Angles; Construction of Angles; Polygons and Interior Angles

CONTENT:

A. ANGLES - DEFINITION AND TYPES

Definition: An angle is the amount of turn between two straight lines that meet at a point (vertex).

Parts of an Angle:

```
     Ray B (Arm)
    /
   /
  /_____ Ray A (Arm)
 O (Vertex)
```

Angle AOB or ∠AOB

Measuring Angles:

- Unit: Degrees (°)
- Tool: Protractor
- Full turn: 360°
- Straight line: 180°
- Right angle: 90°

B. TYPES OF ANGLES

1. Acute Angle

```
 \
  \
```

```
  \___
```

0° < angle < 90°

Examples: 30°, 45°, 60°, 75°

2. Right Angle

```
  |
  |
  |___
```

Angle = 90°

3. Obtuse Angle

```
    /
   /
  /___
```

90° < angle < 180°

Examples: 105°, 120°, 135°, 150°

4. Straight Angle

```
________________
```

Angle = 180°

5. Reflex Angle

(larger turn)

180° < angle < 360°

Examples: 210°, 270°, 315°

6. Complete/Full Angle

(complete turn)

Angle = 360°

C. ANGLE RELATIONSHIPS

1. Complementary Angles Two angles that add up to 90°

Example 1:

30° and 60° are complementary
30° + 60° = 90°

Example 2: Find complement of 35°

Complement = 90° - 35° = 55°

2. Supplementary Angles Two angles that add up to 180°

Example 3:

110° and 70° are supplementary
110° + 70° = 180°

Example 4: Find supplement of 125°

Supplement = 180° - 125° = 55°

3. Vertically Opposite Angles When two lines cross, opposite angles are equal

```
 a° \ / b°
     X
 c° / \ d°
```

a° = c° (opposite)
b° = d° (opposite)
a° + b° = 180° (supplementary)

4. Adjacent Angles Angles that share a common arm and vertex

```
   /
  /b°
 /____a°
```

a° and b° are adjacent

D. CONSTRUCTING ANGLES USING PROTRACTOR AND COMPASS

Using Protractor (Direct Method):

Example 5: Construct 75°

Steps:

6. Draw a base line AB
7. Place protractor center on A
8. Mark 75° from base line
9. Join A to mark
10. Label angle

Using Compass and Ruler (Construction Method):

Basic Angles You Can Construct:

- 60° (equilateral triangle)
- 90° (perpendicular)
- 120° (supplement of 60°)
- 30° (bisect 60°)
- 45° (bisect 90°)
- Others by combination

Constructing 60°:

Steps:

1. Draw line AB

2. With A as center, draw arc

3. With same radius from arc intersection, draw another arc
4. Join A to intersection point
Result: ∠ = 60°

Constructing 90° (Perpendicular):

Steps:
1. Draw line AB
2. From point A, draw arcs above and below line
3. From both intersections, draw arcs to intersect
4. Join A to intersection point
Result: ∠ = 90°

Constructing 30°:

Construct 60°, then bisect it
Result: 30°

Constructing 45°:

Construct 90°, then bisect it
Result: 45°

Constructing 75°:

Method 1: 60° + 15°

- Construct 60°
- Bisect 30° to get 15°
- Add 60° + 15° = 75°

Method 2: 90° - 15°

- Construct 90°
- Construct 30° then bisect for 15°

- Subtract 15° from 90°

Constructing 105°:

Method: 90° + 15° OR 60° + 45°

Construct 90° and 15°, combine them
OR
Construct 60° and 45°, combine them
Result: 105°

Constructing 120°:

Method: Supplement of 60°

Construct 60° on opposite side of 180° line
OR
Construct 60° + 60°
Result: 120°

Constructing 135°:

Method: 90° + 45°

Construct 90°
Construct 45°
Combine them
Result: 135°

E. POLYGONS

Definition: A polygon is a closed plane figure made up of three or more straight line segments.

Parts of a Polygon:

- **Sides:** The straight line segments
- **Vertices:** Points where sides meet

- **Interior angles:** Angles inside the polygon
- **Exterior angles:** Angles outside the polygon
- **Diagonals:** Lines joining non-adjacent vertices

Types of Polygons by Number of Sides:

Name	Sides	Example
Triangle	3	△
Quadrilateral	4	▭
Pentagon	5	⬠
Hexagon	6	⬡
Heptagon	7	
Octagon	8	⯃
Nonagon	9	
Decagon	10	
Dodecagon	12	
n-gon	n	

Types of Polygons by Regularity:

1. Regular Polygon:

- All sides equal
- All angles equal
- Examples: Equilateral triangle, square, regular pentagon

2. Irregular Polygon:

- Sides and/or angles not all equal
- Examples: Rectangle, scalene triangle

3. Convex Polygon:

- All interior angles < 180°
- All vertices point outward

4. Concave Polygon:

- At least one interior angle > 180°
- At least one vertex points inward

F. SUM OF INTERIOR ANGLES

Formula: Sum of interior angles = (n - 2) × 180° OR **Sum = (2n - 4) right angles** (since 1 right angle = 90°)

Where n = number of sides

Derivation: A polygon with n sides can be divided into (n - 2) triangles from one vertex. Each triangle has angle sum = 180° Therefore, total = (n - 2) × 180°

Example 6: Triangle (n = 3)

Sum = (3 - 2) × 180°
 = 1 × 180°
 = 180°

Example 7: Quadrilateral (n = 4)

Sum = (4 - 2) × 180°
 = 2 × 180°
 = 360°

Example 8: Pentagon (n = 5)

Sum = (5 - 2) × 180°

= 3 × 180°

= 540°

Example 9: Hexagon (n = 6)

Sum = (6 - 2) × 180°

= 4 × 180°

= 720°

Example 10: Octagon (n = 8)

Sum = (8 - 2) × 180°

= 6 × 180°

= 1,080°

G. EACH INTERIOR ANGLE OF REGULAR POLYGON

Formula: Each interior angle = [(n - 2) × 180°] / n

OR

Each interior angle = [(2n - 4) × 90°] / n

Example 11: Each angle in regular pentagon

n = 5

Each angle = [(5 - 2) × 180°] / 5

= [3 × 180°] / 5

= 540° / 5

= 108°

Example 12: Each angle in regular hexagon

n = 6

Each angle = [(6 - 2) × 180°] / 6

= 720° / 6

= 120°

Example 13: Each angle in regular octagon

n = 8

Each angle = [(8 - 2) × 180°] / 8

= 1,080° / 8

= 135°

H. EXTERIOR ANGLES

Properties:

11. Exterior angle + Adjacent interior angle = 180°
12. Sum of all exterior angles = 360° (for any convex polygon)

For Regular Polygon: Each exterior angle = 360° / n

Example 14: Each exterior angle of regular pentagon

n = 5

Each exterior angle = 360° / 5 = 72°

Check: Interior + Exterior = 180°

108° + 72° = 180° ✓

Example 15: Each exterior angle of regular hexagon

Each exterior angle = 360° / 6 = 60°

I. FINDING NUMBER OF SIDES

If given each interior angle:

Example 16: Each interior angle is 140°. Find number of sides.

Method 1: Using formula

Each angle = $[(n - 2) \times 180°] / n$

$140 = [(n - 2) \times 180] / n$

$140n = 180n - 360$

$40n = 360$

$n = 9$

The polygon has 9 sides (nonagon)

Example 17: Using exterior angle Each interior angle = 140°

Each exterior angle = 180° - 140° = 40°

$n = 360° / 40° = 9$ sides

J. PROBLEM SOLVING

Example 18: A polygon has 12 sides. Find: a) Sum of interior angles b) Each interior angle if regular c) Each exterior angle if regular

Solution:

a) Sum = $(12 - 2) \times 180°$

$= 10 \times 180°$

$= 1{,}800°$

b) Each interior angle = 1,800° / 12

$= 150°$

c) Each exterior angle = 360° / 12

$= 30°$

OR: 180° - 150° = 30°

Example 19: Four angles of a pentagon are 100°, 110°, 105°, and 95°. Find the fifth angle.

Solution:

Sum of interior angles = (5 - 2) × 180° = 540°

Sum of 4 angles = 100° + 110° + 105° + 95° = 410°

Fifth angle = 540° - 410° = 130°

Example 20: In a hexagon, three angles are each 130° and two are each 110°. Find the sixth angle.

Solution:

Sum = (6 - 2) × 180° = 720°

Known angles = 3(130°) + 2(110°)

= 390° + 220°

= 610°

Sixth angle = 720° - 610° = 110°

Example 21: Each exterior angle of a regular polygon is 24°. How many sides does it have?

Solution:

n = 360° / exterior angle

n = 360° / 24°

n = 15 sides

Example 22: Can a regular polygon have each interior angle of 140°?

Check:

Exterior angle = 180° - 140° = 40°

n = 360° / 40° = 9 ✓

Yes, it's a regular nonagon (9 sides)

Example 23: Can a regular polygon have each interior angle of 170°?

Check:

Exterior angle = 180° - 170° = 10°

n = 360° / 10° = 36 ✓

Yes, it's a 36-sided polygon

EVALUATION:

1. Define an angle
2. What is an acute angle? Give 2 examples
3. Two angles are complementary. One is 37°. Find the other
4. Find supplement of 115°
5. State the formula for sum of interior angles of polygon
6. Find sum of interior angles of heptagon (7 sides)
7. Each interior angle of regular polygon is 120°. How many sides?
8. Pentagon has angles 100°, 110°, 108°, 112°. Find fifth angle
9. Each exterior angle of regular polygon is 30°. Find number of sides
10. Construct angle of 75° (describe steps)

REVISION QUESTIONS:

1. List 5 types of angles with examples
2. What's the difference between complementary and supplementary?
3. When are angles vertically opposite?
4. Define a polygon
5. List first 10 polygons by number of sides
6. What's the difference between regular and irregular polygon?
7. Formula for each interior angle of regular polygon?

8. Why does sum of exterior angles always equal 360°?
9. How do you construct 60° using compass?
10. Can a polygon have each interior angle of 100°? Explain
11. Hexagon means ___ sides
12. Sum of angles in quadrilateral?
13. How many triangles in a pentagon when divided from one vertex?
14. Each exterior angle of square?
15. If each interior angle is 144°, find number of sides

ASSIGNMENT:

Section A: Angle Types and Relationships (20 marks)

1. Classify each angle:
    - a) 45°
    - b) 95°
    - c) 180°
    - d) 270°
2. Find complement of:
    - a) 25°
    - b) 48°
    - c) 62°
    - d) 71°
3. Find supplement of:
    - a) 65°
    - b) 118°
    - c) 90°
    - d) 145°
4. Two angles are complementary. One is twice the other. Find both angles

5. Two angles are supplementary. One is 40° more than the other. Find both angles

Section B: Angle Construction (15 marks)

Describe detailed steps to construct:

1. 60° angle using compass
2. 90° (perpendicular) angle
3. 45° angle
4. 30° angle
5. 75° angle
6. 105° angle
7. 120° angle
8. 135° angle

Section C: Polygons - Basic (20 marks)

9. Complete the table:

Polygon	Sides	Sum of Interior Angles	Each Interior Angle (if regular)	Each Exterior Angle (if regular)
Triangle	3			
Quadrilateral	4			
Pentagon				
Hexagon				
Octagon				
Decagon				

Section D: Problem Solving (25 marks)

1. Hexagon has angles: 115°, 120°, 130°, 110°, 125°. Find sixth angle
2. Regular polygon has 15 sides. Find:
    - a) Sum of interior angles

- b) Each interior angle
- c) Each exterior angle

3. Each interior angle of regular polygon is 156°. Find number of sides
4. Pentagon has 3 angles each 108° and other 2 equal. Find the equal angles
5. Can a regular polygon have:
   - a) Each interior angle 125°?
   - b) Each exterior angle 50°?
   - c) Each interior angle 165°? Explain each answer
6. A polygon has sum of interior angles 1,620°. How many sides?
7. Regular polygon has each exterior angle 36°. Find:
   - a) Number of sides
   - b) Each interior angle
   - c) Sum of interior angles

Section E: Mixed Problems (20 marks)

1. Three angles of quadrilateral are 85°, 95°, 100°. Find fourth angle
2. Octagon has 5 angles each 140° and others equal. Find the equal angles
3. Each exterior angle of regular polygon is 45°. Name the polygon
4. If one angle of triangle is 60° and another is 75°, find third angle
5. A regular polygon has 20 sides. Find:
   - a) Sum of interior angles
   - b) Each interior angle
   - c) Each exterior angle
6. Pentagon has angles in ratio 1:2:3:4:5. Find each angle
7. If each interior angle of regular polygon is 165°, how many sides?

8. Prove: Sum of exterior angles of any polygon is 360°

WEEK 3: ANGLES OF ELEVATION AND DEPRESSION

TOPIC: Definition and Applications; Solving Problems Using Scale Drawings

CONTENT:

A. ANGLE OF ELEVATION

Definition: The angle of elevation is the angle between the horizontal line and the line of sight when looking UP at an object.

Key Components:

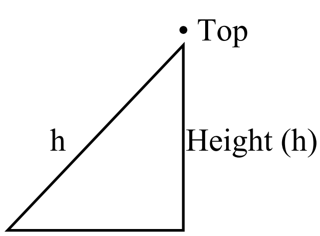
```
       • Object (top of building, plane, etc.)
      /|
     / |
    /  | Height
   /θ  |
  /————|
Observer  Horizontal line
```

θ = Angle of elevation

Important Points:

1. Measured from horizontal upward
2. Always between 0° and 90°
3. Observer looks UP
4. Forms right-angled triangle

Example 1: Standing 50 m from a building, you look up at 30° to see the top.



30°

50 m

Angle of elevation = 30°

Distance = 50 m

Real-Life Examples:

- Looking up at a tall building
- Watching an airplane in the sky
- Observing a bird on a tree
- Viewing the top of a mountain
- Looking at a flag on a pole

B. ANGLE OF DEPRESSION

Definition: The angle of depression is the angle between the horizontal line and the line of sight when looking DOWN at an object.

Key Components:

```
Observer • ————————— Horizontal line
(on top)  \θ
           \
            \
             \
              \
               • Object (on ground)
```

θ = Angle of depression

Important Points:

1. Measured from horizontal downward
2. Always between 0° and 90°
3. Observer looks DOWN
4. Equal to angle of elevation from object to observer

Example 2: From a 40 m high tower, you look down at 25° to see a car.

Tower top • ———————— Horizontal

\25°

h=\40m

\

\

• Car

Angle of depression = 25°

Height = 40 m

Real-Life Examples:

- Looking down from a building
- Viewing ground from an airplane
- Watching from a lighthouse
- Observing from a bridge
- Looking down from a hill

C. RELATIONSHIP BETWEEN ANGLES

Key Principle: Angle of elevation = Angle of depression (alternate angles)

Observer B (high) • ———————— Horizontal

\\θ_2

\

\

\θ_1

• Observer A (low) ———— Horizontal

θ_1 = Angle of elevation (from A to B)

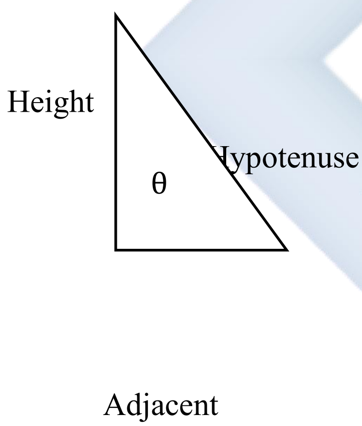
θ_2 = Angle of depression (from B to A)

$\theta_1 = \theta_2$ (alternate angles on parallel lines)

Example 3: A person on ground looks up at plane at 40°. What's the angle of depression from plane to person?

Answer: 40° (they are equal)

D. COMPONENTS OF RIGHT-ANGLED TRIANGLE



Opposite = Height (opposite to angle)

Adjacent = Horizontal distance

Hypotenuse = Line of sight

θ = Angle of elevation or depression

Trigonometric Ratios (Introduction):

- **tan θ = Opposite / Adjacent = Height / Distance**
- sin θ = Opposite / Hypotenuse

- $\cos \theta$ = Adjacent / Hypotenuse

E. SOLVING PROBLEMS USING SCALE DRAWING

Method:

1. Understand the problem
2. Draw a diagram (not to scale)
3. Choose appropriate scale
4. Draw accurate scale diagram
5. Measure required angle/distance
6. Calculate actual value

Example 4: A person stands 30 m from a building. The angle of elevation to the top is 35°. Find the height using scale drawing.

Solution:

Step 1: Draw rough diagram

```
          • Top
        / |
      /   | h
    /     |
  /35°    |
/———————|
 30 m
```

Step 2: Choose scale Scale: 1 cm = 5 m

Step 3: Calculate drawing dimensions

Horizontal distance = 30 m
Drawing distance = 30 ÷ 5 = 6 cm

Step 4: Construct scale drawing

1. Draw horizontal line 6 cm
2. Use protractor to mark 35° angle
3. Draw line from angle
4. Draw vertical from end of horizontal to meet angled line
5. Measure vertical height in drawing

Step 5: Measure and calculate

Suppose vertical measures 4.2 cm
Actual height = 4.2 × 5 = 21 m

Example 5: From top of 50 m building, angle of depression to a car is 28°. Find distance of car from building using scale drawing.

Solution:

Diagram:

```
Building • ————————
          \28°
      50m\
             \
              • Car
       d = ?
```

Scale: 1 cm = 10 m

Steps:

1. Vertical line 5 cm (represents 50 m)
2. Draw horizontal from top
3. Mark 28° below horizontal
4. Draw line to ground level

5. Measure horizontal distance
6. Calculate actual distance

If measured distance = 9.4 cm

Actual distance = 9.4 × 10 = 94 m

Example 6: A ladder 10 m long leans against a wall. The angle it makes with ground is 65°. Find height reached on wall.

Solution:

Diagram:

```
Wall  |\
      | \  10 m (ladder)
    h |   \
      |    \
      |65°   \
      |_____\
```


Scale: 1 cm = 2 m

Steps:

7. Draw line 5 cm (represents 10 m ladder)
8. Make 65° angle with horizontal
9. Draw vertical wall
10. Measure height reached
11. Calculate actual height

If measured = 4.5 cm

Actual height = 4.5 × 2 = 9 m

F. PRACTICAL APPLICATIONS

Application 1: Measuring Tree Height

Stand at known distance from tree, measure angle of elevation to top, calculate height.

Example 7: Standing 20 m from tree, angle of elevation is 42°. Estimate height.

Application 2: Navigation

Ships and planes use angles of depression to find distances to landmarks.

Example 8: From lighthouse 60 m high, ship is spotted at angle of depression 15°. How far is ship?

Application 3: Construction

Architects use angles to design ramps, roofs, staircases.

Example 9: A ramp must rise 2 m over 8 m horizontal distance. Find angle.

Application 4: Surveying

Land surveyors measure distances and heights using angles.

G. STEP-BY-STEP PROBLEM SOLVING

Problem Type 1: Finding Height

Example 10: Distance from building = 40 m, angle of elevation = 38°. Find height.

Given:
- Adjacent (distance) = 40 m
- Angle = 38°
- Find: Opposite (height)

Scale Drawing:
1. Choose scale: 1 cm = 5 m

2. Draw horizontal: 40 ÷ 5 = 8 cm
3. Construct 38° angle
4. Complete triangle
5. Measure height: suppose 6.2 cm
6. Calculate: 6.2 × 5 = 31 m

Height = 31 m (approximately)

Problem Type 2: Finding Distance

Example 11: Building height = 60 m, angle of depression = 32°. Find distance from building.

Given:
- Opposite (height) = 60 m
- Angle of depression = 32° (= angle of elevation from ground)
- Find: Adjacent (distance)

Scale Drawing:
1. Scale: 1 cm = 10 m
2. Draw vertical: 60 ÷ 10 = 6 cm
3. From top, draw 32° below horizontal
4. Complete triangle
5. Measure horizontal: suppose 9.5 cm
6. Calculate: 9.5 × 10 = 95 m

Distance = 95 m (approximately)

Problem Type 3: Finding Angle

Example 12: Person 50 m from building sees top at height 30 m. Find angle of elevation.

Given:

- Adjacent = 50 m
- Opposite = 30 m
- Find: Angle

Scale Drawing:

1. Scale: 1 cm = 10 m
2. Horizontal: 5 cm
3. Vertical: 3 cm
4. Complete triangle
5. Measure angle with protractor: suppose 31°

Angle of elevation = 31° (approximately)

H. COMBINED PROBLEMS

Example 13: Two buildings are 80 m apart. From top of first building (40 m high), angle of depression to top of second building is 15°. Find height of second building.

Diagram:

Building 1 (40m) • ————————
\15°
\\
• Top of Building 2
|
| h_2
|
Ground ——————————————————————
80 m

Solution using scale drawing:

1. Scale: 1 cm = 10 m
2. Draw horizontal 8 cm (80 m)
3. Vertical line 4 cm at left (40 m)
4. From top, draw 15° depression angle
5. Mark where it intersects above second building base
6. Measure this height from ground
7. Calculate actual height

Suppose measured 3.5 cm from ground
Height of second building = 3.5 × 10 = 35 m

Example 14: A plane is flying at 5,000 m altitude. Ground observer sees it at angle of elevation 62°. How far horizontally is plane from observer?

Given:
- Height = 5,000 m
- Angle of elevation = 62°
- Find: Horizontal distance

Scale: 1 cm = 1,000 m

Steps:
1. Vertical: 5 cm (5,000 m)
2. Construct 62° at base
3. Complete triangle
4. Measure horizontal: suppose 2.7 cm
5. Calculate: 2.7 × 1,000 = 2,700 m

Horizontal distance = 2,700 m or 2.7 km

I. QUANTITATIVE APTITUDE PROBLEMS

Example 15: A man walks 100 m away from base of tower. His angle of elevation to top decreases from 45° to 30°. Find height of tower.

Solution Strategy:

- Draw both positions
- Use same height for both triangles
- Find relationship
- Calculate height

Example 16: From ship, lighthouse top is seen at 35° elevation. After sailing 500 m directly toward it, angle is 55°. Find lighthouse height.

Solution: Two positions create two triangles sharing same height. Use scale drawing for each position.

J. ACCURACY AND ESTIMATION

Tips for Accurate Scale Drawings:

1. **Use sharp pencil**
2. **Measure carefully**
3. **Use ruler and protractor properly**
4. **Choose convenient scale**
5. **Draw large enough for accuracy**
6. **Check measurements**
7. **Label clearly**

Common Errors to Avoid:

1. Wrong angle (elevation vs depression)
2. Wrong side as opposite/adjacent

3. Scale conversion mistakes
4. Inaccurate protractor reading
5. Forgetting to convert back to actual units

EVALUATION:

1. Define angle of elevation
2. Define angle of depression
3. Are angle of elevation and depression always equal? Explain
4. Standing 25 m from pole, angle of elevation to top is 40°. Draw scale diagram (1 cm = 5 m) and find height
5. From 80 m high cliff, angle of depression to boat is 25°. Find distance of boat from cliff base
6. What instruments do you need for scale drawings?
7. Give 3 real-life situations where angles of elevation are used
8. If you look up at 35°, someone at that point looks down at you. What angle do they use?
9. Why must we use scale for such problems?
10. A ladder makes 70° with ground. What does this tell you about its position?

REVISION QUESTIONS:

1. Difference between elevation and depression?
2. Which angle is measured upward from horizontal?
3. Draw diagram showing angle of elevation
4. Draw diagram showing angle of depression
5. In right triangle for elevation problems, which is the opposite side?
6. Why are alternate angles important here?
7. List steps for solving using scale drawing
8. What scale would you choose for 100 m distance?
9. How do you convert drawing measurement to actual?

10. Can angle of elevation be 95°? Why or why not?
11. Give formula relating height, distance, and angle
12. What is hypotenuse in these problems?
13. Why is protractor important?
14. Name 3 professions that use these angles
15. How do you check if your answer is reasonable?

ASSIGNMENT:

Section A: Definitions and Diagrams (15 marks)

1. Define and draw diagram:
    - a) Angle of elevation
    - b) Angle of depression
2. Label all parts of your diagrams
3. Explain why angle of elevation from A to B equals angle of depression from B to A
4. Give 5 real-life examples of:
    - a) Angle of elevation
    - b) Angle of depression

Section B: Scale Drawing Problems - Heights (25 marks)

Use appropriate scale for each problem:

5. Standing 30 m from building, angle of elevation to top is 45°. Find height.
6. 40 m from tree base, angle of elevation to top is 38°. Find tree height.
7. Person 50 m from tower sees top at angle of elevation 35°. Find tower height.
8. Kite string is 80 m long at angle of elevation 55° from horizontal. Find kite's height above ground.

9. From 20 m distance, angle to top of pole is 60°. Find pole height.

Section C: Scale Drawing Problems - Distances (25 marks)

1. From 70 m high building, angle of depression to car is 30°. Find distance of car from building.
2. Lighthouse is 50 m high. Ship is spotted at angle of depression 18°. How far is ship?
3. Plane at 8,000 m altitude sees airport at angle of depression 42°. Find horizontal distance to airport.
4. From top of 100 m cliff, boat seen at angle of depression 25°. Find distance of boat from cliff base.
5. Person on 40 m high bridge sees car below at angle of depression 52°. How far is car from bridge base?

Section D: Finding Angles (15 marks)

1. Person 60 m from building sees top at height 45 m. Find angle of elevation.
2. From 80 m high tower, object 150 m away is spotted. Find angle of depression.
3. Ladder 12 m long reaches 10 m up wall. Find angle ladder makes with ground.

Section E: Combined Problems (20 marks)

1. Two people stand 100 m apart on same side of tower. Their angles of elevation to top are 30° and 45°. Find tower height using each angle and check consistency.
2. Person walks toward building. At first position (60 m away), angle of elevation is 25°. At second position (30 m away), angle is 42°. Find height using both positions.
3. From ship, lighthouse top seen at 32° elevation. Ship moves 400 m closer, angle becomes 48°. Find lighthouse height.
4. A hill is 500 m high. From its top, village is seen at angle of depression 15°. From village, hill top is seen at angle of elevation 15°. Find distance from village to hill base.

5. From top of 60 m building, foot of another building is at angle of depression 40°, while its top is at angle of elevation 35°. Find:
    - a) Distance between buildings
    - b) Height of second building

WEEK 4: BEARINGS AND DISTANCES I

TOPIC: Cardinal Points; Locating Positions; Finding Distances and Bearings Using Scale Drawing

CONTENT:

A. CARDINAL POINTS

Definition: Cardinal points are the four main directions used for navigation.

The Four Cardinal Points:

```
         N (North)
         ↑
         |
         |
W (West) ←———•———→ E (East)
         |
         |
         ↓
       S (South)
```

Key Facts:

- **North (N):** 0° or 360°
- **East (E):** 90°
- **South (S):** 180°
- **West (W):** 270°

- Measured **clockwise** from North

B. INTERCARDINAL (ORDINAL) POINTS

Secondary Directions:

```
NW    N    NE
  \   |   /
   \  |  /
W ——————— • ——————— E
   /  |  \
  /   |   \
SW    S    SE
```

The Eight Main Directions:

Direction	Abbreviation	Bearing from N	Description
North	N	0° (360°)	Cardinal
Northeast	NE	45°	Intercardinal
East	E	90°	Cardinal
Southeast	SE	135°	Intercardinal
South	S	180°	Cardinal
Southwest	SW	225°	Intercardinal
West	W	270°	Cardinal
Northwest	NW	315°	Intercardinal

C. THREE-FIGURE BEARINGS

Definition: A bearing is a direction measured clockwise from North, expressed as a three-figure number (000° to 360°).

Rules:

1. Always measured from North
2. Always clockwise
3. Always three figures (use leading zeros)
4. In degrees

Examples:

Example 1: 045° (Northeast direction)
Example 2: 090° (Due East)
Example 3: 180° (Due South)
Example 4: 270° (Due West)
Example 5: 315° (Northwest)

Converting Descriptions to Bearings:

Example 6:

- North = 000° or 360°
- East = 090°
- South = 180°
- West = 270°

Example 7: Northeast

- Halfway between N (0°) and E (90°)
- Bearing = 045°

Example 8: Southeast

- Halfway between E (90°) and S (180°)
- Bearing = 135°

Example 9: Southwest

- Halfway between S (180°) and W (270°)

- Bearing = 225°

Example 10: Northwest

- Halfway between W (270°) and N (360°)
- Bearing = 315°

D. MEASURING BEARINGS

Method 1: Using Compass Directions

Example 11: "30° East of North"

```
N
|   /
|  /30°
|/____
•   E
```

Bearing = 030°

Example 12: "40° South of East"

```
   N
   |
   |
W ———•——— E
     \
      \40°
       \
        S
```

From North clockwise:

90° + 40° = 130°

Bearing = 130°

Example 13: "25° West of South"

From North clockwise:

180° + 25° = 205°

Bearing = 205°

Example 14: "50° North of West"

From North clockwise:

270° + 50° = 320°

Wait - this goes past North!

Actually: 270° - 50° = 220°

OR: From North: 360° - 50° = 310°

Check the direction description carefully!

"50° North of West" means starting from West, go 50° toward North

270° + 50° = 320°... but that's past North

Correct: 270° - 50° = 220°? No...

Actually, "North of West" means from West going toward North

West = 270°, going toward North (which would decrease the angle going clockwise, or we think counterclockwise from West)

Better: 360° - 50° = 310° ✓

Bearing = 310°

Method 2: Back Bearings

Definition: The back bearing (or return bearing) is the bearing from B to A when you know the bearing from A to B.

Rule:

- If bearing from A to B < 180°: Back bearing = Bearing + 180°
- If bearing from A to B > 180°: Back bearing = Bearing - 180°

Example 15: Bearing from A to B is 060°. Find bearing from B to A.

```
    N
    |    /
  A |   /60°
    | /
    •__________
    |B
    |
```

Bearing A to B = 060°

Since 060° < 180°:

Bearing B to A = 060° + 180° = 240°

Example 16: Bearing from P to Q is 215°. Find bearing from Q to P.

Since 215° > 180°:

Bearing Q to P = 215° - 180° = 035°

Example 17: Bearing from X to Y is 130°. Find bearing from Y to X.

Since 130° < 180°:

Bearing Y to X = 130° + 180° = 310°

E. LOCATING POSITIONS

Using Bearings and Distances:

To locate a point, you need:

1. Starting point (reference)
2. Bearing (direction)
3. Distance

Example 18: From point A, point B is on bearing 070° and 50 km away. Locate B.

Steps:

1. Mark point A
2. Draw North line from A
3. Measure 70° clockwise from North
4. Mark scale (e.g., 1 cm = 10 km)
5. Measure 5 cm in direction of 070°
6. Mark point B

N

| /

A | /70°

|/___

•

/ B (50km away)

/5cm on scale 1cm=10km

Example 19: From town T, village V is on bearing 215° and distance 30 km. Locate V.

Choose scale: 1 cm = 5 km

Drawing distance = 30 ÷ 5 = 6 cm

Steps:

1. Mark T
2. Draw North line
3. Measure 215° clockwise
4. Mark 6 cm along this direction
5. Mark point V

F. FINDING DISTANCES USING SCALE DRAWING

Example 20: Town A is 40 km North of town B. Town C is 30 km East of B. Find distance from A to C.

Solution:

Scale: 1 cm = 10 km

Steps:

1. Mark point B
2. Draw North line
3. Mark A 4 cm North of B (40 km)
4. Mark C 3 cm East of B (30 km)
5. Join A to C
6. Measure AC in drawing
7. Calculate actual distance

```
A •
  |
  |4cm (40km)
  |
```

B •————————• C

3cm (30km)

By measurement: AC ≈ 5 cm (on drawing)

Actual distance = 5 × 10 = 50 km

OR by calculation (Pythagoras):

$AC^2 = 40^2 + 30^2 = 1{,}600 + 900 = 2{,}500$

AC = 50 km ✓

Example 21: Ship A is 60 km on bearing 030° from port P. Ship B is 80 km on bearing 120° from P. Find distance between ships using scale drawing.

Solution:

Scale: 1 cm = 20 km

Ship A: 60 ÷ 20 = 3 cm at 030°

Ship B: 80 ÷ 20 = 4 cm at 120°

Steps:

1. Mark port P
2. Draw North line
3. Draw line at 030°, mark A at 3 cm
4. Draw line at 120°, mark B at 4 cm
5. Join A to B
6. Measure AB
7. Calculate actual distance

Suppose AB measures 6.5 cm

Actual distance = 6.5 × 20 = 130 km

G. FINDING BEARINGS USING SCALE DRAWING

Example 22: Town P is 50 km North of town Q. Town R is 40 km East of Q. Find bearing of R from P.

Solution:

Scale: 1 cm = 10 km

P •

|

|5cm

|

Q •————————• R

4cm

Steps:

1. Draw the configuration
2. Draw North line from P
3. Join P to R
4. Measure angle from North (clockwise) to PR
5. Suppose angle measures 129°

Bearing of R from P = 129°

Example 23: From lighthouse L, boat A is 25 km on bearing 040°. Boat B is 35 km on bearing 150°. Find bearing of B from A.

Solution:

Scale: 1 cm = 5 km

Steps:

1. Mark L, draw North
2. Mark A at 5 cm on 040°
3. Mark B at 7 cm on 150°
4. Draw North line from A
5. Join A to B
6. Measure angle from North at A to AB (clockwise)
7. Read bearing

Suppose angle = 168°

Bearing of B from A = 168°

H. PRACTICAL NAVIGATION PROBLEMS

Problem Type 1: Three-Point Navigation

Example 24: A hiker walks 8 km on bearing 060° from camp C to point P, then 6 km on bearing 150° to point Q. Find: a) Distance from C to Q b) Bearing of Q from C

Solution:

Scale: 1 cm = 2 km

Steps:

1. Mark C
2. Draw North, mark P at 4 cm on 060°
3. From P, draw North, mark Q at 3 cm on 150°
4. Join C to Q, measure distance and bearing

Results (by measurement):

a) Suppose CQ = 6 cm → Actual = 12 km

b) Suppose bearing = 094° → Bearing of Q from C = 094°

Problem Type 2: Return Journey

Example 25: Airplane flies 200 km on bearing 135° from airport A to city B. What bearing should it use to return to A?

Bearing A to B = 135°
Since 135° < 180°:
Bearing B to A = 135° + 180° = 315°

Problem Type 3: Shortest Distance

Example 26: Town A is 50 km due North of town B. Town C is 40 km due East of A. A person at B wants to reach C. Find shortest distance.

Solution:

C •
|40km
|
B •———————• A
50km

By scale drawing or Pythagoras:
$BC^2 = BA^2 + AC^2 = 50^2 + 40^2 = 2{,}500 + 1{,}600 = 4{,}100$
$BC = \sqrt{4{,}100} \approx 64$ km

I. COMPASS BEARINGS VS THREE-FIGURE BEARINGS

Compass Bearings (Old Method): Examples: N30°E, S40°W, N60°W

Three-Figure Bearings (Modern Method): Examples: 030°, 220°, 300°

Conversion:

Example 27: Convert N30°E to three-figure bearing

Start from North, go 30° toward East

Three-figure bearing = 030°

Example 28: Convert S40°E to three-figure bearing

Start from South (180°), go 40° toward East

Three-figure bearing = 180° - 40° = 140°

Example 29: Convert N50°W to three-figure bearing

Start from North, go 50° toward West

Three-figure bearing = 360° - 50° = 310°

Example 30: Convert three-figure bearing 240° to compass bearing

240° is past South (180°) toward West (270°)

240° - 180° = 60°

Compass bearing = S60°W

EVALUATION:

1. Name the four cardinal points
2. What bearing represents due East?
3. Convert NE to three-figure bearing
4. If bearing from A to B is 075°, find bearing from B to A
5. Town X is 40 km North of Y. Town Z is 30 km East of Y. Find distance XZ using scale drawing (Scale: 1 cm = 10 km)
6. What is a back bearing?
7. How are bearings always measured?
8. Convert S35°E to three-figure bearing
9. What is the bearing of Southwest?
10. Draw diagram showing bearing of 125° from point A

REVISION QUESTIONS:

1. Define bearing
2. Why are bearings written with three figures?
3. List all eight main compass directions with their bearings
4. How do you find back bearing?
5. What does "clockwise from North" mean?
6. Difference between 045° and 45°?
7. What is the bearing of due South?
8. If you walk on bearing 180°, which direction are you going?
9. Can bearing be 400°? Why or why not?
10. What scale would you use for 100 km distance?
11. Why do we need North line in bearing problems?
12. Convert: a) N45°E b) S30°W to three-figure bearings
13. What instrument measures bearings in real life?
14. Give 3 professions that use bearings
15. If bearing from P to Q is 090°, what is bearing from Q to P?

ASSIGNMENT:

Section A: Cardinal Points and Bearings (20 marks)

1. State three-figure bearings for:
    - a) North
    - b) Northeast
    - c) East
    - d) Southeast
    - e) South
    - f) Southwest

- g) West
- h) Northwest

2. Convert to three-figure bearings:

    - a) N40°E
    - b) S25°E
    - c) S60°W
    - d) N35°W

3. Find back bearings:

    - a) 045°
    - b) 120°
    - c) 210°
    - d) 285°

Section B: Locating Positions (20 marks)

Using scale 1 cm = 10 km, locate these points:

1. From A, B is 50 km on bearing 030°
2. From C, D is 40 km on bearing 135°
3. From E, F is 60 km on bearing 225°
4. From G, H is 35 km on bearing 310°

Section C: Finding Distances (25 marks)

Use appropriate scale for each:

1. Town P is 40 km North of Q. Town R is 30 km East of Q. Find distance PR.
2. Ship A is 50 km on bearing 060° from port. Ship B is 70 km on bearing 150° from same port. Find distance between ships.
3. From camp C, checkpoint A is 25 km on bearing 045°. Checkpoint B is 35 km on bearing 120°. Find distance AB.

4. Village X is 60 km North of Y. Village Z is 80 km East of Y. Find distance XZ.
5. Plane flies 120 km on bearing 030° from airport A to city B, then 100 km on bearing 120° to city C. Find distance AC.

Section D: Finding Bearings (20 marks)

Use scale drawings to find bearings:

1. Town A is 50 km North of B. Town C is 40 km East of B. Find bearing of C from A.
2. From point P, Q is 30 km North and 40 km East. Find bearing of Q from P.
3. Ship sails 80 km on bearing 050° from port P to point A, then 60 km on bearing 140° to point B. Find bearing of B from P.
4. Point X is 45 km on bearing 075° from O. Point Y is 55 km on bearing 160° from O. Find bearing of Y from X.

Section E: Combined Problems (15 marks)

1. A hiker walks from camp C:
   - 12 km on bearing 040° to point A
   - Then 15 km on bearing 130° to point B Find:
   - a) Distance from C to B
   - b) Bearing of B from C
   - c) Bearing to return from B to C directly
2. Three towns A, B, C form a triangle:
   - B is 50 km on bearing 060° from A
   - C is 70 km on bearing 150° from A Find:
   - a) Distance BC
   - b) Bearing of C from B
   - c) Bearing of A from C
3. Boat travels from port P:

- 40 km on bearing 075° to lighthouse L
- Then 50 km on bearing 165° to island I Find:
- a) Distance PI
- b) Bearing of I from P
- c) Direct bearing from I back to P

WEEK 6: BEARINGS AND DISTANCES II

TOPIC: Constructing Triangles; Bisecting Angles

CONTENT:

A. TRIANGLE CONSTRUCTIONS

Definition: Triangle construction means drawing an accurate triangle given certain measurements.

Three Standard Cases:

1. Three sides (SSS)
2. Two sides and included angle (SAS)
3. Two angles and included side (ASA)

B. CONSTRUCTION CASE 1: THREE SIDES (SSS)

When Given: All three side lengths

Example 1: Construct triangle ABC where AB = 7 cm, BC = 6 cm, AC = 5 cm

Steps:

1. Draw base AB = 7 cm

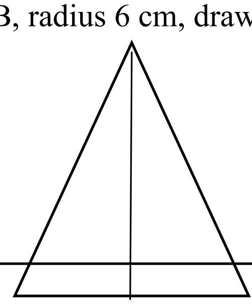
A •————————————————• B

7 cm

2. With compass point at A, radius 5 cm, draw arc above AB

A •~~~~arc (5cm radius)

3. With compass point at B, radius 6 cm, draw arc to intersect first arc



B •~~~~arc (6cm radius)

4. Mark intersection point as C

• C

5cm 6cm

A B

7 cm

5. Join A to C and B to C

Complete triangle ABC

Example 2: Construct triangle PQR with sides 8 cm, 6 cm, 4 cm

Steps:

1. Draw longest side as base: PQ = 8 cm
2. From P: arc radius 6 cm
3. From Q: arc radius 4 cm
4. Mark intersection R
5. Join to complete triangle

Triangle Inequality Check: Before construction, verify:

- Sum of any two sides > third side
- $8 + 6 > 4$ ✓
- $8 + 4 > 6$ ✓

- 6 + 4 > 8 ✓ Valid triangle

Example 3: Can you construct triangle with sides 3 cm, 4 cm, 8 cm?

Check: 3 + 4 = 7, which is NOT > 8
Invalid - cannot construct this triangle
The two shorter sides must add to more than the longest side

C. CONSTRUCTION CASE 2: TWO SIDES AND INCLUDED ANGLE (SAS)

When Given: Two sides and the angle between them

Example 4: Construct triangle ABC where AB = 6 cm, angle BAC = 50°, AC = 5 cm

Steps:

1. Draw AB = 6 cm

```
A •——————————————————• B
     6 cm
```

2. At A, construct angle of 50° using protractor

```
A •———————— (horizontal)
   \
    \ 50°
     \
```

3. On this angle line, mark AC = 5 cm

```
        • C (5cm from A)
```

```
        /
       /50°
      /
  A •————————————————————• B
      6 cm
```

4. Join C to B

```
      • C
     /|\
    / | \
   /  |  \
  5cm 50° 6cm
  /   |   \
 /    |    \
A ————————————————————— B
```

Complete triangle ABC

Example 5: Construct triangle XYZ where XY = 7 cm, angle XYZ = 65°, YZ = 5 cm

Steps:

1. Draw XY = 7 cm
2. At Y, construct 65° angle
3. Mark Z at 5 cm on this angle line
4. Join X to Z

Example 6: Triangle PQR: PQ = 8 cm, angle QPR = 40°, PR = 6 cm

Steps:

1. Draw PQ = 8 cm
2. At P, make 40° angle
3. Mark R at 6 cm from P
4. Join Q to R

D. CONSTRUCTION CASE 3: TWO ANGLES AND INCLUDED SIDE (ASA)

When Given: Two angles and the side between them

Example 7: Construct triangle ABC where AB = 7 cm, angle CAB = 60°, angle CBA = 50°

Steps:

1. Draw AB = 7 cm

```
A •———————————————————• B
     7 cm
```

2. At A, construct angle of 60°

```
A •———————— (direction of C)
  \60°
   \
```

3. At B, construct angle of 50°

```
    (direction of C)
   /50°
  /
B •——————
```

4. Mark intersection point as C

```
     • C
    /|\
   / | \
 60° | 50°
  /  |  \
 /   |   \
A ————————————— B
     7 cm
```

5. Complete triangle

The two angle lines meet at C

Example 8: Triangle PQR: PQ = 8 cm, angle P = 55°, angle Q = 70°

Steps:

1. Draw PQ = 8 cm
2. At P, construct 55° angle
3. At Q, construct 70° angle
4. Lines meet at R

Check: Third angle = 180° - 55° - 70° = 55°

Example 9: Can you construct triangle with angles 80°, 60°, 50°?

Check: 80° + 60° + 50° = 190° ≠ 180°

Invalid - cannot construct this triangle

Sum of angles must equal 180°

E. SPECIAL TRIANGLES

Equilateral Triangle: All sides equal, all angles 60°

Construction (given side 6 cm):

1. Draw base 6 cm
2. From each end, draw arcs radius 6 cm
3. Mark intersection
4. Complete triangle

Isosceles Triangle: Two sides equal, two angles equal

Right-Angled Triangle: One angle = 90°

Example 10: Construct right-angled triangle with legs 5 cm and 4 cm

Steps:

5. Draw one leg horizontal: 5 cm
6. At one end, construct 90° angle (perpendicular)
7. Mark 4 cm on perpendicular
8. Join to complete triangle

F. BISECTING ANGLES

Definition: Angle bisection means dividing an angle into two equal parts.

Method Using Compass:

Example 11: Bisect angle of 80°

Steps:

1. Draw angle ABC = 80°

A •
\

\80°
\
B •————• C

2. With B as center, any convenient radius, draw arc cutting both arms

A •
\ /~~~arc
\ /
\/
B •————• C
arc

3. With compass at both intersections, same radius, draw arcs intersecting inside angle

A •
\ × (intersection)
\/
B •————• C

4. Draw line from B through intersection point

A •
\|
|\40° each side
| \
B •—|——• C
bisector

Each angle = 40°

Example 12: Bisect right angle (90°)

Result: Two angles of 45° each

Steps:

9. Draw 90° angle
10. Place compass at vertex
11. Draw arc cutting both arms
12. From arc intersections, draw arcs meeting inside
13. Join vertex to intersection
14. Result: 45° + 45°

Example 13: Bisect angle of 120°

Result: Two angles of 60° each

Verification: Measure each resulting angle = 60°

G. CONSTRUCTING ANGLES USING BISECTION

Example 14: Construct 45° angle

Method:

1. Construct 90° angle (perpendicular)
2. Bisect it
3. Result: 45°

Example 15: Construct 30° angle

Method:

1. Construct 60° angle (equilateral triangle method)
2. Bisect it

3. Result: 30°

Example 16: Construct 15° angle

Method:

1. Construct 30° angle
2. Bisect it
3. Result: 15°

OR:

1. Construct 60°
2. Bisect to get 30°
3. Bisect 30° to get 15°

Example 17: Construct 22.5° angle

Method:

1. Construct 90°
2. Bisect to get 45°
3. Bisect 45° to get 22.5°

H. PERPENDICULAR BISECTOR OF A LINE

Definition: A line that divides another line into two equal parts at 90°.

Construction:

Example 18: Bisect line segment AB = 8 cm

Steps:

1. Draw AB

A •————————————————• B

8 cm

2. With compass at A, radius > half of AB, draw arcs above and below

A •~~~~arcs

3. With same radius, compass at B, draw arcs intersecting the first arcs

B •~~~~arcs intersecting above and below

4. Join the two intersection points

```
  •
  |
A •——+——• B (perpendicular bisector)
  |
  •
```

This line:

- Passes through midpoint of AB (4 cm from each end)
- Is perpendicular to AB

Example 19: Find midpoint of 10 cm line segment using construction

Result: Mark where perpendicular bisector crosses the line = midpoint (5 cm from each end)

I. APPLICATIONS IN BEARINGS

Example 20: From point P, construct:

- Point A on bearing 070° at distance 5 cm
- Point B on bearing 145° at distance 6 cm
- Then find bearing of B from A

Steps:

4. Mark P, draw North line
5. Construct 70° angle from North, mark A at 5 cm
6. Construct 145° angle from North, mark B at 6 cm
7. Draw North line from A
8. Measure angle from North (at A) to AB
9. This is bearing of B from A

J. ACCURACY IN CONSTRUCTION

Tips for Accurate Work:

1. **Sharp pencil** - for fine lines
2. **Good compass** - holds setting
3. **Clear ruler** - for measurement
4. **Quality protractor** - for angles
5. **Light construction lines** - easy to erase
6. **Check measurements** - verify before finalizing
7. **Label clearly** - mark all points

Common Mistakes:

1. Compass slipping - tighten screw
2. Protractor misalignment - check zero line
3. Wrong scale on ruler - verify
4. Arcs too faint - draw clear arcs
5. Lines not straight - use ruler properly

EVALUATION:

1. State three cases for triangle construction
2. What must you check before constructing triangle with given three sides?
3. Construct triangle with sides 7 cm, 5 cm, 4 cm
4. Describe steps to bisect an angle
5. Construct triangle: AB = 6 cm, angle A = 50°, AC = 5 cm
6. Can you construct triangle with sides 2 cm, 3 cm, 6 cm? Why?
7. Bisect angle of 100°. What is each resulting angle?
8. Construct 75° angle (describe method)
9. What is perpendicular bisector?
10. Construct triangle: PQ = 7 cm, angle P = 60°, angle Q = 50°

REVISION QUESTIONS:

1. Name three standard triangle construction cases
2. What is SSS construction?
3. What is SAS construction?
4. What is ASA construction?
5. Triangle inequality rule?
6. Sum of angles in triangle?
7. How do you bisect angle using compass?
8. What angle results from bisecting 90°?
9. How do you find midpoint using construction?
10. What is included angle in SAS?
11. Can triangle have angles 50°, 60°, 80°?
12. Difference between bisecting angle and bisecting line?
13. How do you construct perpendicular?
14. Name instruments needed for constructions

15. Why use compass instead of protractor for bisection?

ASSIGNMENT:

Section A: Triangle Construction - SSS (20 marks)

Construct these triangles accurately:

1. Sides: 7 cm, 6 cm, 5 cm
2. Sides: 8 cm, 6 cm, 4 cm
3. Equilateral triangle, side 6 cm
4. Isosceles triangle: two sides 7 cm, base 5 cm

For each, label all vertices and write measurements.

Section B: Triangle Construction - SAS (20 marks)

Construct:

1. AB = 7 cm, angle A = 60°, AC = 5 cm
2. PQ = 8 cm, angle P = 45°, PR = 6 cm
3. XY = 6 cm, angle X = 75°, XZ = 6 cm (isosceles)
4. LM = 9 cm, angle M = 90°, MN = 5 cm (right-angled)

Section C: Triangle Construction - ASA (20 marks)

Construct:

1. AB = 8 cm, angle A = 55°, angle B = 65°
2. PQ = 7 cm, angle P = 50°, angle Q = 70°
3. XY = 6 cm, angle X = 60°, angle Y = 60° (equilateral check)
4. MN = 9 cm, angle M = 45°, angle N = 45° (isosceles right-angled)

Section D: Angle Bisection (20 marks)

1. Draw angle 100° and bisect it. Measure each part.

2. Draw angle 140° and bisect it. Verify each part = 70°.

3. Construct right angle (90°) and bisect to get 45°.

4. Construct 60° angle and bisect repeatedly to get:

    - a) 30°
    - b) 15°

5. Draw angle 120° and bisect it. What is each angle?

Section E: Line Bisection (10 marks)

1. Draw line 10 cm and construct perpendicular bisector. Verify midpoint.

2. Draw line 12 cm and find midpoint using construction.

3. Explain why perpendicular bisector method finds exact midpoint.

Section F: Combined Problems (10 marks)

1. Construct triangle ABC: AB = 7 cm, angle A = 50°, angle B = 60° Then:

    - a) Bisect angle C
    - b) Find perpendicular bisector of AB

2. Construct triangle PQR: sides 8 cm, 7 cm, 5 cm Then:

    - a) Bisect each angle
    - b) What do you notice about the three bisectors?

WEEK 8: STATISTICS I – DATA PRESENTATION

TOPIC: Collecting Data; Presenting Data in Ordered Forms and Frequency Tables

CONTENT:

A. INTRODUCTION TO STATISTICS

Definition: Statistics is the branch of mathematics dealing with collection, organization, analysis, interpretation, and presentation of data.

Why Study Statistics?

1. Make informed decisions
2. Understand trends and patterns
3. Predict future outcomes
4. Compare information
5. Solve real-world problems

Real-Life Applications:

- Business: Sales analysis, market research
- Education: Test scores, performance tracking
- Health: Disease tracking, treatment effectiveness
- Sports: Player statistics, team performance
- Government: Census, economic data

B. DATA AND TYPES

Data: Facts, figures, or information collected for reference or analysis.

Types of Data:

1. Qualitative (Categorical) Data

- Descriptive, non-numerical
- Examples:

- Colors (red, blue, green)
- Gender (male, female)
- Favorite subject (Math, English, Science)
- Types of transport (car, bus, bicycle)

2. Quantitative (Numerical) Data

- Measured or counted in numbers
- Two subtypes:

a) Discrete Data:

- Counted, whole numbers only
- Examples:
  - Number of students (25, 30, 45)
  - Number of books (10, 15, 20)
  - Number of goals (0, 1, 2, 3)
  - Family size (3, 4, 5 people)

b) Continuous Data:

- Measured, can have decimal values
- Examples:
  - Height (1.65 m, 1.72 m)
  - Weight (45.5 kg, 60.2 kg)
  - Temperature (25.5°C, 30.2°C)
  - Time (2.5 hours, 3.25 hours)

C. COLLECTING DATA

Data Collection Methods:

1. Direct Observation

- Watch and record
- Example: Counting vehicles passing by

2. Questionnaires/Surveys

- Written questions
- Example: Favorite food survey

3. Interviews

- Face-to-face questions
- Example: Career interest interviews

4. Experiments

- Controlled tests
- Example: Plant growth under different conditions

5. Secondary Sources

- Books, internet, reports
- Example: Historical temperature data

Sources of Data:

Primary Data:

- Collected firsthand by researcher
- Examples from different settings:

From Home:

- Family members' ages
- Monthly expenses
- Water/electricity usage
- Family favorite TV shows

- Meal preferences

From School:

- Class test scores
- Students' heights/weights
- Attendance records
- Favorite subjects
- Transportation modes

From Church/Mosque:

- Attendance numbers
- Age groups
- Service participation
- Donation amounts

From Market:

- Item prices
- Number of sellers
- Customer traffic
- Popular products
- Price variations

Secondary Data:

- Collected by someone else
- Examples:
    - Government census
    - Weather reports
    - Published research

Example 1: Collecting data on favorite fruits in class

Method: Survey

Ask each student: "What is your favorite fruit?"
Record responses:
Apple, Orange, Banana, Apple, Mango, Apple, Orange,
Banana, Banana, Apple, Orange, Mango, Apple, Banana,
Orange, Apple, Banana, Orange, Banana, Apple

Example 2: Collecting data on family sizes

Method: Questionnaire

Ask students to count family members living in their home
Record numbers:
5, 4, 6, 3, 5, 5, 4, 6, 4, 5, 7, 4, 5, 3, 4

D. RAW DATA

Definition: Raw data is collected data in its original, unorganized form.

Example 3: Test scores of 20 students (raw data)

45, 67, 52, 78, 56, 67, 89, 45, 72, 56,
61, 78, 67, 53, 89, 72, 56, 45, 67, 78

Problem with Raw Data:

- Difficult to understand patterns
- Hard to identify trends
- Cannot easily compare
- Not suitable for analysis

Solution: Organize the data!

D. ORGANIZING DATA - ORDERED ARRAY

Definition: Arranging data in ascending (increasing) or descending (decreasing) order.

Example 4: Arrange test scores in ascending order

Raw: 45, 67, 52, 78, 56, 67, 89, 45, 72, 56

Ordered (Ascending): 45, 45, 52, 56, 56, 67, 67, 72, 78, 89

Benefits:

- Easy to find lowest and highest values
- Easy to find middle value
- Quick to spot repeating values
- Beginning of organization

Example 5: Ages of family members (raw data)

Raw: 35, 8, 12, 5, 32, 38, 10, 68
Ordered: 5, 8, 10, 12, 32, 35, 38, 68

From ordered data, we easily see:

- Youngest: 5 years
- Oldest: 68 years
- Range: 68 - 5 = 63 years

F. FREQUENCY DISTRIBUTION

Frequency: The number of times a value or item appears in data.

Tally Marks: Used for counting:

- | = 1
- || = 2
- ||| = 3
- |||| = 4
- ~~||||~~ = 5 (cross through four)
- ~~||||~~ | = 6
- ~~||||~~ || = 7

Example 6: Favorite sports in class

Raw data (responses from 20 students): Football, Basketball, Football, Tennis, Football, Basketball, Football, Tennis, Football, Basketball, Football, Tennis, Football, Basketball, Football, Football, Tennis, Basketball, Football, Football

Frequency Table:

Sport	Tally	Frequency
Football	~~	
Basketball	~~	
Tennis		
Total		**20**

Analysis:

- Most popular: Football (12 students)
- Least popular: Tennis (4 students)
- Total students surveyed: 20

Example 7: Number of siblings

Raw data (30 students): 2, 3, 1, 2, 4, 3, 2, 1, 3, 2, 5, 2, 3, 4, 2, 1, 2, 3, 2, 4, 2, 3, 1, 2, 3, 2, 1, 3, 2, 4

Frequency Table:

Number of Siblings	Tally	Frequency
1	~~	
2	~~	
3	~~	
4		
5		
Total		**30**

Interpretation:

- Most common: 2 siblings (12 students)
- Only 1 student has 5 siblings
- Average appears to be around 2-3 siblings

Example 8: Exam scores (grouped data)

Raw data (25 students): 45, 67, 52, 78, 56, 67, 89, 45, 72, 56, 61, 78, 67, 53, 89, 72, 56, 45, 67, 78, 55, 66, 77, 54, 68

Grouped Frequency Table:

Score Range	Tally	Frequency
40-49		
50-59	~~	
60-69	~~	
70-79	~~	

Score Range	Tally	Frequency
80-89		
Total		**25**

Analysis:

- Most students scored 60-69 (7 students)
- Fewer students in extreme ranges
- Performance is fairly distributed

G. CLASS INTERVALS (GROUPED DATA)

When to Group:

- Large range of values
- Many different values
- Need simpler presentation

Class Interval: A range of values (e.g., 10-19, 20-29)

Class Width: Size of each interval (e.g., 10-19 has width 10)

Class Boundaries: Actual limits (9.5-19.5 for class 10-19)

Example 9: Heights of students (cm)

Raw data: 142, 155, 148, 167, 152, 159, 145, 162, 157, 150, 165, 147, 154, 160, 149, 168, 153, 156, 151, 164

Grouped Frequency Table:

Height Range (cm)	Tally	Frequency
140-144		

Height Range (cm)	Tally	Frequency
145-149		
150-154	~~	
155-159	~~	
160-164		
165-169		
Total		**20**

Most common range: 150-154 cm (6 students)

H. STEPS TO CREATE FREQUENCY TABLE

Steps:

1. **Collect data**
2. **Determine range** (Highest - Lowest)
3. **Decide on intervals** (usually 5-10 classes)
4. **Create table structure**
5. **Use tally marks** to count
6. **Write frequencies**
7. **Check total** matches number of data values

Example 10: Create frequency table for ages

Data: 12, 14, 13, 15, 12, 14, 13, 15, 16, 12, 14, 13, 15, 12, 14

Step 1: Range = 16 - 12 = 4 years

Step 2: Small range, so no grouping needed

Frequency Table:

Age	Tally	Frequency
12		
13		
14		
15		
16		
Total		**15**

I. PRACTICAL DATA COLLECTION EXERCISES

Exercise 1: From Home

Collect data on:

- Number of rooms in houses of 10 classmates
- Monthly water/electricity bill (rounded to nearest ₦1000)
- Number of meals eaten per day
- Hours of TV watched daily

Exercise 2: From School

Collect data on:

- Students' math test scores
- Favorite subject (30 students)
- Mode of transport to school
- Shoe sizes in class

Exercise 3: From Market

Collect data on:

- Price of rice (per kg) from 5 sellers

- Price of tomatoes from 5 sellers
- Number of customers in 30 minutes
- Types of products sold (at 10 stalls)

EVALUATION:

1. Define statistics
2. What is data?
3. Give 3 examples each of qualitative and quantitative data
4. Differentiate between discrete and continuous data
5. What is raw data?
6. Why is it important to organize data?
7. Create frequency table for: Red, Blue, Red, Green, Red, Blue, Red, Green, Blue, Red
8. What are tally marks used for?
9. The data 23, 45, 12, 34, 23, 12, 45, 23 has range = ?
10. Why group data into intervals?

REVISION QUESTIONS:

1. Define statistics
2. Name 4 real-life applications of statistics
3. What's the difference between qualitative and quantitative data?
4. Give 3 examples of discrete data
5. Give 3 examples of continuous data
6. List 5 methods of collecting data
7. What is primary data?
8. What is secondary data?
9. Why organize raw data?
10. What is frequency?
11. How do you represent 7 using tally marks?

12. What information does a frequency table show?
13. When should data be grouped?
14. What is class interval?
15. How do you find range?

ASSIGNMENT:

Section A: Types of Data (15 marks)

1. Classify each as qualitative or quantitative:
   - a) Eye color
   - b) Number of books
   - c) Temperature
   - d) Favorite food
   - e) Height
2. Classify each quantitative data as discrete or continuous:
   - a) Number of students
   - b) Weight in kg
   - c) Age in years
   - d) Distance in meters
   - e) Number of cars

Section B: Data Collection (20 marks)

1. Design a questionnaire to collect data about:
   - Favorite subject (from 20 classmates)
   - Include at least 5 subject options
2. Collect data from 10 family members or neighbors:
   - Ages
   - Number of siblings

- Favorite color

3. Visit a local market and collect:
    - Prices of bread from 5 sellers
    - Prices of eggs (per dozen) from 5 sellers

Section C: Organizing Data (25 marks)

1. Arrange in ascending order:
    - 45, 23, 67, 34, 23, 56, 45, 89, 34, 67
2. The following are test scores:

    56, 78, 45, 67, 56, 89, 45, 78, 56, 67,
    72, 56, 89, 78, 67, 56, 45, 78, 67, 56

    - a) Arrange in ascending order
    - b) Find highest score
    - c) Find lowest score
    - d) Find range

Section D: Frequency Tables - Ungrouped (20 marks)

1. Create frequency table for:

    Red, Blue, Red, Green, Red, Blue, Red, Green,
    Blue, Red, Green, Blue, Red, Red, Blue, Green,
    Red, Blue, Red, Green

2. Number of children in families:

    3, 2, 4, 3, 2, 5, 3, 2, 4, 3,
    2, 3, 4, 2, 3, 4, 2, 3, 2, 4

    Create frequency table with tally marks

3. Shoe sizes:

38, 40, 39, 38, 41, 40, 39, 38, 40, 39,
41, 38, 40, 39, 38, 40, 41, 39, 40, 38

- a) Create frequency table
- b) Which size is most common?
- c) Which size is least common?

Section E: Frequency Tables - Grouped (20 marks)

4. Test scores of 30 students:

   45, 67, 52, 78, 56, 67, 89, 45, 72, 56,
   61, 78, 67, 53, 89, 72, 56, 45, 67, 78,
   55, 82, 48, 63, 71, 85, 59, 74, 68, 51

   Create grouped frequency table using intervals: 40-49, 50-59, 60-69, 70-79, 80-89

5. Ages of people at a community center:

   25, 45, 32, 18, 52, 28, 41, 19, 56, 33,
   47, 22, 38, 29, 51, 24, 43, 20, 49, 35

   Create grouped frequency table using intervals: 15-24, 25-34, 35-44, 45-54, 55-64

WEEK 9: STATISTICS II – GRAPHICAL REPRESENTATION

TOPIC: Plotting Pie Charts; Interpreting Data and Stating Their Usefulness in Everyday Life

CONTENT:

A. INTRODUCTION TO GRAPHICAL REPRESENTATION

Why Use Graphs?

1. Visual presentation - easy to understand
2. Quick comparison of data
3. Shows patterns and trends clearly
4. More appealing than tables
5. Effective communication tool

Types of Statistical Graphs:

1. **Pie Charts** (focus of this week)
2. Bar Charts/Bar Graphs
3. Line Graphs
4. Pictographs
5. Histograms

B. PIE CHARTS

Definition: A pie chart (or circle graph) is a circular diagram divided into sectors (slices) to show the relative sizes of different parts of a whole.

Key Features:

- Circle represents 100% or the total
- Each sector represents a part of the whole
- Size of sector proportional to its value
- Total angle = 360°

When to Use Pie Charts:

- Showing parts of a whole
- Comparing proportions
- Data with few categories (usually 2-8)
- When percentages are important

Structure:

```
   ___
  /   \
 |  A  | ← Sector A
 |-------|
  \ B | C/  ← Sectors B and C
   ---
```

Total circle = 360°

C. STEPS TO DRAW A PIE CHART

Step-by-Step Process:

Step 1: Collect or organize data in frequency table

Step 2: Find total frequency

Step 3: Calculate angle for each category **Formula:** Angle = (Frequency / Total Frequency) × 360°

Step 4: Draw circle using compass

Step 5: Mark center and draw radius

Step 6: Use protractor to measure and draw angles

Step 7: Label each sector

Step 8: Add title and legend (if needed)

D. WORKED EXAMPLES

Example 1: Students' Favorite Subjects

Subject	Frequency
Mathematics	8
English	6
Science	5
Social Studies	3
Total	**22**

Solution:

Step 1: Total frequency = 22 students

Step 2: Calculate angles

Mathematics: (8/22) × 360° = 130.9° ≈ 131° English: (6/22) × 360° = 98.2° ≈ 98° Science: (5/22) × 360° = 81.8° ≈ 82° Social Studies: (3/22) × 360° = 49.1° ≈ 49°

Check: 131° + 98° + 82° + 49° = 360° ✓

Step 3: Draw pie chart

Sector Sizes:

Mathematics: 131°

(largest)

Math / Science

\ | / 82°

\ | /

```
    \|/
  Eng--+--SS
    98°  49°
```

Step 4: Label clearly

- Title: "Students' Favorite Subjects"
- Each sector labeled with subject name
- Can include frequencies or percentages

Example 2: Monthly Budget (₦60,000)

Item	Amount (₦)
Food	24,000
Rent	18,000
Transport	9,000
Others	9,000
Total	**60,000**

Solution:

Calculate angles:

Food: (24,000/60,000) × 360° = 144° Rent: (18,000/60,000) × 360° = 108° Transport: (9,000/60,000) × 360° = 54° Others: (9,000/60,000) × 360° = 54°

Verification: 144° + 108° + 54° + 54° = 360° ✓

Alternative - Using Fractions:

Food: 24,000/60,000 = 2/5
2/5 × 360° = 144°

Rent: 18,000/60,000 = 3/10

3/10 × 360° = 108°

Transport: 9,000/60,000 = 3/20

3/20 × 360° = 54°

Others: 9,000/60,000 = 3/20

3/20 × 360° = 54°

Example 3: Transportation Modes to School (40 students)

Mode	Number of Students
Walk	12
Bus	15
Car	8
Bicycle	5
Total	**40**

Solution:

Calculate angles:

Walk: (12/40) × 360° = 108° Bus: (15/40) × 360° = 135° Car: (8/40) × 360° = 72° Bicycle: (5/40) × 360° = 45°

Check: 108° + 135° + 72° + 45° = 360° ✓

Draw and label pie chart

Example 4: Types of Pets

Pet	Frequency

Pet	Frequency
Dog	18
Cat	12
Fish	6
Bird	4
Total	**40**

Calculate angles:

Dog: (18/40) × 360° = 162° Cat: (12/40) × 360° = 108° Fish: (6/40) × 360° = 54° Bird: (4/40) × 360° = 36°

Percentages (optional for labels): Dog: 18/40 = 45% Cat: 12/40 = 30% Fish: 6/40 = 15% Bird: 4/40 = 10%

E. INTERPRETING PIE CHARTS

Skills for Interpretation:

6. **Identify largest sector** - most common category
7. **Identify smallest sector** - least common category
8. **Compare sectors** - which is bigger/smaller
9. **Calculate fractions/percentages** - what proportion
10. **Make conclusions** - what does it tell us?

Example 5: Given pie chart of fruit sales

```
   Apple
  (144°)
   / \
  /   \
```

Orange   Banana
(108°)   (72°)
\   /
Mango
(36°)

Interpretation:

Total angle: 144° + 108° + 72° + 36° = 360°

Questions and Answers:

a) **Which fruit sold most?** Apple (largest sector, 144°)

b) **Which fruit sold least?** Mango (smallest sector, 36°)

c) **What fraction of sales was Apple?** 144°/360° = 2/5 or 40%

d) **If 120 fruits sold in total, how many were Apples?** (144°/360°) × 120 = 48 apples

e) **How many Oranges?** (108°/360°) × 120 = 36 oranges

f) **Compare Apple to Mango sales** Apple: 144° ÷ Mango: 36° = 4:1 Apple sold 4 times more than Mango

---

Example 6: School Club Membership (pie chart given)

Sectors:
- Drama: 90°
- Sports: 120°
- Science: 75°
- Art: 75°

Total students = 240

Find membership in each club:

Drama: (90/360) × 240 = 60 students Sports: (120/360) × 240 = 80 students Science: (75/360) × 240 = 50 students Art: (75/360) × 240 = 50 students

Interpretation:

- Most popular: Sports (80 students, 1/3 of total)
- Least popular: Science and Art (equal, 50 each)
- Drama: 60 students (1/4 of total)
- Sports has 30 more members than Drama

F. PIE CHARTS WITH PERCENTAGES

Example 7: Family Expenses

Category	Percentage
Food	40%
Housing	30%
Education	20%
Others	10%
Total	**100%**

Calculate angles:

Food: 40% of 360° = 0.40 × 360° = 144° Housing: 30% of 360° = 0.30 × 360° = 108° Education: 20% of 360° = 0.20 × 360° = 72° Others: 10% of 360° = 0.10 × 360° = 36°

If monthly income is ₦100,000: Food: ₦40,000 Housing: ₦30,000 Education: ₦20,000 Others: ₦10,000

Example 8: Time Allocation in a Day

Activity	Hours	Angle

Activity	Hours	Angle
Sleep	8	?
School	7	?
Study	3	?
Leisure	4	?
Others	2	?
Total	**24**	**360°**

Calculate angles:

Sleep: (8/24) × 360° = 120° School: (7/24) × 360° = 105° Study: (3/24) × 360° = 45° Leisure: (4/24) × 360° = 60° Others: (2/24) × 360° = 30°

Interpretation:

- 1/3 of day spent sleeping
- School + Study = 10 hours (42% of day)
- Leisure = 1/6 of day

G. COMPARING DATA USING PIE CHARTS

Example 9: Comparing Two Schools

School A (200 students):

- Math Club: 80 students (144°)
- Science Club: 60 students (108°)
- Arts Club: 40 students (72°)
- Sports: 20 students (36°)

School B (200 students):

- Math Club: 50 students (90°)

- Science Club: 50 students (90°)
- Arts Club: 60 students (108°)
- Sports: 40 students (72°)

Comparison:

- School A: More students in Math and Science
- School B: More balanced distribution
- School B: More students in Arts and Sports
- Both: Same total students

H. USEFULNESS IN EVERYDAY LIFE

1. Business Applications:

- Market share analysis
- Sales distribution
- Customer demographics
- Budget allocation
- Product popularity

Example: Company uses pie chart to show:

- 40% customers from online
- 35% from stores
- 25% from wholesalers

2. Personal Finance:

- Budget planning
- Expense tracking
- Savings allocation
- Investment portfolio

Example: Family budget:

- 35% Food
- 30% Rent
- 15% Transport
- 10% Utilities
- 10% Savings

3. Education:

- Student performance tracking
- Subject preferences
- Resource allocation
- Time management

Example: Student's study time:

- 30% Mathematics
- 25% English
- 20% Sciences
- 25% Other subjects

4. Health and Nutrition:

- Daily calorie intake
- Food groups consumption
- Exercise time allocation
- Sleep patterns

5. Government and Planning:

- Budget allocation
- Population distribution
- Resource planning

- Election results

6. Agriculture:

- Crop distribution
- Land use
- Harvest yield by crop
- Seasonal production

I. ADVANTAGES AND LIMITATIONS

Advantages:

1. Easy to understand visually
2. Shows proportions clearly
3. Good for presentations
4. Compares parts to whole effectively
5. Attractive and engaging
6. Works well with percentages

Limitations:

1. Not suitable for too many categories
2. Difficult to show exact values
3. Hard to compare multiple pie charts
4. Requires all data to sum to 100%
5. Small sectors difficult to label
6. Not good for showing trends over time

When NOT to use:

- More than 8-10 categories
- Comparing multiple datasets

- Showing trends over time
- When exact values are critical
- When sectors would be too small

Better alternatives:

- Bar chart: For many categories or comparisons
- Line graph: For trends over time
- Table: For exact values

J. DRAWING TECHNIQUES AND TIPS

Tools Needed:

1. Compass (for circle)
2. Protractor (for angles)
3. Ruler (for radius lines)
4. Pencil and eraser
5. Calculator
6. Colored pencils (optional)

Tips for Accuracy:

1. **Use sharp pencil** - clear lines
2. **Draw large circle** - easier to measure angles (radius 5-7 cm)
3. **Mark center clearly** - all radii from same point
4. **Measure angles carefully** - check protractor alignment
5. **Start from 12 o'clock** - conventional starting point
6. **Draw clockwise** - standard convention
7. **Label clearly** - inside or outside with lines
8. **Check total** - angles must sum to 360°
9. **Use colors** - different color for each sector

10. **Add legend** - explain colors/patterns

Common Mistakes to Avoid:

1. Angles don't add to 360°
2. Protractor misaligned
3. Circle too small
4. Poor labeling
5. Incorrect calculations
6. Sectors not clearly divided
7. Missing title

Example 10: Complete Problem

Question: A survey of 60 students asked about favorite season:

Season	Students
Summer	24
Winter	18
Spring	12
Autumn	6

Tasks: a) Draw pie chart b) What fraction prefer Summer? c) How many times more students prefer Summer than Autumn? d) What percentage prefer Winter?

Solution:

a) Calculate angles:

Summer: (24/60) × 360° = 144° Winter: (18/60) × 360° = 108° Spring: (12/60) × 360° = 72° Autumn: (6/60) × 360° = 36°

Check: 144° + 108° + 72° + 36° = 360° ✓

[Draw pie chart with sectors clearly labeled]

b) Fraction preferring Summer: 24/60 = 2/5 or 40%

c) Comparison: Summer: 24 students Autumn: 6 students 24 ÷ 6 = 4 times more

d) Percentage preferring Winter: (18/60) × 100% = 30%

EVALUATION:

1. What is a pie chart?
2. What does a full circle represent in degrees?
3. If a sector has angle 90°, what fraction of the circle is it?
4. Calculate the angle for a category with frequency 15 out of total 60
5. Why are pie charts useful?
6. Give 3 everyday uses of pie charts
7. What tools do you need to draw a pie chart?
8. If pie chart shows Food = 120°, what percentage is this?
9. A class of 40 students: 20 like Math, 15 like English, 5 like Science. Draw pie chart
10. What are limitations of pie charts?

REVISION QUESTIONS:

1. Define pie chart
2. Why is it called a "pie" chart?
3. What angle represents the whole circle?
4. How do you calculate angle for each sector?
5. What's the first step in drawing pie chart?
6. Can pie chart show trends over time?
7. What's better for comparing many categories: pie chart or bar chart?
8. If sector is 72°, what fraction of circle is it?
9. List 5 real-life uses of pie charts
10. What does the largest sector show?

11. How do you interpret a pie chart?
12. Name 3 tools needed to draw pie chart
13. Why check that angles total 360°?
14. Can a sector be 400°? Why not?
15. What's the advantage of using percentages in pie charts?

ASSIGNMENT:

Section A: Calculating Angles (20 marks)

Calculate the angle for each sector:

1. Frequency 12 out of total 60
2. Frequency 25 out of total 100
3. Frequency 8 out of total 32
4. Frequency 15 out of total 40
5. Amount ₦30,000 out of total ₦120,000

For the following data, calculate all angles:

6. Dogs: 20, Cats: 15, Birds: 5 (Total: 40)
7. Math: 30, English: 25, Science: 20, Others: 15 (Total: 90)

Section B: Drawing Pie Charts (30 marks)

Draw pie charts for the following data (show all calculations):

1. **Favorite Colors (40 students):**
   - Red: 12
   - Blue: 15
   - Green: 8
   - Yellow: 5
2. **Monthly Budget (₦80,000):**

- Food: ₦32,000
- Rent: ₦24,000
- Transport: ₦16,000
- Others: ₦8,000

3. **Types of Books in Library (360 books):**
   - Fiction: 120
   - Science: 90
   - History: 80
   - Biography: 70
4. **Time Spent Daily (12 hours awake time):**
   - School: 6 hours
   - Homework: 2 hours
   - Play: 3 hours
   - Others: 1 hour

Section C: Interpreting Pie Charts (25 marks)

A pie chart shows distribution of 200 students in different clubs:

- Sports: 100°
- Music: 80°
- Drama: 70°
- Art: 60°
- Science: 50°

Answer:

1. Which club has most members?
2. Which club has least members?
3. How many students in Sports club?
4. How many students in Music club?

5. What fraction of students are in Drama club?
6. What percentage are in Art club?
7. How many more students in Sports than Science?
8. Are there more students in Music or Drama?
9. What's the ratio of Sports to Science members?
10. If 20 students leave Art club, what would be the new angle?

Section D: Real-Life Application (15 marks)

1. Collect data from 30 classmates about:
    - Favorite sport (Football, Basketball, Tennis, Volleyball, Others)
    - Create frequency table
    - Calculate angles
    - Draw pie chart
    - Write 3 interpretations
2. From your family's monthly expenses, create:
    - List of expense categories with approximate amounts
    - Frequency table
    - Pie chart
    - Analysis of spending patterns

Section E: Problem Solving (10 marks)

1. A pie chart shows angle for Mathematics = 120°. If total students = 90, how many study Mathematics?
2. In a pie chart, Food sector is 40%. If total budget is ₦150,000, how much is spent on food? What is the angle of Food sector?
3. Three sectors of pie chart have angles 90°, 100°, and 110°. What is the angle of the fourth sector?
4. If a sector represents 25% of the data, what is its angle?

5. In a class of 48 students, 1/3 like Math. What angle would Math sector have in a pie chart?

WEEK 10: PROBABILITY

TOPIC: Definition of Probability; Importance and Usefulness; Examples of Chances/Events; Generating Events; Solving Problems; Calculating Probability

CONTENT:

A. INTRODUCTION TO PROBABILITY

Definition: Probability is the measure of how likely an event is to occur. It is the study of chance.

Key Terms:

Experiment: An action that produces a result

- Example: Tossing a coin, rolling a die, picking a card

Outcome: A possible result of an experiment

- Example: Heads or Tails (coin toss)

Event: A set of one or more outcomes

- Example: Getting an even number when rolling a die

Sample Space: All possible outcomes of an experiment

- Example: Sample space for coin toss = {Heads, Tails}

B. PROBABILITY SCALE

Range: Probability values range from 0 to 1

0 0.5 1

|————————————————|————————————————|

Impossible Even Chance Certain

0 = Impossible (will never happen)

0.5 = Even chance (50-50)

1 = Certain (will definitely happen)

Examples on Scale:

- **0 (Impossible):**
  - Sun rising in the west
  - Rolling a 7 on a standard die
  - Picking a red ball from bag of only blue balls
- **0.5 (Even chance):**
  - Tossing heads on fair coin
  - Drawing red or black card from deck
- **1 (Certain):**
  - Sun rising in the east
  - Picking a colored ball from bag with only colored balls
  - Getting a number from 1-6 when rolling a die
- **Between 0 and 1:**
  - Most real-life events
  - Raining tomorrow (maybe 0.3)
  - Winning a game (maybe 0.7)

C. EXPRESSING PROBABILITY

Probability can be expressed as:

1. **Fraction:** 1/2, 3/4, 2/5
2. **Decimal:** 0.5, 0.75, 0.4
3. **Percentage:** 50%, 75%, 40%

Conversions:

Example 1:

Fraction: 1/2

Decimal: $1 \div 2 = 0.5$

Percentage: $0.5 \times 100\% = 50\%$

Example 2:

Fraction: 3/4

Decimal: $3 \div 4 = 0.75$

Percentage: $0.75 \times 100\% = 75\%$

D. PROBABILITY FORMULA

Basic Formula:

P(Event) = Number of favorable outcomes / Total number of possible outcomes

Or:

$P(E) = n(E) / n(S)$

Where:

- $P(E)$ = Probability of event E
- $n(E)$ = Number of outcomes in event E
- $n(S)$ = Total number of outcomes in sample space S

E. EXAMPLES WITH COINS

Experiment: Tossing One Coin

Sample Space: $S = \{H, T\}$ where H = Heads, T = Tails **Total outcomes:** $n(S) = 2$

Example 3: Probability of getting Heads

Favorable outcomes = 1 (just H)

Total outcomes = 2 (H or T)

P(Heads) = 1/2 = 0.5 = 50%

Example 4: Probability of getting Tails

P(Tails) = 1/2 = 0.5 = 50%

Experiment: Tossing Two Coins

Sample Space: S = {HH, HT, TH, TT} **Total outcomes:** n(S) = 4

Example 5: Probability of getting 2 Heads

Favorable outcomes = 1 (HH only)

P(2 Heads) = 1/4 = 0.25 = 25%

Example 6: Probability of getting at least 1 Head

Favorable outcomes = 3 (HH, HT, TH)

P(at least 1 Head) = 3/4 = 0.75 = 75%

Example 7: Probability of getting exactly 1 Head

Favorable outcomes = 2 (HT, TH)

P(exactly 1 Head) = 2/4 = 1/2 = 50%

Example 8: Probability of getting no Heads (all Tails)

Favorable outcomes = 1 (TT)

P(no Heads) = 1/4 = 0.25 = 25%

F. EXAMPLES WITH DICE

Experiment: Rolling One Die

Sample Space: S = {1, 2, 3, 4, 5, 6} **Total outcomes:** n(S) = 6

Example 9: Probability of rolling a 4

Favorable outcomes = 1 (just 4)

P(rolling 4) = 1/6 ≈ 0.167 ≈ 16.7%

Example 10: Probability of rolling an even number

Even numbers = {2, 4, 6}

Favorable outcomes = 3

P(even) = 3/6 = 1/2 = 0.5 = 50%

Example 11: Probability of rolling an odd number

Odd numbers = {1, 3, 5}

Favorable outcomes = 3

P(odd) = 3/6 = 1/2 = 50%

Example 12: Probability of rolling a number less than 4

Numbers less than 4 = {1, 2, 3}

Favorable outcomes = 3

P(less than 4) = 3/6 = 1/2 = 50%

Example 13: Probability of rolling a number greater than 4

Numbers greater than 4 = {5, 6}

Favorable outcomes = 2

P(greater than 4) = 2/6 = 1/3 ≈ 33.3%

Example 14: Probability of rolling a 7

No 7 on standard die

Favorable outcomes = 0

P(rolling 7) = 0/6 = 0 (Impossible)

Example 15: Probability of rolling a number from 1 to 6

All outcomes are 1-6

Favorable outcomes = 6

P(1 to 6) = 6/6 = 1 (Certain)

Experiment: Rolling Two Dice

Sample Space: 36 possible outcomes

(1,1) (1,2) (1,3) (1,4) (1,5) (1,6)

(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)

(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)

(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)

(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)

(6,1) (6,2) (6,3) (6,4) (6,5) (6,6)

Example 16: Probability of getting sum of 7

Favorable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)

Number of favorable outcomes = 6

P(sum = 7) = 6/36 = 1/6 ≈ 16.7%

Example 17: Probability of getting doubles (same number on both dice)

Doubles: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6)

Favorable outcomes = 6

P(doubles) = 6/36 = 1/6 ≈ 16.7%

G. EXAMPLES WITH CARDS

Standard Deck: 52 cards

- 4 suits: Hearts (♥), Diamonds (♦), Clubs (♣), Spades (♠)
- Each suit: 13 cards (Ace, 2-10, Jack, Queen, King)

- Red cards: 26 (Hearts and Diamonds)
- Black cards: 26 (Clubs and Spades)

Example 18: Probability of drawing a Heart

Number of Hearts = 13
Total cards = 52
P(Heart) = 13/52 = 1/4 = 0.25 = 25%

Example 19: Probability of drawing a red card

Red cards = 26
P(red) = 26/52 = 1/2 = 50%

Example 20: Probability of drawing a King

Number of Kings = 4
P(King) = 4/52 = 1/13 ≈ 7.7%

Example 21: Probability of drawing an Ace or a King

Aces = 4, Kings = 4
Total favorable = 8
P(Ace or King) = 8/52 = 2/13 ≈ 15.4%

H. PROBABILITY WITH COLORS/BALLS

Example 22: A bag contains 5 red balls, 3 blue balls, and 2 green balls.

Total balls = 10

a) Probability of picking red ball:

P(red) = 5/10 = 1/2 = 50%

b) Probability of picking blue ball:

P(blue) = 3/10 = 30%

c) Probability of picking green ball:

P(green) = 2/10 = 1/5 = 20%

d) Probability of picking red or blue:

Red or blue = 5 + 3 = 8
P(red or blue) = 8/10 = 4/5 = 80%

e) Probability of not picking green:

Not green = 10 - 2 = 8
P(not green) = 8/10 = 4/5 = 80%

Example 23: A box contains 12 marbles: 4 red, 5 blue, 3 yellow

Total = 12

a) P(red) = 4/12 = 1/3 ≈ 33.3% b) P(blue) = 5/12 ≈ 41.7% c) P(yellow) = 3/12 = 1/4 = 25% d) P(not yellow) = 9/12 = 3/4 = 75%

I. COMPLEMENTARY EVENTS

Definition: The complement of an event E is the event "not E", denoted as E' or Ē.

Rule: P(E) + P(E') = 1

Or: **P(E') = 1 - P(E)**

Example 24: If P(rain) = 0.3, find P(no rain)

P(no rain) = 1 - P(rain)
= 1 - 0.3
= 0.7 = 70%

Example 25: If P(passing exam) = 4/5, find P(failing)

P(failing) = 1 - 4/5

= 5/5 - 4/5

= 1/5 = 20%

J. EXPERIMENTAL PROBABILITY

Definition: Probability based on actual experiments or past observations.

Formula: P(Event) = Number of times event occurred / Total number of trials

Example 26: A coin is tossed 50 times. Heads appears 28 times.

Experimental probability of Heads:

P(Heads) = 28/50 = 14/25 = 0.56 = 56%

Note: This differs from theoretical probability (50%), but approaches it with more trials.

Example 27: A die is rolled 60 times with results:

Number	Frequency
1	8
2	12
3	9
4	11
5	10
6	10

Experimental probabilities:

P(1) = 8/60 = 2/15 ≈ 13.3%

P(2) = 12/60 = 1/5 = 20%

P(3) = 9/60 = 3/20 = 15%

P(4) = 11/60 ≈ 18.3%

P(5) = 10/60 = 1/6 ≈ 16.7%

P(6) = 10/60 = 1/6 ≈ 16.7%

Theoretical probability for each number = 1/6 ≈ 16.7%

K. USING LUDO FOR PROBABILITY

Ludo Die: Same as standard die (1-6)

Activity: Roll die multiple times and record

Example 28: Class experiment - Rolling Ludo die

Each student rolls 10 times, class combines data:

Number	Frequency (out of 300 rolls)
1	48
2	52
3	49
4	51
5	50
6	50

Experimental probabilities: All close to 1/6 (theoretical) = 50/300 ≈ 16.7%

Observation: More trials → experimental closer to theoretical

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L. IMPORTANCE AND USEFULNESS OF PROBABILITY

1. Weather Forecasting:

- "70% chance of rain tomorrow"

- Helps people plan activities

2. Insurance:

- Calculate risk
- Determine premiums
- Life, health, car insurance based on probability

3. Medical Diagnosis:

- Probability of disease given symptoms
- Treatment success rates
- Drug effectiveness

4. Sports:

- Predicting match outcomes
- Player performance statistics
- Betting odds

5. Business Decisions:

- Risk assessment
- Investment decisions
- Market predictions
- Quality control

6. Games and Gambling:

- Casino games designed using probability
- Lottery odds
- Card game strategies

7. Agriculture:

- Crop yield predictions

- Weather impact on farming
- Pest control planning

8. Education:

- Test design
- Grading systems
- Predicting student success

9. Daily Life:

- Traffic planning
- Shopping decisions
- Time management

10. Science and Research:

- Hypothesis testing
- Experimental design
- Data analysis

M. PROBABILITY PROBLEMS

Example 29: A bag has 15 balls: 6 red, 5 blue, 4 green

a) P(red) = ?

P(red) = 6/15 = 2/5 = 0.4 = 40%

b) P(not red) = ?

P(not red) = 1 - 2/5 = 3/5 = 60%
OR: (5 + 4)/15 = 9/15 = 3/5

c) P(blue or green) = ?

P(blue or green) = (5 + 4)/15 = 9/15 = 3/5 = 60%

d) If 3 red balls are removed, what is new P(red)?

New: 3 red, 5 blue, 4 green = 12 total

P(red) = 3/12 = 1/4 = 25%

Example 30: Two coins are tossed

a) List all possible outcomes

Sample space = {HH, HT, TH, TT}

b) P(2 heads) = ?

P(HH) = 1/4 = 25%

c) P(1 head and 1 tail) = ?

Outcomes: HT, TH

P = 2/4 = 1/2 = 50%

d) P(at least 1 tail) = ?

Outcomes: HT, TH, TT

P = 3/4 = 75%

EVALUATION:

1. Define probability
2. What is sample space?
3. What probability value means "impossible"?
4. What probability value means "certain"?
5. A die is rolled. Find P(rolling 5)
6. A coin is tossed. Find P(Heads)
7. A bag has 3 red and 7 blue balls. Find P(red)

8. What is complementary event?
9. If P(event) = 0.6, find P(not event)
10. Give 3 real-life uses of probability

REVISION QUESTIONS:

1. What is probability?
2. Give 3 examples of experiments in probability
3. What is an outcome?
4. What is sample space?
5. What values can probability take?
6. Express 3/4 as decimal and percentage
7. State the probability formula
8. What is experimental probability?
9. How many outcomes when tossing 2 coins?
10. How many outcomes when rolling 1 die?
11. In a deck of cards, how many red cards?
12. What is complementary probability?
13. Can probability be greater than 1? Why?
14. If event is certain, its probability = ?
15. Name 5 fields that use probability

ASSIGNMENT:

Section A: Basic Concepts (15 marks)

1. Define:
    - a) Probability
    - b) Sample space
    - c) Event

- d) Outcome

2. State whether each probability is impossible (0), unlikely (0-0.5), even chance (0.5), likely (0.5-1), or certain (1):
   - a) 0.2
   - b) 1
   - c) 0.5
   - d) 0.9
   - e) 0
3. Convert:
   - a) 1/4 to decimal and percentage
   - b) 0.75 to fraction and percentage
   - c) 60% to fraction and decimal

Section B: Coin Tossing (20 marks)

1. One coin is tossed. Find:
   - a) P(Heads)
   - b) P(Tails)
2. Two coins are tossed. List sample space and find:
   - a) P(2 Heads)
   - b) P(2 Tails)
   - c) P(exactly 1 Head)
   - d) P(at least 1 Head)
   - e) P(no Heads)
3. Conduct experiment: Toss a coin 30 times
   - Record results
   - Calculate experimental P(Heads)
   - Compare with theoretical probability

Section C: Die Rolling (25 marks)

1. One die is rolled. Find:
    - a) P(rolling 3)
    - b) P(rolling even number)
    - c) P(rolling odd number)
    - d) P(rolling number > 4)
    - e) P(rolling number ≤ 3)
    - f) P(rolling 7)
    - g) P(rolling 1, 2, 3, 4, 5, or 6)
2. Two dice are rolled. Find:
    - a) Total possible outcomes
    - b) P(sum = 7)
    - c) P(doubles)
    - d) P(sum > 10)
3. Conduct experiment: Roll a die 40 times
    - Record frequency of each number
    - Calculate experimental probability for each
    - Compare with theoretical

Section D: Cards (15 marks)

A card is drawn from standard 52-card deck. Find:

1. P(Heart)
2. P(red card)
3. P(black card)
4. P(King)
5. P(Ace)
6. P(face card) - Jack, Queen, or King

7. P(number card 2-10)
8. P(red King)
9. P(not a Heart)
10. P(Ace or King)

Section E: Colors/Balls (20 marks)

1. A bag contains 8 red, 5 blue, 7 green balls. Find:
   - a) Total balls
   - b) P(red)
   - c) P(blue)
   - d) P(green)
   - e) P(not green)
   - f) P(red or blue)
2. A box has 15 marbles: 6 yellow, 4 white, 5 black. Find:
   - a) P(yellow)
   - b) P(white)
   - c) P(black)
   - d) P(not black)
   - e) If 2 yellow marbles are removed, find new P(yellow)
3. Create your own: Use colored paper/objects
   - Count different colors
   - Calculate probability for each
   - Draw one randomly 20 times
   - Calculate experimental probability
   - Compare results

Section F: Complementary Events (10 marks)

1. If P(rain) = 0.4, find P(no rain)

2. If P(winning) = 3/5, find P(losing)

3. If P(passing) = 0.85, find P(failing)

4. If P(event A) = 2/7, find P(not A)

5. If P(sunny) = 60%, find P(not sunny)

Section G: Real-Life Application (15 marks)

6. Design a simple game using:
   - Coins, or
   - Dice, or
   - Colored cards Write rules and calculate probability of winning
7. Research and explain how probability is used in:
   - Weather forecasting
   - Sports
   - Your chosen career field
8. Explain why probability is important in making decisions

WEEK 11: REVISION

COMPREHENSIVE REVISION - THIRD TERM

REVISION TOPICS SUMMARY:

Week 1: Graphical Representation of Real-Life Situations

- Distance-time graphs
- Cost graphs
- Temperature graphs
- Interpretation and prediction

Week 2: Angles and Polygons

- Types of angles
- Angle relationships
- Constructing angles
- Polygons and interior angles
- Exterior angles

Week 3: Angles of Elevation and Depression

- Definitions and differences
- Right-angled triangles
- Scale drawings
- Finding heights and distances

Week 4: Bearings and Distances I

- Cardinal points
- Three-figure bearings
- Back bearings
- Locating positions
- Finding distances and bearings

Week 6: Bearings and Distances II (Triangle Construction)

- SSS, SAS, ASA constructions
- Bisecting angles
- Perpendicular bisectors

Week 8: Statistics I - Data Presentation

- Types of data
- Collecting data
- Frequency tables

- Organizing data

Week 9: Statistics II - Pie Charts

- Drawing pie charts
- Calculating angles
- Interpreting pie charts
- Real-life applications

Week 10: Probability

- Basic concepts
- Sample space
- Calculating probability
- Complementary events
- Applications

REVISION EXERCISES:

Exercise Set 1: Angles and Polygons

1. Find sum of interior angles of:
   - a) Pentagon
   - b) Octagon
   - c) Decagon
2. Each interior angle of regular polygon is 140°. Find number of sides.
3. Hexagon has angles: 110°, 120°, 125°, 115°, 130°. Find sixth angle.
4. Each exterior angle of regular polygon is 36°. Find:
   - a) Number of sides
   - b) Each interior angle
5. Construct:

- a) Angle of 75°
- b) Triangle with sides 7 cm, 6 cm, 5 cm
- c) Triangle: AB = 8 cm, angle A = 60°, AC = 6 cm

Exercise Set 2: Angles of Elevation and Depression

1. Standing 40 m from building, angle of elevation to top is 35°. Using scale 1 cm = 10 m, find height.
2. From 60 m high tower, car seen at angle of depression 28°. Find distance from tower base.
3. Ladder 12 m long leans at angle 65° with ground. Find height reached on wall.
4. From lighthouse 80 m high, ship spotted at angle of depression 18°. Find distance of ship.

Exercise Set 3: Bearings

1. Convert to three-figure bearings:
   - a) N40°E
   - b) S35°W
   - c) Northwest
2. If bearing from A to B is 075°, find bearing from B to A.
3. Town P is 50 km North of Q. Town R is 40 km East of Q. Using scale 1 cm = 10 km:
   - a) Find distance PR
   - b) Find bearing of R from P
4. Ship sails 60 km on bearing 040° from port P to point A, then 80 km on bearing 130° to port Q. Find:
   - a) Distance PQ
   - b) Bearing of Q from P

Exercise Set 4: Statistics - Frequency Tables

1. Create frequency table for:

   Red, Blue, Red, Green, Red, Blue, Red, Green,
   Blue, Red, Green, Blue, Red, Red, Blue

2. Test scores:

   56, 78, 45, 67, 56, 89, 45, 78, 56, 67,
   72, 56, 89, 78, 67, 56, 45, 78, 67, 56

   Create grouped frequency table using intervals: 40-49, 50-59, 60-69, 70-79, 80-89

Exercise Set 5: Pie Charts

1. Draw pie chart for favorite fruits (40 students):
   - Apple: 15
   - Orange: 12
   - Banana: 8
   - Mango: 5
2. A pie chart shows students in clubs:
   - Sports: 120°
   - Music: 90°
   - Drama: 80°
   - Art: 70°

   If total = 180 students, find:

   - a) Students in each club
   - b) Which club has most members
   - c) Percentage in Sports

Exercise Set 6: Probability

1. A die is rolled. Find:
    - a) P(rolling 4)
    - b) P(even number)
    - c) P(number > 4)
2. Two coins tossed. Find:
    - a) P(2 Heads)
    - b) P(at least 1 Tail)
3. Bag has 12 balls: 5 red, 4 blue, 3 green. Find:
    - a) P(red)
    - b) P(not green)
    - c) P(blue or green)
4. If P(sunny) = 0.65, find P(not sunny)
5. Card drawn from deck. Find:
    - a) P(Heart)
    - b) P(red card)
    - c) P(King)

MIXED PRACTICE PROBLEMS:

Problem 1: A regular polygon has each interior angle 156°. Find:

- a) Number of sides
- b) Each exterior angle
- c) Sum of interior angles

Problem 2: From point A, point B is 45 km on bearing 125°. From A, point C is 60 km on bearing 215°. Using scale drawing:

- a) Locate B and C
- b) Find distance BC
- c) Find bearing of C from B

Problem 3: A class of 30 students scored:

45, 67, 52, 78, 56, 67, 89, 45, 72, 56,
61, 78, 67, 53, 89, 72, 56, 45, 67, 78,
55, 82, 48, 63, 71, 85, 59, 74, 68, 51

- a) Create grouped frequency table
- b) Draw pie chart
- c) What percentage scored 70-79?

Problem 4: Two dice are rolled. Find:

- a) P(sum = 8)
- b) P(doubles)
- c) P(sum > 9)

Problem 5: Triangle construction: AB = 9 cm, angle A = 50°, angle B = 60°

- a) Construct triangle
- b) Measure AC and BC
- c) Bisect angle C
- d) What is angle C?
